Shift-Aligned Reduced-Order Models and Weak-Form PDE Discovery for Interacting-Agent Densities from Noisy Trajectories

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Abstract

Can interpretable macroscopic PDEs be recovered from noisy interacting-agent trajectories? This thesis investigates the question for Vicsek–Morse particle systems by pairing shift-aligned Proper Orthogonal Decomposition (POD) forecasting (Multivariate Autoregression (MVAR) and LSTM latent time-steppers) with weak-form PDE discovery (Weak-form Sparse Identification of Nonlinear Dynamics (WSINDy)) on density and polarisation flux fields across nine dynamical regimes and four initial-condition families.

WSINDy recovers sparse, regime-consistent effective PDEs—matching the mechanism structure anticipated from kinetic theory (advection, diffusion, Morse forcing) with density weak-form $R^2 > 0.99$ in eight of nine regimes (the ninth at 0.95)—yet the identified coefficients often reside outside the regime of numerical stability for autonomous forward-integration. This *closure gap* between identification and rollout quality is itself a diagnostic: the recovered coefficients are effective finite-N parameters at the chosen KDE bandwidth, and the gap reveals which closure ingredients the effective description still lacks. In particular, the three-field library (ρ, p_x, p_y) appears insufficient to close the mean-field description when agent speed is state-dependent, suggesting that variable-speed regimes require additional closure fields beyond density and polarisation flux.

On the forecasting side, the 116× parameter-count gap between MVAR (1,824 parameters) and LSTM (211,091) does not yield proportional accuracy gains: MVAR achieves forecast $R^2 > 0.99$ on the strongest constant-speed regimes (CS + PV median 0.995), while variable-speed variants degrade sharply (VS median 0.425 for MVAR, -0.052 for LSTM), exposing a shift-predictability bottleneck that neither model class overcomes. Together, these findings delineate the boundary between what is identifiable and what is predictable from coarse-grained collective-motion data.[1]

[1] Code and configurations: https://github.com/eugenio-macias/wsindy-manifold.

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Notation and Symbols

Principal mathematical symbols used throughout this thesis.

Geometry and Domains

$\Omega = [0, L_x] \times [0, L_y]$	Computational domain
Γ	Boundary of Ω
n_x, n_y	Density-grid resolution

Simulation Parameters

N	Number of particles
T_{sim}	Simulation duration (s)
Δt	Simulation time-step
v_0	Particle speed
η	Angular-noise strength
R	Alignment interaction radius

Fields and Densities

$\rho(t, x, y)$	Agent density field
$p_x,\ p_y$	Polarisation flux
$\Phi(t, x, y)$	Morse potential field
$W(r)$	Morse interaction kernel
$C_a,\ C_r$	Attraction / repulsion strengths
$\ell_a,\ \ell_r$	Attraction / repulsion scales

Differential Operators

∂_t	Time derivative
$\nabla,\ \nabla\cdot$	Gradient / divergence
Δ	Laplacian
Δ^2	Biharmonic operator

Shift Alignment

ρ_{ref}	Reference density
$(\Delta y_t, \Delta x_t)$	Per-frame pixel shifts
$\tilde{\rho}$	Aligned density field

Data and Density Estimation

$\{\mathbf{x}_i(t)\}$	Agent trajectories
$\{\mathbf{v}_i(t)\}$	Agent velocities
σ_h	KDE bandwidth
$g(\cdot)$	Density transform
ϵ	Numerical offset

POD / SVD

$\mathbf{X}$	Snapshot matrix
$\overline{\mathbf{X}}$	Temporal mean field
$\mathbf{U\Sigma V}^\top$	SVD factorisation
$\mathbf{U}_r$	POD basis (first r columns)
r	Truncation rank
ξ	Energy threshold (0.995)
$\mathbf{y}(t)$	Latent coordinates
M	Number of training runs
Δ_r	Temporal subsampling factor

MVAR Forecasting

p	VAR lag order
α	Ridge regularisation
$\mathbf{A}^{(1)}, \ldots, \mathbf{A}^{(p)}$	VAR coefficient matrices
$\mathbf{c}$	VAR intercept / bias
$\mathbf{C}$	Companion matrix ($pd \times pd$)
R^2_{train}	Training R^2

WSINDy (Weak-form PDE Discovery)

$\psi(t, x, y)$	Compact test function
$\ell_t,\ \ell_x,\ \ell_y$	Test-function half-widths
$q_t,\ q_x,\ q_y$	Test-function exponents
$s_t,\ s_x,\ s_y$	Query-point strides
$\mathbf{G}$	Weak-form design matrix
$\mathbf{b}$	Weak-form RHS
$\mathbf{w}$	Sparse coefficient vector
$\hat{\lambda}$	MSTLS threshold
$\mathbf{s}$	Column-norm scaling

WSINDy Library

$(\mathcal{L}_j,\ f_j)$	(Operator, feature) pair
$f_j(U)$	Nonlinear feature
$\mathcal{L}_j$	Differential operator
$p_{\max}$	Max polynomial power
$\mathbf{m}$	Active-term mask
$\tilde{\Pi}_m$	Dominant-balance ratio
$S(\boldsymbol{\ell})$	Model-selection score
P_j	Cross-scale inclusion freq.

ETDRK4 Time Integration

$\hat{L}(k_x, k_y)$	Linear Fourier multiplier
$\hat{N}(u)$	Nonlinear Fourier multiplier
$k_x,\ k_y$	Wavenumber grids
$E = e^{\Delta t\, \hat{L}}$	Full-step exponential
$E_2 = e^{\Delta t\, \hat{L}/2}$	Half-step exponential
Q	φ_1 weight
$f_1,\ f_2,\ f_3$	Cox–Matthews weights

Metrics and Diagnostics

Ψ^n_{pol}	Polarisation order parameter
R^2	Coeff. of determination
R^2_{fc}	Forecast-window mean R^2
R^2_{wf}	Weak-form R^2
$R^2_{\text{wf},\rho},\ R^2_{\text{wf},p}$	Per-equation wf-R^2
$R^2(t)$	Time-resolved R^2
RMSE	Root-mean-squared error
σ_{rel}	Relative error
ΔM	Mass conservation error
$\mathcal{T}_{\text{train}},\ \mathcal{T}_{\text{test}}$	Train / test intervals

1 Introduction

We study whether interpretable macroscopic PDE models can be identified from noisy interacting-agent trajectories, and whether the discovered equations reflect the distinct physical mechanisms of each dynamical regime. The thesis pairs WSINDy [6, 26] with two POD-based [2] latent forecasters that have deliberately different inductive biases: a linear multivariate autoregression (MVAR) and a nonlinear long short-term memory (LSTM) network. Starting from trajectories $\{x_i(t)\}_{i=1}^N \subset \Omega \subset \mathbb{R}^2$ generated by alignment and attraction–repulsion interactions, we coarse-grain to a smooth density field $\rho(x,t)$ and seek an effective coarse-grained evolution law that is both predictive and physically interpretable. At the macroscopic level, density evolution obeys the conservation law

$$\partial_t \rho + \nabla \cdot \mathbf{j} = 0, \tag{1.0.1}$$

where $\mathbf{j}(x,t)$ is the agent flux. The modeling task is to identify an effective parameterised closure $\mathbf{j}_{\boldsymbol{\theta}}[\rho, \mathbf{p}, \Phi]$ from candidate mechanisms so that

$$\partial_t \rho(x,t) + \nabla \cdot \mathbf{j}_{\boldsymbol{\theta}}[\rho, \mathbf{p}, \Phi](x,t) = 0 \tag{1.0.2}$$

holds on the observed data, where $\mathbf{p}$ is the polarisation flux and $\Phi = W * \rho$ the Morse interaction potential (both defined in Chapter 2). The multi-field dependence reflects the fact that the scalar density equation is not expected to close on its own. In practice, we work in weak form to avoid unstable pointwise differentiation of noisy density fields.

We develop two complementary pipelines. Both start from the same upstream data: particle simulations, KDE-based macroscopic fields, and regime-specific train/test splits. The first is an equation-free reduced-order model (EF-ROM) [1] that compresses shift-aligned density snapshots via POD, evolves the latent coordinates with a learned time-stepper (MVAR or LSTM), and lifts back to the physical field. Because coherent flock translation inflates the effective rank under standard POD [5, 7, 8, 23, 30, 31, 33], we register each frame to a common reference via FFT-based cross-correlation on the periodic torus, yielding integer-pixel shifts that remove translational drift without interpolation. Comparing MVAR and LSTM reveals whether the latent dynamics induced by Vicsek [35] alignment and D'Orsogna [14] attraction–repulsion remain effectively linear after reduction.

The second pipeline targets interpretability via weak-form sparse identification (WSINDy). Here the learning object differs from the EF-ROM branch: WSINDy does not operate on the POD latent dataset, but instead remains in physical field space and identifies sparse continuous-time equations directly from weak-form systems built on the raw density field and auxiliary multi-fields. We formulate a coupled multi-field system: density ρ, polarization fluxes p_x, p_y (from velocity-weighted KDE), and a Morse interaction potential $\Phi = W * \rho$ (via FFT convolution). The WSINDy library includes physically motivated candidate terms (continuity, diffusion, Morse forcing, nonlinear diffusion, alignment self-saturation, and density-driven pressure), yielding a three-equation discovery problem for (ρ_t, $p_{x,t}$, $p_{y,t}$). Compactly supported polynomial test functions and integration by parts replace pointwise derivatives; sparsity is enforced via modified sequential thresholded least squares (MSTLS) [29] with a dominant-balance criterion [9] and automatic test-function width selection. Discovered PDEs are integrated forward using ETDRK4 [12, 22], which treats stiff linear terms exactly in Fourier space, as a targeted closure stress test on selected identified equations rather than as the primary evaluation.

The central question is not whether the discovered PDEs can "beat" the EF-ROM as a forecaster; the two branches answer different questions. The EF-ROM branch (Part I, Chapters 2–7) asks *what is forecastable* from the coarse-grained density movies; the WSINDy branch (Part II, Chapters 8–9) asks *what effective mechanisms are identifiable* from the same data. These are related but not identical questions: a model can forecast accurately without revealing mechanism, and a correctly identified PDE can fail at autonomous rollout because its library does not close the effective law. Throughout, WSINDy is used *primarily as an identifier:* the closure problem for mean-field particle PDEs (Section 9.5) means that even a correctly identified sparse PDE may be unclosed, so direct forecast comparisons would be confounded by closure error rather than identification quality. MVAR and LSTM serve as the forecasting benchmarks; WSINDy contributes complementary mechanistic information about which library terms are active in each dynamical regime.

The central empirical finding is a *closure gap*. Throughout, we use 'closure gap' to mean that the identified PDE,

while matching the data in weak form, lacks the mathematical ingredients needed to close the mean-field hierarchy and remain bounded under autonomous integration. WSINDy recovers sparse, regime-consistent PDE structures with density weak-form R^2 exceeding 0.99 in eight of nine regimes, yet the identified equations frequently cannot be stably integrated forward in time. In attractive regimes, the discovered Morse aggregation term creates a positive feedback loop (attraction increases local density, which strengthens attraction) that the candidate library cannot arrest because it lacks a saturation mechanism corresponding to the particle system's short-range repulsion. This decoupling of identification quality from rollout quality is itself a diagnostic: it reveals which closure ingredients are missing from the effective description at the chosen coarse-graining scale.

Because rollout alone is an insufficient success criterion, we define a composite notion of identification quality. We use *regime-consistent* throughout to mean that the identified PDE recovers the expected mechanism pattern in a way that is stable, sign-plausible, and well fit in weak form; operationally, the per-regime verdicts in Table 9.1 are based on four observable proxies: continuity-backbone recovery, physically plausible sign structure, cross-scale stability across six support-scale triplets, and strong weak-form fit, with momentum quality used to distinguish strong from moderate or partial outcomes.

Contributions. The thesis makes four contributions.

1. Multi-field WSINDy formulation. We formulate a coupled weak-form identification problem for the density and polarisation flux equations on finite-N Vicsek–Morse data, incorporating nonlocal Morse interaction features, automatic test-function width selection, and a dominant-balance sparsity criterion. This extends the scalar WSINDy framework of Messenger and Bortz [27] to a coupled three-field setting with regime-specific libraries.

2. Closure interpretation. We show that weak-form identification and autonomous PDE rollout can decouple: WSINDy recovers regime-consistent sparse support patterns and sign structures even where the identified PDE cannot be stably integrated, and we interpret this gap as evidence of closure insufficiency at the selected coarse-graining scale.

3. Shift-aligned POD pipeline. We assemble an FFT cross-correlation registration step with $\sqrt{\cdot}$-transformed POD that removes coherent translational drift, enabling faithful low-rank compression of transport-dominated density movies and defining the latent space in which the forecasting benchmark operates.

4. Forecasting benchmark across nine regimes. We compare linear (MVAR) and nonlinear (LSTM) latent time-steppers across nine dynamical regimes and four initial-condition families, establishing quantitative forecast ceilings and showing that variable-speed dynamics create a shift-predictability bottleneck that limits both model classes.

The thesis does not claim recovery of a unique continuum-limit PDE independent of coarse-graining; the recovered coefficients are effective parameters at the chosen data scale, and the closure gap itself is a central diagnostic.

Software. The complete implementation accompanying this thesis (the agent simulator, KDE coarse-graining, shift-aligned POD pipeline, MVAR and LSTM latent surrogates, WSINDy discovery engine, ETDRK4 integrator, and all YAML experiment configurations) is released as an open-source Python package at `https://github.com/eugenio-macias/wsindy-manifold`. Every figure and table in the following chapters can be reproduced from the scripts and configuration files therein. A mapping from the mathematical symbols used in the main text to their corresponding configuration keys and source modules is provided in Appendix A (Table A.3).

Relation to prior work. The EF-ROM component follows the restriction–lifting philosophy of Alvarez et al. [1]; the alignment step draws on shifted POD and symmetry-reduction methods for transport-dominated problems [5, 7, 8, 23, 31, 33], of which we adopt the simplest variant (single-reference FFT cross-correlation on a periodic domain). The PDE discovery component builds on Rudy et al.'s pointwise sparse regression approach (PDE-FIND) [32] and the subsequent weak-form extension of Messenger and Bortz [26], which replaces pointwise derivatives with integral test-function projections. Messenger and Bortz [27] subsequently applied WSINDy to learn scalar mean-field equations from particle data; our contribution extends this to a coupled multi-field identification problem that jointly discovers density and polarisation-flux equations with regime-specific Morse libraries, and systematically applies it across nine regimes. Concurrent work by Minor et al. [28] demonstrates that WSINDy can also recover interpretable PDEs for atmospheric phenomena, underscoring the generality of the weak-form

identification framework. The present thesis is also informed by growing interest in geometry-aware modelling, in which macroscopic laws must be consistent with the underlying spatial geometry on which collective motion occurs. The kinetic-theory literature on Vicsek-type models [3, 13, 21, 34] provides the external validation baseline used in Section 9.3.4.

Reader roadmap. Chapter 2 sets the microscopic modeling and coarse-graining background. Chapters 3–7 develop and evaluate the EF-ROM forecasting branch. Chapters 8–9 turn to weak-form PDE identification and closure analysis. Chapter 10 synthesises both branches. Figure 1.1 previews the pipeline structure; the headline empirical result, contrasting forecast R^2 with weak-form identification R^2 across all nine regimes, is presented in Figure 9.1 (Chapter 9).

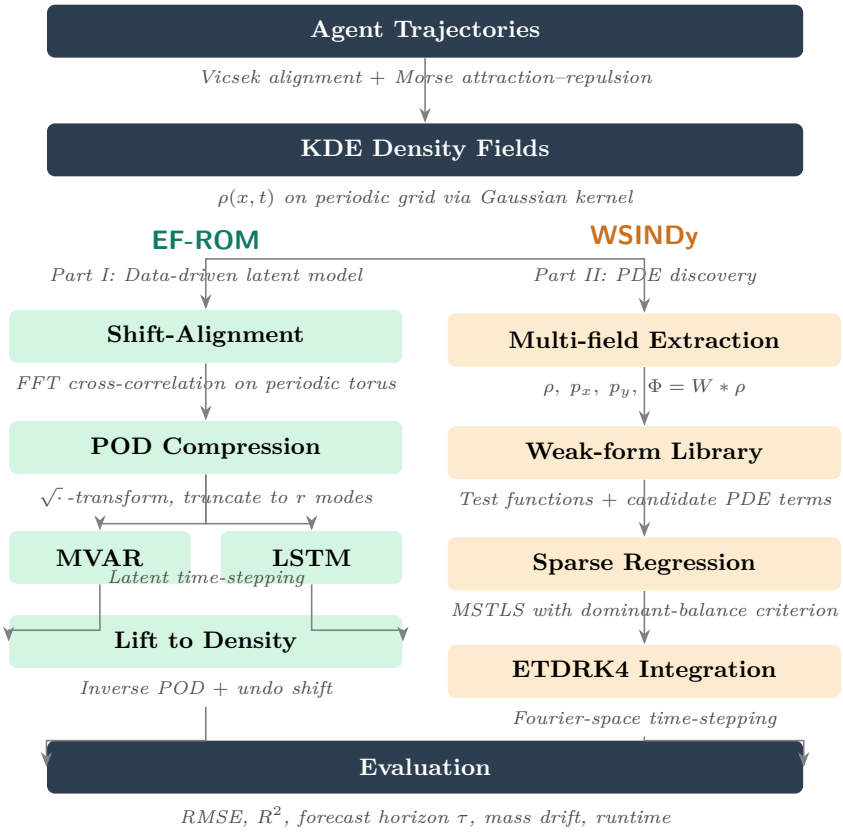


Figure 1.1: Overview of the two complementary modeling pipelines developed in this thesis. Both branches share the same microscopic-to-macroscopic data generation (top) and are evaluated on common metrics (bottom). Left: the equation-free reduced-order model compresses aligned density snapshots into a low-dimensional latent space and evolves them with learned time-steppers. Right: WSINDy discovers interpretable PDE flux laws from multi-field data via weak-form sparse regression.

Part I

Equation-Free Reduced Order Modeling from Agent Simulations

2 Related Work and Background

This chapter defines the two ingredients shared by every experiment in the thesis: the microscopic particle simulator (Section 2.1) and the KDE coarse-graining operator that maps discrete trajectories to the smooth macroscopic fields consumed by both the EF-ROM and WSINDy branches (Section 2.2 and its multi-field extension in Section 2.2.1). The coarse-graining step is not mere preprocessing; it is a modeling choice that determines which physical structures are visible at the field level and therefore which effective mechanisms can be identified or forecast.

2.1 Collective motion models

This section defines the microscopic simulator that generates all training and test data. The model extends the discrete-time Vicsek alignment framework studied in [1] with Morse attraction–repulsion forces [14] and variable-speed dynamics, producing a family of regimes whose macroscopic behavior ranges from gentle drift to near-singular collapse. Each microscopic ingredient (alignment, attraction–repulsion, and speed mode) motivates a specific candidate term in the WSINDy library (Chapter 8): alignment maps to self-advection and relaxation terms, Morse forces to nonlocal density-weighted gradients, and variable speed to flux–density coupling that the present library does not close.

2.1.1 Vicsek-style noisy alignment The alignment rule is the minimal ingredient shared by every regime in the benchmark and gives rise to the self-advection and relaxation terms in the WSINDy momentum library. We consider a discrete-time Vicsek-type alignment model for self-driven agents. At discrete times $t_n = n\Delta t$, agent i has position $x_i^n \in \Omega \subset \mathbb{R}^2$ and unit heading $p_i^n := (\cos\theta_i^n,\ \sin\theta_i^n) \in \mathbb{S}^1$, with constant-speed velocity $v_i^n = v_0\, p_i^n$ for a fixed $v_0 > 0$. Given an interaction radius $R > 0$, the neighbor set is $\mathcal{N}_i^n := \{\, j \in \{1,\dots,N\} : d_\Omega(x_i^n, x_j^n) \le R \,\}$, where d_Ω denotes the domain distance. The local mean heading direction is obtained by averaging neighbors' headings and renormalizing:

$$s_i^n := \sum_{j\in\mathcal{N}_i^n} p_j^n, \qquad \bar{p}_i^n := \begin{cases} s_i^n/\|s_i^n\|, & \|s_i^n\| > \varepsilon, \\ p_i^n, & \text{otherwise}, \end{cases}$$

with a small $\varepsilon > 0$ to avoid division by zero. Equivalently, one may write $\bar{\theta}_i^n = \mathrm{Arg}(\bar{p}_i^n) = \mathrm{atan2}((\bar{p}_i^n)_2, (\bar{p}_i^n)_1)$ and update angles modulo 2π. The noisy Vicsek update rotates the mean direction by an angular perturbation $\Delta\theta_i^n$:

$$p_i^{n+1} = R(\Delta\theta_i^n)\,\bar{p}_i^n, \qquad R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}, \tag{2.1.1}$$

followed by the forward Euler position update

$$x_i^{n+1} = x_i^n + \Delta t\, v_0\, p_i^{n+1}. \tag{2.1.2}$$

This is the standard Vicsek alignment mechanism [17, 35]; its mean-field limit and the continuum equations that arise from it are studied in [3, 13, 34]. The noise model is specified per regime (uniform or Gaussian angular noise); the original model uses uniform angular noise $\Delta\theta_i^n \sim \mathrm{Unif}(-\eta/2,\ \eta/2)$ with strength η; in addition, we also consider Gaussian noise $\Delta\theta_i^n \sim \mathcal{N}(0, \sigma^2)$. When variances are matched, $\sigma = \eta/\sqrt{12}$. The nature of the order–disorder transition can depend on whether scalar (intrinsic) or vectorial (extrinsic) noise is used [18].

2.1.2 Boundary conditions We consider a rectangular domain $\Omega = [0, L_x) \times [0, L_y)$ with periodic boundary conditions throughout this thesis. Distances are computed via the standard minimal-image convention, and periodicity is enforced componentwise after each position update: $x_{i,\alpha}^{n+1} \leftarrow x_{i,\alpha}^{n+1} \bmod L_\alpha$ for $\alpha \in \{1, 2\}$.

2.1.3 Attraction–repulsion interactions with Morse potentials Beyond pure alignment, we incorporate short-range repulsion and longer-range attraction via a Morse-type interaction force [14]. Let $r_{ij} := x_i - x_j$, $r := \|r_{ij}\|$, and $\hat{r}_{ij} := r_{ij}/r$. The radial Morse force magnitude is

$$f(r) = \frac{C_r}{\ell_r} e^{-r/\ell_r} - \frac{C_a}{\ell_a} e^{-r/\ell_a}, \tag{2.1.3}$$

where $C_r, C_a > 0$ denote repulsion and attraction strengths and $\ell_r, \ell_a > 0$ their interaction lengths. The pairwise force is $F_{ij} = f(\|x_i - x_j\|)\,\hat{r}_{ij}$, summed over neighbours within a cutoff radius $r_{\text{cut}} = 5\,\max(\ell_r, \ell_a)$. A cohesive regime is obtained when $\ell_r < \ell_a$ (short-range repulsion, longer-range attraction).

Constant-speed mechanism. Morse forces steer headings without changing speed:

$$\tilde{p}_i^n = \bar{p}_i^n + \mu_t \frac{F_i^n}{\|F_i^n\| + \varepsilon}, \qquad p_i^{n+1} = \frac{\tilde{p}_i^n}{\|\tilde{p}_i^n\|}, \qquad v_i^{n+1} = v_0\, p_i^{n+1}, \tag{2.1.4}$$

where μ_t is the translational mobility. This yields clean latent dynamics while producing milling, clustering, and ring formation [4].

Variable-speed dynamics (this thesis). As a stress test for the ROM pipeline, we extend the Morse mechanism to allow force-driven dynamics with variable speed: $v_i(t + \Delta t) = v_i(t) + \Delta t\, \mu_t\, F_i(t)$, $x_i(t + \Delta t) = x_i(t) + \Delta t\, v_i(t + \Delta t)$. When speed is no longer slaved to heading, the effective phase-space dimension increases and the POD latent coordinates may no longer capture the dynamics faithfully, making VS regimes a stress test for low-dimensional ROM assumptions. The full update pseudocode is given in Algorithm 3 (Appendix F); the key step is the sequential application of alignment averaging, force steering, noise injection, and position update on the periodic torus.

The Vicsek update therefore defines a nonlinear one-step map on the microscopic state, owing to normalization in the alignment rule, composition of normalization with rotational updates, discontinuous changes in neighbor sets at threshold distance R, and the nonlinear Morse forces combined with directional normalization. Together, these features produce a high-dimensional, piecewise-smooth dynamical system.

2.1.4 Order parameters for emergent behavior

These diagnostics are not part of the dynamical model itself; they are regime-level validation observables used to test whether reduced models preserve qualitative collective behavior. To characterize emergent collective behavior and to validate reduced-order models, we compute a set of order parameters over time. Closely related diagnostics for attraction–repulsion systems appear in [4]. Let the center of mass and relative positions be defined by $x_{\text{cm}}^n := \frac{1}{N}\sum_{i=1}^N x_i^n, \qquad r_i^n := x_i^n - x_{\text{cm}}^n$, Global alignment is quantified by the *polarization* order parameter:

$$\Psi_{\text{pol}}^n := \left\| \frac{1}{N} \sum_{i=1}^N \frac{v_i^n}{\|v_i^n\|} \right\| \in [0, 1], \tag{2.1.5}$$

Angular momentum (milling): measures rotational motion about the center of mass

$$M_{\text{ang}}^n := \frac{\left|\sum_{i=1}^N r_i^n \times v_i^n\right|}{N\,\langle\|r\|\rangle\,\langle\|v\|\rangle}, \qquad \langle\|r\|\rangle := \frac{1}{N}\sum_{i=1}^N \|r_i^n\|, \quad \langle\|v\|\rangle := \frac{1}{N}\sum_{i=1}^N \|v_i^n\|. \tag{2.1.6}$$

The *mean speed* $\langle\|v\|\rangle^n := N^{-1}\sum_{i=1}^N \|v_i^n\|$ serves as a diagnostic for numerical stability. *Nematic order* (axial alignment) Axial, head–tail symmetric alignment is quantified by the nematic tensor

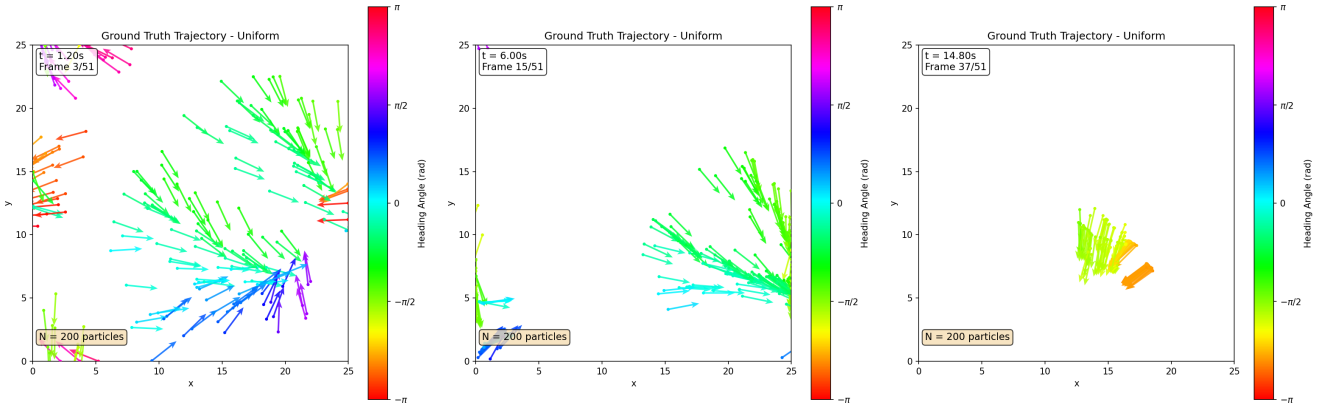
$$Q^n := \frac{1}{N}\sum_{i=1}^N (n_i^n \otimes n_i^n) - \frac{1}{2}I, \qquad n_i^n := \frac{v_i^n}{\|v_i^n\|}, \qquad S^n := \lambda_{\max}(Q^n) \tag{2.1.7}$$

where $\lambda_{\max}$ denotes the largest eigenvalue. Let $\rho^n \in \mathbb{R}^{N_x \times N_y}$ denote the discretized density field at time t_n. We define the *spatial order* parameter (density heterogeneity) as

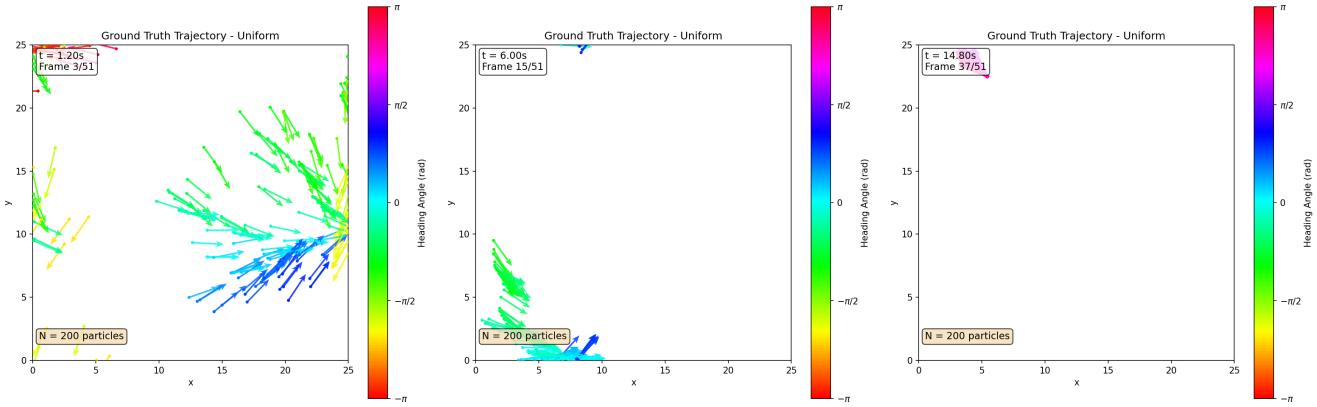
$$\mathcal{O}_{\text{spatial}}^n := \text{std}(\rho^n) = \sqrt{\frac{1}{N_x N_y}\sum_{a,b}\left(\rho_{a,b}^n - \bar{\rho}^n\right)^2}, \qquad \bar{\rho}^n := \frac{1}{N_x N_y}\sum_{a,b}\rho_{a,b}^n. \tag{2.1.8}$$

Large values indicate clustered or heterogeneous spatial structure, while values near zero correspond to near-uniform density distributions. This diagnostic is used to compare true and predicted density fields in ROM evaluations.

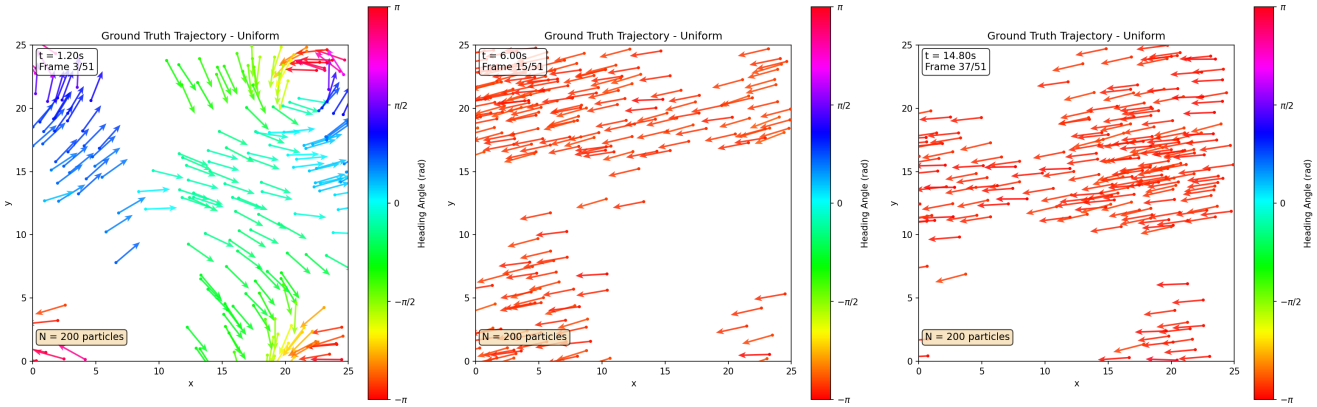
These quantities serve as regime-level validation targets in Chapters 7 and 9. In particular, they distinguish failures of transport, clustering, or milling structure that may not be fully visible from density-space R^2 alone: a forecast can achieve high pixelwise R^2 yet produce qualitatively wrong order-parameter trajectories, or conversely preserve the expected regime-level structure despite moderate density error.



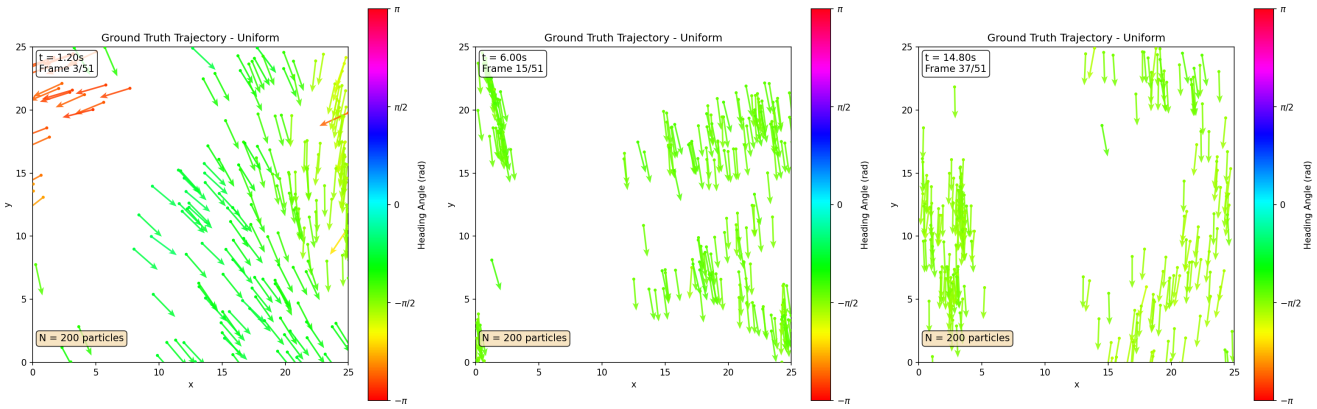
(a) Gentle crawl regime (weak Morse forces, low speed)



(b) Blackhole regime (strong attraction, near-singular collapse)



(c) Pure Vicsek regime (alignment only, no Morse forces)



(d) Blackhole VS regime (variable speed, erratic trajectories)

$t \approx 0\,\mathrm{s}$ $t \approx 6\,\mathrm{s}$ $t \approx 15\,\mathrm{s}$

Figure 2.1: Particle trajectories generated by Algorithm 3 under four contrasting parameter regimes, each starting from a uniform initial condition. Arrows indicate heading directions. (a) Gentle crawl: slow drift with gradual clustering from weak Morse attraction. (b) Blackhole: rapid collapse into a dense aggregate driven by strong attraction (C_a=20). (c) Pure Vicsek: alignment-only dynamics (no forces) producing coherent flocking without spatial aggregation. (d) Blackhole VS: same attraction as (b) but with variable speed; stochastic speed fluctuations produce erratic, space-filling trajectories that are visually and quantitatively harder to forecast (Section 7.1).

2.2 Kernel density estimation and coarse-graining scale

KDE is not simply a visualisation convenience: it defines the macroscopic field on which both the EF-ROM and WSINDy branches operate. Because both branches consume the same KDE output, the comparison between them is fair, but both inherit the same upstream smoothing bias, and every PDE coefficient recovered in later stages must be interpreted relative to the chosen bandwidth.

Following the equation-free framework of [1], the KDE step serves as a coarse-graining operator that maps discrete positions $\{x_i(t)\}_{i=1}^N$ to a smooth *number density* field $\rho(\cdot, t)$. In continuous form, we define

$$\rho(x,t) = \sum_{i=1}^{N} K_h\big(x - x_i(t)\big), \tag{2.2.1}$$

where K_h is a smooth, nonnegative kernel with bandwidth σ specified in grid-cell units (so the physical smoothing length is $h = \sigma\,\Delta x$) controlling the smoothing level and normalized so that $\int_{\mathbb{R}^2} K_h(z)\,dz = 1$. With this convention, ρ has units of *particles per unit area* and satisfies the mass constraint $\int_\Omega \rho(x,t)\,dx = N$ (up to discretization), which is essential for subsequent POD-based compression and mass-preserving lifting.

We use an isotropic Gaussian kernel,

$$K_h(z) = \frac{1}{2\pi h^2} \exp\left(-\frac{\|z\|^2}{2h^2}\right), \tag{2.2.2}$$

which provides smooth coarse-graining appropriate for periodic rectangular domains.

Remark 2.1 (Histogram plus Gaussian smoothing approximates gridded Gaussian KDE). Let $\tilde{\rho}(\cdot, t)$ be the piecewise-constant histogram number density on a uniform grid with spacings $\Delta x = L_x/n_x$ and $\Delta y = L_y/n_y$. Let G_σ denote the discrete Gaussian convolution operator (standard deviation σ in *grid-cell units*) applied to the gridded field $\tilde{\rho}$. If the grid is square ($\Delta x = \Delta y$) and the same σ is used in both axes, then the smoothed histogram

$$\rho(\cdot, t) \;=\; G_\sigma * \tilde{\rho}(\cdot, t) \tag{2.2.3}$$

approximates the *cell-averaged* continuous Gaussian KDE (2.2.1) with physical bandwidth $h = \sigma\,\Delta x$. This follows from standard properties of Gaussian convolution.

Remark 2.2 (Mass conservation in periodic domains). For periodic boundary conditions, Gaussian smoothing with wrapped boundaries preserves discrete mass: $\sum_{a,b}(G_\sigma * \tilde{\rho})_{ab} = \sum_{a,b} \tilde{\rho}_{ab}$, since the discrete Gaussian filter weights sum to 1. The total mass $M(t) := \sum_{a,b} \rho_{ab}(t)\,\Delta x\,\Delta y$ therefore equals N up to floating-point error.

The full pseudocode is given in Algorithm 4 (Appendix F).

Implementation details: We specify the bandwidth σ in grid-cell units, so the physical smoothing length is $h = \sigma\,\Delta x$ (for a square grid $\Delta x = \Delta y$). All experiments use $\sigma = 5.0$ grid cells (so $h = 5\,\Delta x$) with $n_x = n_y = 64$. We record metadata (bandwidth, grid resolution, domain size, boundary condition) alongside the resulting density movie $\rho(t) \in \mathbb{R}^{T \times n_y \times n_x}$, which serves as input to POD-based compression and subsequent reduced-order modeling.

The bandwidth is part of the coarse-grained model, not merely a numerical tuning parameter: it sets the spatial scale at which the macroscopic field is defined, and thereby determines which microscopic structures are visible to the regression stage. Smaller bandwidths retain more localised structure but increase variance and may worsen derivative-sensitive tasks such as weak-form regression; larger bandwidths reduce noise but risk erasing clustering and nonlocal interaction signatures that the WSINDy library is designed to detect. The chosen value $\sigma = 5$ therefore reflects a trade-off between preserving emergent spatial structure and obtaining stable reduced and weak-form models; at this scale, short-range interaction structure (sub-five-cell features) may be unrecoverable, even if longer-range alignment and attraction signatures remain visible. No systematic bandwidth sensitivity study was performed; the values used here were selected empirically and held fixed across all regimes. All recovered PDE coefficients, and the ROM forecast quality, are conditional on this fixed coarse-graining choice.

2.2.1 Multi-field coarse-graining

Beyond the scalar density ρ, the multi-field WSINDy formulation (Chapter 8) requires two additional coarse-grained fields and, optionally, a nonlocal interaction field. All three are computed using the same histogram-plus-Gaussian-smoothing approach as the density KDE above.

Polarisation flux fields. The velocity-weighted *polarisation flux* fields p_x, p_y are defined by replacing the unit weight in (2.2.1) with the corresponding velocity component:

$$p_\alpha(x,t) = \sum_{i=1}^{N} K_h\big(x - x_i(t)\big)\, v_{i,\alpha}(t), \qquad \alpha \in \{x, y\}, \tag{2.2.4}$$

where K_h is the same Gaussian kernel used for ρ. In practice, velocity-weighted counts are deposited into a 2-D histogram and smoothed with the same Gaussian kernel and periodic boundary handling used for the density, yielding $p_x, p_y \in \mathbb{R}^{T \times n_y \times n_x}$.

Morse potential field. The Morse interaction potential field Φ is the convolution of the density with the radial Morse kernel $W(r) = -C_r\, e^{-r/\ell_r} + C_a\, e^{-r/\ell_a}$:

$$\Phi(x,t) = (W * \rho)(x,t) = \mathcal{F}^{-1}\big[\hat{W} \cdot \hat{\rho}\big](x,t), \tag{2.2.5}$$

evaluated efficiently via 2-D FFT convolution. The combined field set (ρ, p_x, p_y), together with their spatial derivatives and, when forces are active, Φ, forms the input to the weak-form PDE library constructed in Chapter 8.

These auxiliary fields are introduced because the macroscopic flux is not expected to close as a function of density alone. The polarization fields encode directional alignment information that is lost by scalar KDE, while the Morse potential provides a nonlocal summary of attraction–repulsion interactions. In the WSINDy branch, these fields are treated as coarse observables supporting a coupled closure model, rather than as exact mean-field state variables derived from first principles.

Summary. The microscopic Vicsek–Morse simulator generates the trajectory data from which all results in this thesis derive. KDE with a fixed Gaussian bandwidth converts those trajectories into shared macroscopic fields (ρ, p_x, p_y, Φ) on a periodic grid. The chosen bandwidth fixes the coarse-graining scale; both the EF-ROM forecast quality and the WSINDy-recovered PDE coefficients must be interpreted relative to that field representation. This shared upstream ensures a fair comparison between the two branches, while also implying that any smoothing bias propagates to every subsequent result.

3 Experimental Protocol and Metrics

This chapter defines the evaluation protocol shared by every experiment in this thesis. Section 3.1 outlines the eight-stage pipeline from configuration to evaluation. Section 3.2 defines the train/test split. Section 3.3 motivates the MVAR lag order via an information-theoretic study. Section 3.4 specifies all evaluation metrics and the forecast horizon.

3.1 Pipeline orchestration

Every experiment is driven by a single YAML configuration file (one per regime) and executed end-to-end by the pipeline described in Figure 3.1. The pipeline proceeds through the eight sequential stages shown in Figure 3.1.

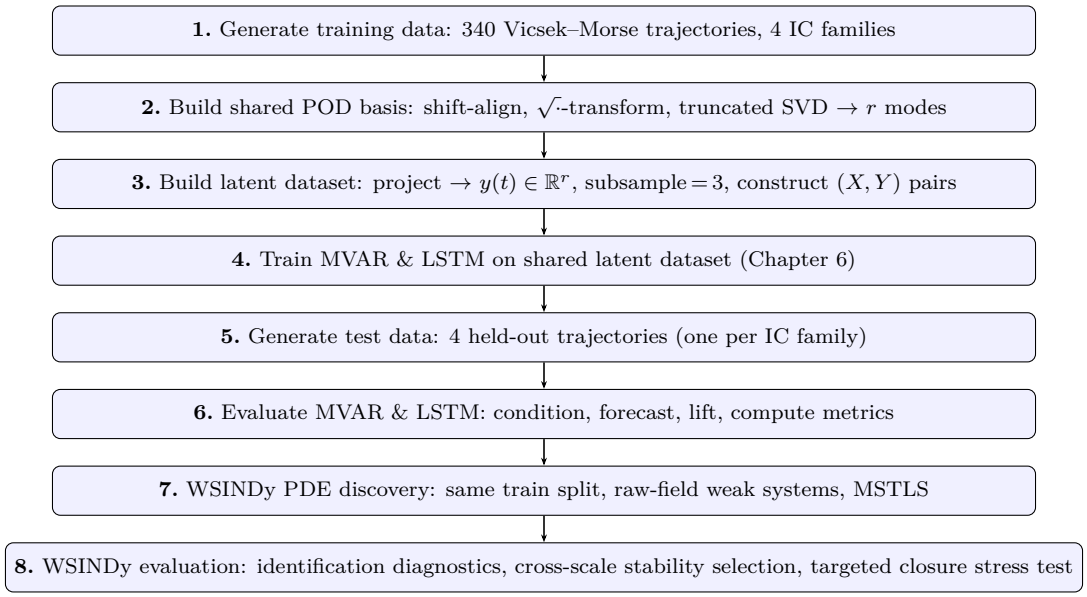


Figure 3.1: Eight-stage pipeline. Stages 1–6 run the EF-ROM branch [1]; stages 7–8 run WSINDy [26] PDE discovery and evaluation.

Stages 1 and 5 share a common upstream: the same Vicsek–Morse [14, 35] simulations, KDE-based coarse-grained fields, and train/test split. Stages 2–6 build a shift-aligned [31] POD latent dataset and learn MVAR and LSTM evolution maps; stages 7–8 bypass POD and apply MSTLS [29]/stability-selection [25] weak-form regression on the same trajectories, with ETDRK4 [22] rollout performed only as a targeted closure diagnostic on selected identified PDEs (not as the primary evaluation protocol). All artefacts and metrics are written to a self-contained experiment directory.

3.2 Train/test splits

Training and test data are drawn from the same parameter regime but use disjoint initial conditions.

Training set. Each regime generates 340 runs from four IC families (Table 4.3, Chapter 4). For the EF-ROM branch, all runs contribute to the shared POD basis and latent dataset. For WSINDy, the same training trajectories are used in physical space: density and auxiliary fields are computed on each run and then converted into weak-form regression constraints.

Test set. Four runs (one per IC family) are generated at held-out IC parameters (e.g. a Gaussian centred at $(12.5, 12.5)$ rather than the training grid positions) and simulated for T_{test} (Table 4.1). They are never seen during POD construction, latent-model fitting, or WSINDy regression. The same held-out trajectories define the test set for both branches.

Forecasting protocol. All models are evaluated in closed-loop (iterated single-step) mode. The effective forecast start is $t_{\text{fc}} = \max(t_{\text{fc}}^{\text{req}}, w_{\max}\Delta t_{\text{rom}})$, where $w_{\max}$ is the largest enabled latent lag. In baseline configs $w_{\text{MVAR}} = w_{\text{LSTM}} = 5$, giving $t_{\text{fc}} = 0.6\,\text{s}$. WSINDy evaluation uses the same t_{fc} but initialises from physical fields rather than a lagged latent window.

3.3 Lag selection

The MVAR lag order w determines how many past latent frames the model uses for prediction. We evaluate BIC, HQIC, and AIC on unrestricted VAR(w) models for $w = 1, \ldots, 100$ on every regime. For reference, the information-theoretically optimal lags on a representative DO catalogue regime (DO_CS01, collective swarm; see Appendix B) range from $w^*_{\mathrm{BIC}} = 23$ to $w^*_{\mathrm{AIC}} = 50$. Despite these large information-criterion optima, we fix $w_{\mathrm{MVAR}} = 5$ for the baseline systematic comparison. This is a deliberate design trade-off between three considerations:

1. **Cross-regime comparability.** A single lag order applied uniformly to all nine benchmark regimes ensures that performance differences reflect the dynamics rather than per-regime hyperparameter tuning.
2. **Computational cost.** The parameter count of an MVAR(w) model scales as wr^2; at $r = 19$, increasing w from 5 to 23 would multiply the number of coefficients by 4.6×.
3. **Diminishing returns after preprocessing.** After shift-alignment and the $\sqrt{\cdot}$-transform in our pipeline, the latent trajectories become approximately linear, so additional lags contribute less new information than in the raw (unconditioned) setting where the BIC/AIC criteria were computed.

No formal lag sweep over the deployed ridge-regularised MVAR is presented in this thesis; validating whether w>5 yields measurable forecast gains is left to future work (Section 10.3). We note that the BIC analysis was performed on a single DO catalogue regime; the information-criterion optima may differ for the NDYN benchmark regimes, so w=5 is a conservative lower bound rather than a per-regime optimum. The choice w=5 should therefore be interpreted as a practical, not individually optimal, setting. The LSTM lag is treated as a separate hyperparameter; whenever $w_{\mathrm{LSTM}} \neq w_{\mathrm{MVAR}}$, the common forecast start is shifted according to the largest enabled lag.

3.4 Evaluation metrics

All metrics compare predicted densities $\hat{\boldsymbol{\rho}}^n$ against ground truth $\boldsymbol{\rho}^n$ over the forecast window $\mathcal{T}_{\mathrm{fc}} = \{n_0, n_0{+}1, \ldots, n_1\}$, where n_0 is the forecast start index and n_1 is the terminal index. Let $d = n_x n_y$ denote the number of spatial grid points.

Coefficient of determination (R^2). The per-snapshot R^2 is

$$R^2(n) \;=\; 1 \;-\; \frac{\sum_{j=1}^{d}\big(\rho_j^n - \hat{\rho}_j^n\big)^2}{\sum_{j=1}^{d}\big(\rho_j^n - \bar{\rho}^n\big)^2}, \qquad \bar{\rho}^n = \frac{1}{d}\sum_{j=1}^{d}\rho_j^n, \tag{3.4.1}$$

where the denominator uses the spatial mean of the ground-truth snapshot at time n. We report the mean over the forecast window, $R^2_{\mathrm{fc}} = |\mathcal{T}_{\mathrm{fc}}|^{-1}\sum_{n\in\mathcal{T}_{\mathrm{fc}}} R^2(n)$, as the primary scalar metric. Because the null model in (3.4.1) is the per-snapshot spatial mean $\bar{\rho}^n$ (a single scalar per frame), so R^2 values near 1 indicate that the predictor tracks the full spatial structure, not merely the global density level. For nearly uniform density fields, however, the denominator becomes small and R^2 can be dominated by numerical noise; the relative-error norms below provide a complementary check in such regimes.

Relative ℓ^p errors. Three relative error norms are averaged over the forecast window:

$$\mathrm{RelErr}_1(n) = \frac{\|\boldsymbol{e}^n\|_1}{\|\boldsymbol{\rho}^n\|_1 + \epsilon}, \qquad \mathrm{RelErr}_2(n) = \frac{\|\boldsymbol{e}^n\|_2}{\|\boldsymbol{\rho}^n\|_2 + \epsilon}, \qquad \mathrm{RelErr}_\infty(n) = \frac{\|\boldsymbol{e}^n\|_\infty}{\|\boldsymbol{\rho}^n\|_\infty + \epsilon}, \tag{3.4.2}$$

where $\boldsymbol{e}^n = \boldsymbol{\rho}^n - \hat{\boldsymbol{\rho}}^n$ and $\epsilon = 10^{-12}$ guards against division by zero.

Mass conservation. The discrete total mass at time n is $m^n = (\Delta x\,\Delta y)\,\mathbf{1}^\top\boldsymbol{\rho}^n$ (and analogously $\hat{m}^n$ for the prediction). We track m^n and $\hat{m}^n$ over the forecast window and report the maximum relative mass violation:

$$\Delta m_{\max} \;=\; \max_{n\in\mathcal{T}_{\mathrm{fc}}} \frac{|\hat{m}^n - m^n|}{m^{n_0}}. \tag{3.4.3}$$

The pipeline optionally applies a post-hoc mass correction step, either the global rescaling (5.5.3) or the simplex projection (5.5.2), to match the initial mass. In the preprocessing ablations this had little effect on MVAR but improved LSTM rollout robustness and reduced $\Delta m_{\max}$ in practice.

Spatial order parameter. To assess whether the model preserves the spatial structure of the density field, we track the spatial standard deviation $\sigma_\rho(n) = \mathrm{std}(\boldsymbol{\rho}^n)$ defined in Eq. (2.1.8) (Chapter 2). A high σ_ρ indicates

concentrated density; a low value indicates a nearly uniform field. Together with the agent-level order parameters defined in Section 2.1.4, this diagnostic helps distinguish regime-level structural failures that may not be fully visible from density-space R^2 alone.

Metric hierarchy by branch. For the EF-ROM branch, forecast accuracy metrics, particularly R^2_{fc} and the relative errors (3.4.2), are primary evaluation targets. For the WSINDy branch, these same metrics are reported for completeness and as consistency checks under ETDRK4 rollout, but the primary criteria are identification-oriented: sparsity of the discovered PDE, coefficient stability under cross-scale selection, agreement with expected regime-specific mechanisms, and weak-form residual quality. Common metrics provide a shared diagnostic vocabulary, but the decision criteria differ between the two branches.

Parameter counts. At the baseline comparison settings (r=19, w_{MVAR}=w_{LSTM}=5, hidden width h=128, L=2 LSTM layers), the MVAR has $p_{\mathrm{MVAR}} = w\,r^2 + r = 1{,}824$ trainable parameters, while the LSTM has $p_{\mathrm{LSTM}} = 211{,}091$, a roughly 116× ratio. This count compares raw learnable parameters, not effective degrees of freedom: LSTM weight sharing across time steps and the recurrent bottleneck mean that the effective capacity gap is smaller than 116×. Full derivations appear in Chapter 6, §6.1.3 and §6.2.

From this point onward, all branch-level results should be read under this common upstream protocol but evaluated against branch-specific decision criteria: forecast accuracy for the EF-ROM, and identification quality for WSINDy.

4 Experimentation

4.1 Parameter choices and dynamics regimes

The experiment catalogue separates four sources of modelling difficulty. Table 4.2 lists the nine *primary benchmark regimes* that carry the main-text analysis in Chapters 7 and 9: five constant-speed (CS) regimes spanning gas, blackhole, supernova, crystal, and pure-Vicsek behaviours, plus four variable-speed (VS) diagnostic variants. For the EF-ROM branch, these regimes test the limits of shift-aligned POD and latent autoregression under increasing structural complexity. For the WSINDy branch, they test whether weak-form identification can recover the correct dominant mechanisms (diffusion, attraction, repulsion, alignment) in each regime. The remaining configurations in the systematic catalogue (Appendix B) provide supplementary coverage of force-parameter sweeps and D'Orsogna outcome-targeted families; they are used for robustness checks but are not central to the main argument.

To keep comparisons across learning models meaningful, we fix a core set of numerical and grid parameters across the full pipeline (Table 4.1) and vary only the physics and stochastic knobs that define each dynamics regime. All configuration files are provided in the accompanying repository.

Table 4.1: Fixed parameters shared by all dynamics regimes and pipeline stages.

Parameter	Value	Code key
Number of particles N	300	`N`
Domain size $L_x = L_y$	25	`Lx`, `Ly`
Boundary conditions	periodic	`bc`
Time step Δt	0.04 s	`dt`
Training horizon T_{train}	20 s (500 steps)	`T`
Test horizon T_{test}	200 s	`test_sim.T`
Interaction radius R	4.0	`R`
KDE grid resolution $n_x \times n_y$	64×64	`nx`, `ny`
KDE bandwidth	5.0 grid cells	`bandwidth`
Snapshot subsampling stride	3	`subsample`
Effective ROM time step Δt_{ROM}	0.12 s	$= \Delta t \times$ subsample
Fixed POD modes m	19	`fixed_modes`
Density transform	$\sqrt{\cdot}$	`density_transform`
Shift alignment	enabled (mean ref)	`shift_align`
Mass post-processing	scale	`mass_postprocess`

The fixed choices in Table 4.1 are shared experimental settings rather than per-regime optima. They are held constant so that later differences in ROM or WSINDy performance can be attributed primarily to the dynamics rather than to retuning of coarse-graining, reduction, or lifting. In particular, the KDE bandwidth σ=5.0 grid cells was selected heuristically to balance feature resolution with noise suppression; a systematic study of the coupling between coarse-graining scale and identified coefficients remains a subject for future inquiry (Section 10.3).

The regime catalogue is designed to separate four distinct sources of difficulty: coherent transport speed, stochastic disorder, strong nonlocal clustering, and state-dependent speed modulation. Each regime family varies one of these axes while holding the others fixed, so that performance differences in later chapters can be attributed to specific physical mechanisms.

4.1.1 NDYN regimes (physics-knob variations) Fourteen *NDYN* regimes isolate individual physics knobs (speed, noise strength, attraction/repulsion amplitude, force length scale, and speed mode) while holding the others at moderate baseline values. The repository defines 14 NDYN configurations, of which 11 non-duplicate entries are retained in the final systematic catalogue (Appendix B). All use Morse forces except NDYN08, which disables forces entirely (pure Vicsek alignment [35]).

Pure-alignment and weak-force constant-speed regimes are expected to be the most favourable for shift-aligned POD and compact latent autoregression, because their density movies are dominated by a single coherent translation that the preprocessing largely removes. Strong-clustering and variable-speed regimes, by contrast, are expected to stress both latent linearity and sparse closure discovery: the former introduces near-singular density spikes, while the latter decouples speed from heading and thereby inflates the effective latent dimension.

4.1.2 DO regimes (outcome-targeted variations) A second set of fourteen *DO* (dynamics-outcome) regimes holds speed v_0=3, noise η=0.2, and interaction parameters fixed except for the repulsion coefficient C_r and repulsion length ℓ_r. This sweep targets specific collective behaviours (collective swarms, single and double mills,

double rings, and escape patterns) following the D'Orsogna classification [14]. The full DO catalogue is provided in Appendix B.

4.1.3 Primary benchmark regimes

The main-text analysis in Chapters 7 and 9 focuses on nine regimes that span the key behavioural axes (disorder, attraction, repulsion, equilibrium, alignment only) in both constant-speed (CS) and variable-speed (VS) variants. Table 4.2 lists these regimes with their governing parameters; the complete systematic catalogue (11 non-duplicate NDYN + 14 DO, most with VS counterparts; 37 produced usable results) is provided in Appendix B.

Table 4.2: Primary benchmark regimes used for the main-text analysis. Speed mode is constant-with-forces for CS variants and variable for VS variants. Shared parameters: μ_t=0.3, r_{cut}=5 $\max(\ell_r, \ell_a)$; see Table 4.1 for the remaining fixed settings.

Label	Nickname	v_0	η	C_a	C_r	ℓ_a	ℓ_r	Behaviour
NDYN04	gas	3.0	1.5	0.5	0.2	1.5	0.5	disordered gas
NDYN04_VS	gas (VS)	3.0	1.5	0.5	0.2	1.5	0.5	variable-speed gas
NDYN05	blackhole	4.0	0.15	20.0	0.05	1.5	0.3	near-singular spike
NDYN05_VS	blackhole (VS)	4.0	0.15	20.0	0.05	1.5	0.3	variable-speed collapse
NDYN06	supernova	3.0	0.15	0.05	15.0	1.5	0.5	explosive repulsion
NDYN06_VS	supernova (VS)	3.0	0.15	0.05	15.0	1.5	0.5	variable-speed explosion
NDYN07	crystal	2.0	0.1	3.0	3.0	2.0	0.5	balanced equilibrium
NDYN07_VS	crystal (VS)	2.0	0.1	3.0	3.0	2.0	0.5	variable-speed equilibrium
NDYN08	pure Vicsek	3.0	0.3	—	—	—	—	alignment only

Variable-speed variants. Most NDYN and DO configurations have a companion *_VS* variant that changes exactly one parameter: the speed mode from constant-with-forces to variable. In variable-speed mode, Morse forces modulate agent *speed* as well as heading, producing non-constant flux fields. NDYN08 (pure Vicsek, no forces) and NDYN14 (already variable-speed) have no _VS counterpart.

4.2 Initial condition families

To probe ROM robustness beyond a single "typical" microscopic state, we generate ensembles of trajectories from four qualitatively different *initial position* distributions. The intent is to (i) span a range of macroscopic organisations (homogeneous, localised, symmetric, multi-modal) and (ii) induce distinct transient regimes under the *same* simulator and numerical settings. All particles evolve on the periodic domain $\Omega = [0, L_x] \times [0, L_y]$.

Common velocity initialisation. Across all IC families we draw random headings so that differences in dynamics are driven by the initial *spatial* configuration:

$$\theta_i \sim \mathcal{U}(0, 2\pi), \qquad \mathbf{v}_i(0) = v_0(\cos\theta_i, \sin\theta_i)^\top, \qquad i = 1, \ldots, N, \tag{4.2.1}$$

so $\|\mathbf{v}_i(0)\| = v_0$ for all particles and the initial polarisation is close to zero in expectation.

Uniform IC. The uniform family represents a maximum-entropy baseline with no imposed spatial structure:

$$\mathbf{x}_i(0) \sim \mathcal{U}(\Omega), \qquad i = 1, \ldots, N, \tag{4.2.2}$$

equivalently $x_i \sim \mathcal{U}(0, L_x)$ and $y_i \sim \mathcal{U}(0, L_y)$ independently. The expected coarse density is spatially constant, $\mathbb{E}[\rho(\mathbf{r}, 0)] = N/(L_x L_y)$. Uniform ICs serve both as training diversity and as the baseline for generalisation comparisons.

Single Gaussian IC. The single-Gaussian family produces a localised "blob" with controllable centre and spread:

$$\mathbf{x}_i(0) \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2 I_2), \qquad i = 1, \ldots, N. \tag{4.2.3}$$

The centre $\boldsymbol{\mu} = (\mu_x, \mu_y)$ is drawn from a 4×4 grid of interior positions and σ^2 is varied across three values, yielding $4 \times 4 \times 3 = 48$ parameter combinations with 2 samples each, giving 96 training runs per regime.

Ring IC. The ring family enforces rotational symmetry at t=0 and typically generates vortex-like macroscopic patterns. Let $\mathbf{c} = (L_x/2,\, L_y/2)$ be the domain centre. Angles and radii are sampled in polar coordinates:

$$\theta_i \sim \mathcal{U}(0, 2\pi), \qquad r_i \sim \mathcal{N}(R_{\text{ring}},\, \sigma_r^2), \tag{4.2.4}$$

and mapped to Cartesian positions $\mathbf{x}_i(0) = \mathbf{c} + r_i(\cos\theta_i, \sin\theta_i)^\top$, then wrapped into Ω via modulo. Training enumerates five radii and four widths with 4 samples each, yielding $5 \times 4 \times 4 = 80$ training runs per regime.

Two-cluster Gaussian IC. The two-cluster family generates a bimodal initial distribution as an equal-weight mixture of two isotropic Gaussians:

$$p(\mathbf{x}) = \tfrac{1}{2}\,\mathcal{N}(\boldsymbol{\mu}_1, \sigma^2 I_2) + \tfrac{1}{2}\,\mathcal{N}(\boldsymbol{\mu}_2, \sigma^2 I_2), \tag{4.2.5}$$

implemented by sampling $N_1 = \lfloor N/2 \rfloor$ particles from the first component and $N_2 = N - N_1$ from the second. The centres are placed symmetrically about the domain midpoint, separated horizontally by distance d:

$$\boldsymbol{\mu}_1 = \left(\tfrac{L_x}{2} - \tfrac{d}{2},\, \tfrac{L_y}{2}\right), \qquad \boldsymbol{\mu}_2 = \left(\tfrac{L_x}{2} + \tfrac{d}{2},\, \tfrac{L_y}{2}\right). \tag{4.2.6}$$

Default values are $d = L_x/3$ and $\sigma = \min(L_x, L_y)/8$. After sampling, positions are wrapped into Ω via modulo. Training enumerates four separations and three widths with 7 samples each, yielding $4 \times 3 \times 7 = 84$ training runs per regime.

Training–test split summary. Table 4.3 summarises the per-regime dataset composition. The four held-out test runs are intended as balanced regime-level diagnostic stress tests across IC families, not as a Monte Carlo estimate of out-of-sample uncertainty; they probe whether each model class preserves fidelity under qualitatively different initial spatial organisations. Every regime (NDYN and DO alike) uses the identical IC schedule, so the total training set comprises $340 \times 25 = 8{,}500$ simulations across the 25 base regimes in the experiment catalogue, each with a companion variable-speed variant where applicable.

Table 4.3: Training and test runs per dynamics regime.

IC family	Training runs	Test runs
Gaussian	96	1
Uniform	80	1
Ring	80	1
Two-clusters	84	1
Total	**340**	**4**

Each run is assigned a unique identifier from which a deterministic RNG seed is derived, ensuring exact reproducibility. The density movies produced by these simulations feed both modeling branches. For the EF-ROM pipeline they are assembled into a global snapshot matrix for POD basis construction, as described in Chapter 5. For WSINDy, the same regime-specific training and test simulations are retained in physical space and converted into density and auxiliary multi-fields for weak-form regression.

These IC families expose specific modelling biases: uniform ICs probe spurious structure formation; Gaussian ICs test localised transport; ring ICs test rotationally organised dynamics not reducible to translation; and two-cluster ICs test interaction between separated peaks. Ring and two-cluster ICs are expected to be harder for the shift-aligned POD pipeline, since translational alignment does not resolve all coherent spatial organisation.

5 Global POD Reduction and Restriction–Lifting Pipeline

This chapter describes the data-driven reduction pipeline that maps high-dimensional density fields to low-dimensional latent trajectories suitable for time-series modelling.

The processing order is shown in Figure 5.1. Steps 2–3 are invertible and are undone during lifting (reconstruction).

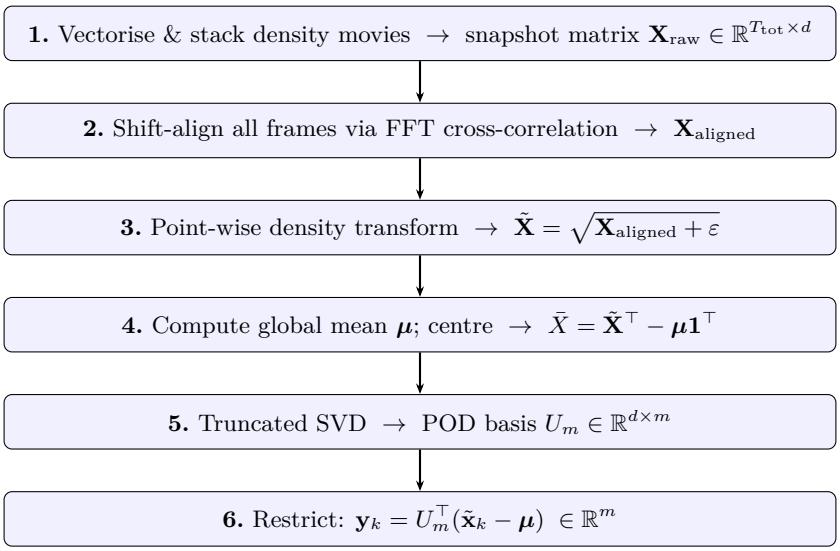


Figure 5.1: POD restriction pipeline: density fields to latent coordinates (six processing stages).

5.1 Snapshot matrix construction

We generate macroscopic density fields from microscopic particle trajectories via KDE (Chapter 2, Algorithm 2) and assemble a global snapshot matrix for POD. Let M denote the number of training runs and T_i the number of saved snapshots for run i (after temporal subsampling by stride $s = 3$).

5.1.1 Vectorisation and global stacking Each density snapshot $\rho^{(i)}(t_k) \in \mathbb{R}^{n_y \times n_x}$ is flattened into a column vector $\mathbf{s}_k^{(i)} := \text{vec}(\rho^{(i)}(t_k)) \in \mathbb{R}^d$, $d := n_x n_y$, and all frames from run i are stacked row-wise into $\mathbf{X}^{(i)} \in \mathbb{R}^{T_i \times d}$. The global training matrix is the vertical concatenation

$$\mathbf{X}_{\text{raw}} := \begin{bmatrix} \mathbf{X}^{(0)} \\ \vdots \\ \mathbf{X}^{(M-1)} \end{bmatrix} \in \mathbb{R}^{T_{\text{tot}} \times d}, \qquad T_{\text{tot}} = \sum_{i=0}^{M-1} T_i.$$

Column-stacked notation. Alvarez et al. define a snapshot matrix whose *columns* are density snapshots: $X = [x_1, \ldots, x_{n_t}] \in \mathbb{R}^{n_c \times n_t}$ with n_c=d, n_t=T_{tot}, and centre using $\bar{X} = XH$ where $H = I - \frac{1}{n_t}\mathbf{1}\mathbf{1}^\top$ [1]. To map our row-stacked implementation to this notation, set $X := \mathbf{X}_{\text{raw}}^\top$.

Scope. The POD pipeline operates exclusively on the scalar density field ρ. The auxiliary fields used by WSINDy (flux p_x, p_y and Morse potential Φ; see Chapter 2, §2.2.1) are computed in physical space and are *not* projected through POD. The POD branch is intentionally density-only so that the latent benchmark remains a pure reduced-order surrogate for the observable density movie, while WSINDy retains access to additional coarse fields needed for closure.

5.2 Preprocessing: shift alignment and density transform

Two preprocessing steps are applied *before* POD to improve the compressibility of the snapshot manifold.

5.2.1 Shift alignment Travelling density patterns (flocks, mills) cause the dominant POD mode to encode rigid translation rather than shape variation. Shift alignment removes this transport by registering every frame to a common reference via circular cross-correlation.

Reference field. The reference is the temporal mean of all (unaligned) training snapshots, reshaped to $n_y \times n_x$:

$$\rho_{\text{ref}} := \frac{1}{T_{\text{tot}}} \sum_{k=1}^{T_{\text{tot}}} \rho_k \;\in\; \mathbb{R}^{n_y \times n_x}. \tag{5.2.1}$$

Registration as inner-product maximisation. The alignment task is to find, for each snapshot ρ_k, the integer translation $\boldsymbol{\delta}_k = (\Delta y_k, \Delta x_k)$ that best registers ρ_k to the reference. Under periodic boundary conditions the natural similarity measure is the periodic cross-correlation

$$C_k(\boldsymbol{\delta}) \;:=\; \sum_{i,j} \rho_{\text{ref}}(i,j)\; \rho_k\big((i-\delta_y) \bmod n_y,\; (j-\delta_x) \bmod n_x\big) \;=\; \langle \rho_{\text{ref}},\; \mathcal{T}_{\boldsymbol{\delta}}\, \rho_k \rangle, \tag{5.2.2}$$

where $\mathcal{T}_{\boldsymbol{\delta}}$ denotes the circular-shift operator defined formally below. $C_k(\boldsymbol{\delta})$ is maximised at the translation that brings ρ_k into closest L^2-alignment with the reference, because for non-negative fields the inner product $\langle \rho_{\text{ref}}, \mathcal{T}_{\boldsymbol{\delta}} \rho_k \rangle$ is largest when the two mass distributions overlap most. The optimal shift is therefore

$$\boldsymbol{\delta}_k^* \;=\; \arg\max_{\boldsymbol{\delta} \in G}\; C_k(\boldsymbol{\delta}). \tag{5.2.3}$$

Since the group $G = \mathbb{Z}_{n_y} \times \mathbb{Z}_{n_x}$ consists of integer-pixel shifts, sub-pixel registration via interpolation is not pursued; the integer constraint is exact on the periodic lattice and avoids interpolation artefacts that would introduce artificial smoothing into the aligned density fields.

Efficient evaluation via the convolution theorem. A naïve search over all $n_y n_x$ candidate shifts would require $\mathcal{O}(n_y^2 n_x^2)$ multiplications per frame. The discrete convolution theorem reduces this to $\mathcal{O}(n_y n_x \log(n_y n_x))$ by observing that periodic cross-correlation is equivalent to element-wise multiplication in the Fourier domain:

$$C_k(\boldsymbol{\delta}) \;=\; \mathcal{F}^{-1}\big[\mathcal{F}[\rho_{\text{ref}}] \;\odot\; \overline{\mathcal{F}[\rho_k]}\big](\boldsymbol{\delta}), \tag{5.2.4}$$

where $\mathcal{F}$ denotes the 2-D DFT, $\odot$ the Hadamard product, and the overline complex conjugation. Because $\mathcal{F}[\mathcal{T}_{\boldsymbol{\delta}}\, \rho_k](\boldsymbol{\xi}) = e^{-2\pi i\, \boldsymbol{\xi}\cdot\boldsymbol{\delta}/\boldsymbol{n}}\, \mathcal{F}[\rho_k](\boldsymbol{\xi})$, the pointwise product in (5.2.4) simultaneously evaluates C_k at every shift in a single forward–inverse FFT pair. Two FFTs of the reference are precomputed once and reused for all frames.

Applying and inverting shifts. Each frame is circularly rolled by $\boldsymbol{\delta}_k^*$ using periodic boundary conditions. The shift vectors are stored so that the inverse operation $(-\Delta y_k, -\Delta x_k)$ can undo the alignment during the reconstruction (lifting) step.

Symmetry group and quotienting. On the discrete periodic grid the integer-pixel translations form the cyclic product group $G = \mathbb{Z}_{n_y} \times \mathbb{Z}_{n_x}$, acting unitarily on density fields via circular shifts. The group action partitions the field space into translation orbits; shift alignment selects one representative per orbit, mapping the snapshot manifold onto a cross-section $\hat{\mathcal{M}} \simeq \mathcal{M}/G$ that collapses translational variation entirely. Quotienting reduces the manifold's Kolmogorov n-width [30], which is why the shifted POD basis decays faster than the unshifted one. Figure 5.2 confirms this empirically across all experiments in our catalogue: the sPOD curves (blue) decay consistently faster than their unaligned POD counterparts (red), reaching 99 % cumulative energy in roughly half as many modes.

Because the continuum density evolution $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$ on a periodic domain is translationally equivariant, the snapshot manifold inherits a continuous family of near-degenerate directions along group orbits; quotienting removes these directions while preserving all shape information. The choice of reference ρ_{ref} determines the cross-section; we use the temporal mean (5.2.1). This construction is the discrete, translational-only instance of the more general *method of slices* [7, 8].

Scope and limitations. The alignment map is restricted to integer-pixel translations. It is therefore appropriate for dominant bulk drift, but it does *not* resolve rotational organisation, multiple simultaneous transport velocities, deformation of the density envelope, or topological changes such as merging and splitting of clusters. This matters especially for variable-speed regimes, where speed fluctuations produce complex, non-translational density dynamics (Section 7.1).

This discrete single-reference alignment is a special case of the *shifted proper orthogonal decomposition* (sPOD) of Reiss et al. [31]; the full sPOD machinery decomposes transport-dominated fields into K co-moving frames via

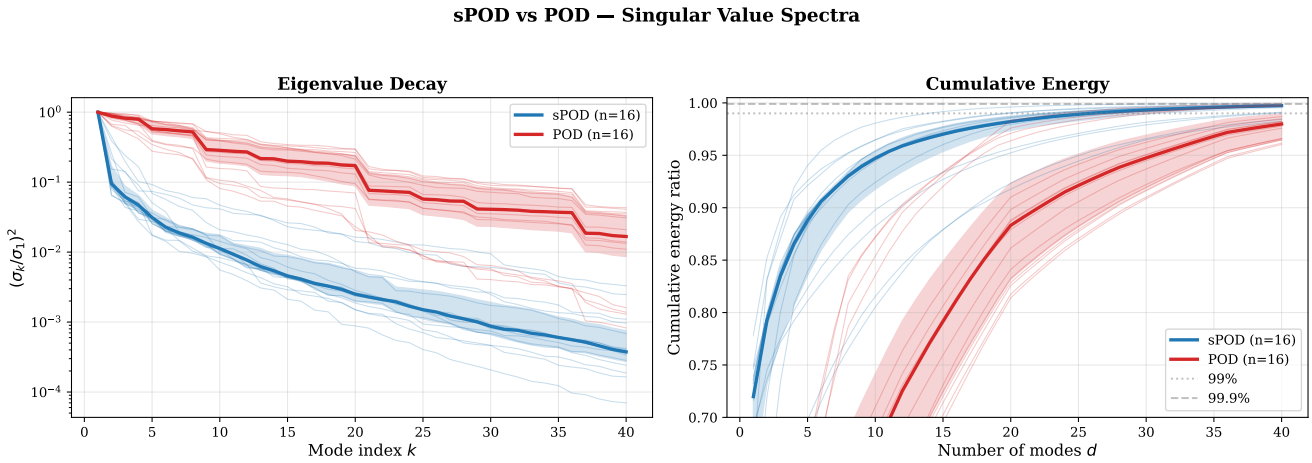


Figure 5.2: sPOD vs. standard POD singular-value spectra aggregated over all experiments in the catalogue. Left: normalised eigenvalue decay $(\sigma_k/\sigma_1)^2$ on a log scale. Right: cumulative energy ratio. Thick lines show the cross-experiment mean; shaded bands span individual runs. Shift alignment accelerates spectral decay uniformly, supporting a compact POD truncation at $m = 19$ modes.

nuclear-norm minimisation [5, 23]. More broadly, it connects to diffusion-map separation of symmetry degrees of freedom [33].

5.2.2 Density transform

After alignment, a point-wise variance-stabilising transform is applied to the (positive) density values:

$$\tilde{\rho} := \sqrt{\rho + \varepsilon}, \qquad \varepsilon = 10^{-10}, \tag{5.2.5}$$

The square root compresses the dynamic range of near-singular density spikes (e.g. NDYN05 blackhole) and improves the spectral decay of the snapshot covariance, so that fewer POD modes are needed to capture a given energy fraction. Because the square-root map is nonlinear, it changes both Euclidean geometry and total transformed mass; exact linear-POD mass-preservation statements (Section 5.6) therefore do not apply prior to inverse transformation.

All subsequent steps (mean computation, centering, SVD, restriction, and lifting) operate in the *transformed* ($\tilde{\rho}$) space. The inverse transform is applied after lifting:

$$\hat{\rho} = \left[\max(0, \hat{\tilde{\rho}})\right]^2 - \varepsilon, \tag{5.2.6}$$

where the clamp to zero guards against small negative values introduced by POD truncation.

Empirical role in the ROM comparison. In the preprocessing ablations, the density transform had little to no effect on MVAR relative to working directly with raw density snapshots, but it improved LSTM rollout stability and time-resolved R^2 more consistently across regimes. Since the thesis compares the mechanistic WSINDy pipeline against black-box EF-ROM benchmarks, and compares MVAR and LSTM on a shared latent representation, we retain the same density transform for both surrogates for like-for-like benchmarking even though the main benefit is observed on the LSTM side.

5.3 POD via truncated SVD

Let $\mathbf{X} \in \mathbb{R}^{T_{\text{tot}} \times d}$ denote the shift-aligned, $\sqrt{\cdot}$-transformed snapshot matrix. Compute the global mean $\boldsymbol{\mu} = \frac{1}{T_{\text{tot}}} \mathbf{X}^\top \mathbf{1} \in \mathbb{R}^d$ and the centred matrix $\bar{X} := \mathbf{X}^\top - \boldsymbol{\mu}\,\mathbf{1}^\top \in \mathbb{R}^{d \times T_{\text{tot}}}$. Compute the thin SVD $\bar{X} = U\,\Sigma\,V^\top$, where $U \in \mathbb{R}^{d \times r}$ contains the spatial POD modes, $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, and $r = \operatorname{rank}(\bar{X})$. This matches the construction in Alvarez et al. [1]; for a foundational treatment of POD see [2].

Energy spectrum and truncation. Define the total energy $E_{\text{tot}} = \sum_{k=1}^{r} \sigma_k^2$ and cumulative explained ratio $\tau(m) := \sum_{k=1}^{m} \sigma_k^2 \,/\, E_{\text{tot}}$, $1 \le m \le r$. In practice we use a *fixed* latent dimension $m = 19$ and retain $U_m \in \mathbb{R}^{d \times m}$. This is a controlled experimental setting for fair cross-regime comparison; individual truncation optima may differ. An automatic alternative is the optimal hard threshold of Gavish and Donoho [15]; the fixed choice adopted here is deliberately conservative.

5.4 Restriction operator $\mathcal{R}$: density → latent coordinates

In the equation-free setting, restriction is the map $\mathcal{R} : \mathbb{R}^d \to \mathbb{R}^m$ that takes a (transformed, centred) snapshot to its latent coordinates [1]:

$$\mathbf{y} = \mathcal{R}(\mathbf{x}) := U_m^\top \big(\mathbf{x} - \boldsymbol{\mu}\big) \ \in \mathbb{R}^m, \tag{5.4.1}$$

where $\mathbf{x} = \mathrm{vec}(\tilde{\rho})$ is the transformed density. Thus the latent coordinates parameterise the aligned-and-transformed snapshot manifold rather than raw physical density fields directly. In batch form, $Y = U_m^\top (X - \boldsymbol{\mu}\,\mathbf{1}^\top) \in \mathbb{R}^{m \times n_t}$.

In the row-stacked implementation, the equivalent operation is $\mathbf{Y}_{\mathrm{row}} = (\mathbf{X} - \mathbf{1}\,\boldsymbol{\mu}^\top)\, U_m \in \mathbb{R}^{T_{\mathrm{tot}} \times m}$, so that each row of $\mathbf{Y}_{\mathrm{row}}$ is a latent snapshot.

5.5 Lifting operator $\mathcal{L}$: latent → reconstructed density

Lifting inverts the restriction by projecting back to the POD subspace and adding the mean:

$$\hat{\mathbf{x}} = \mathcal{L}(\mathbf{y}) := U_m\, \mathbf{y} + \boldsymbol{\mu}. \tag{5.5.1}$$

Since $\boldsymbol{\mu}$ and U_m live in the transformed ($\sqrt{\cdot}$) space, the full reconstruction proceeds in three conceptually distinct stages: a mathematical lifting step, an inverse transformation back to density space, and optional admissibility repairs.

Stage 1: Mathematical lifting. The latent vector is mapped back to the POD subspace: $\hat{\mathbf{x}} = U_m\, \mathbf{y} + \boldsymbol{\mu}$ (in $\sqrt{\rho}$ space).

Stage 2: Inverse transform to density space. The square-root map is inverted to recover physical density: $\hat{\rho} = [\max(0, \hat{\mathbf{x}})]^2 - \varepsilon$ (Eq. 5.2.6), where the clamp guards against small negatives introduced by POD truncation.

Stage 3: Optional admissibility postprocessing. Two optional corrections restore basic physical admissibility, chiefly nonnegativity and total mass, that may be lost under the nonlinear transform and forecast error:

- *Nonnegativity clamp*: $\hat{\rho}^+ \leftarrow \max(\hat{\rho}, 0)$.
- *Mass correction*: either global rescaling $\mathcal{S}_{M_0}$ (Eq. 5.5.3) or simplex projection $\Pi_{\Delta^{d-1}_{M_0}}$ (Eq. 5.5.2), where $M_0 := \sum_{j=1}^d \rho_j(t_{n_0})$ is the true mass at forecast start.

Finally, the spatial shift is undone: $\hat{\rho}_{2\mathrm{D}} \leftarrow \mathrm{roll}(\hat{\rho}_{2\mathrm{D}},\ -\Delta y_k, -\Delta x_k)$.

The two mass-correction operators are defined as

$$\Pi_{\Delta^{d-1}_{M_0}}(\hat{\rho}) := \arg\min_{\rho \in \Delta^{d-1}_{M_0}} \|\rho - \hat{\rho}\|_2^2, \tag{5.5.2}$$

$$\mathcal{S}_{M_0}(\hat{\rho}) := \begin{cases} \hat{\rho}\, \dfrac{M_0}{\sum_{j=1}^d \hat{\rho}_j}, & \text{if } \sum_{j=1}^d \hat{\rho}_j > 0, \\ \hat{\rho}, & \text{otherwise}, \end{cases} \tag{5.5.3}$$

where $\Delta^{d-1}_{M_0} := \{\rho \in \mathbb{R}^d : \rho_j \geq 0,\ \sum_j \rho_j = M_0\}$. In preliminary ablations, both operators had negligible effect on MVAR forecast quality but improved LSTM rollout stability. All reported results use mass rescaling paired with the nonnegativity clamp, which gave slightly higher R^2 in the final ablation round; results with simplex projection were comparable.

5.6 Mass preservation under linear POD

When no density transform is applied (i.e. the pipeline is purely linear), POD lifting preserves total mass exactly. This is a standard property of mean-centred orthogonal projections; a proof for the normalised-density case in the crowd-dynamics setting appears in Alvarez et al. [1, Proposition 1]. We give a self-contained argument below, generalised to arbitrary constant mass M_0 and written in the notation of our pipeline (Sections 5.3–5.5).

Proposition 5.1 (Mass preservation under constant-mass snapshots)**.** *Let $X = [x_1, \ldots, x_{n_t}] \in \mathbb{R}^{d \times n_t}$ with $\mathbf{1}^\top x_k = M_0$ for all k. Let $\bar{X} = XH$, $\bar{X} = U\Sigma V^\top$, and define the reconstruction $\tilde{X} := U_m U_m^\top \bar{X} + \boldsymbol{\mu}\mathbf{1}^\top$. Then $\mathbf{1}^\top \tilde{x}_k = M_0$ for all k.*

Proof. Write the reconstruction column-wise: $\tilde{x}_k = U_m U_m^\top \bar{x}_k + \boldsymbol{\mu}$, where $\boldsymbol{\mu} = \frac{1}{n_t} X\mathbf{1}$ is the snapshot mean. Summing $\boldsymbol{\mu}$ gives $\mathbf{1}^\top \boldsymbol{\mu} = \frac{1}{n_t}\mathbf{1}^\top X\mathbf{1} = \frac{1}{n_t} \cdot n_t M_0 = M_0$. It therefore suffices to show that the projection term contributes zero mass. Because every column of $\bar{X}$ sums to zero ($\mathbf{1}^\top \bar{x}_k = \mathbf{1}^\top x_k - M_0 = 0$), the constant vector $\mathbf{1}$ lies in the left null-space of $\bar{X}$. Every column of U_m is a left singular vector of $\bar{X}$ and thus belongs to its column space, so

$\mathbf{1}^\top U_m = \mathbf{0}^\top$. Hence $\mathbf{1}^\top(U_m U_m^\top \bar{x}_k) = \mathbf{0}^\top \bar{x}_k = 0$ and $\mathbf{1}^\top \tilde{x}_k = 0 + M_0$. □

Remark 5.2 (Effect of the density transform). Proposition 5.1 applies to the purely linear pipeline only and does not extend to the full square-root pipeline. The $\sqrt{\cdot}$ transform is nonlinear: even if $\sum_j \rho_j = M_0$ for all frames, the transformed snapshots $\tilde{\rho}_j = \sqrt{\rho_j + \varepsilon}$ do *not* share a common mass. Proposition 5.1 therefore does not apply in the transformed space, and an optional mass correction operator (rescaling or simplex projection) is used to restore physical mass after inverse transformation. Proposition 5.1 nevertheless motivates the expectation that mass drift should be small even in the nonlinear pipeline, and explains why only mild post-processing corrections are needed in practice.

Algorithm 6 (Appendix F) gives the complete pseudocode for the global POD restriction–lifting pipeline. Algorithm 6 describes the pipeline used in all experiments; Proposition 5.1, by contrast, describes an idealised linear special case that clarifies the source of exact mass preservation.

What the ROM pipeline preserves and what it discards. The restriction–lifting pipeline deliberately quotients out coherent translational drift (via shift alignment) and compresses the aligned density movie to $m = 19$ latent coordinates (via truncated SVD), retaining the dominant spatial modes while discarding fine-scale structure below the truncation threshold. The resulting latent space is therefore a controlled compression of the aligned density manifold, not a physically closed state description: it contains information only about mass-density shape modes, not about the auxiliary fields (polarisation flux, Morse potential) that WSINDy uses for mechanism identification. Whether the latent dynamics are effectively linear under this representation, or whether substantial nonlinear memory remains, is the question addressed by the MVAR/LSTM comparison in Chapter 6.

6 Latent Reduced-Order Model: MVAR and LSTMs in POD Coordinates

This chapter describes the two latent-space time-steppers used in the restrict–evolve–lift pipeline of Chapter 5: a multivariate autoregressive model (MVAR) and a long short-term memory network (LSTM). Both models operate on the r-dimensional POD coefficient vector [2] $y(t_k) \in \mathbb{R}^r$ and produce a one-step-ahead prediction that is iterated in closed loop to generate long-horizon density forecasts. The purpose of this chapter is not to search for the best black-box predictor in the abstract. Rather, it is to test whether shift alignment and POD produce a latent representation whose time evolution is effectively linear at forecast-relevant scales, or whether substantial nonlinear memory remains. If MVAR performs well, that suggests alignment has removed a major source of apparent complexity; if LSTM materially outperforms MVAR, genuine nonlinear latent structure persists; and if both fail, the issue may be low-dimensionality itself rather than model class.

6.1 MVAR(w) model

Let $x(t)$ denote the macroscopic density field on a fixed grid, sampled at discrete times $t_k := t_0 + k\,\Delta t$ ($k = 0, 1, 2, \dots$) with effective sampling interval $\Delta t = 0.04 \times 3 = 0.12\,\mathrm{s}$. The restriction operator $\mathcal{R}$ (Chapter 5) maps $x(t_k)$ to a latent coordinate $y(t_k) \in \mathbb{R}^r$ with $r = 19$. In this chapter we focus on learning a surrogate time-stepper for y.

6.1.1 Model class We model the latent coordinates as a discrete-time vector autoregression with w lags. For each training trajectory ("case") c we write

$$y^{(c)}(t_k) \;=\; A_0 \;+\; \sum_{j=1}^{w} A_j\, y^{(c)}(t_{k-j}) \;+\; \varepsilon(t_k), \tag{6.1.1}$$

where $A_0 \in \mathbb{R}^r$ is an intercept vector and $A_j \in \mathbb{R}^{r\times r}$ are lag-j regression matrices [1]. The residual $\varepsilon(t_k) \in \mathbb{R}^r$ is modelled as zero-mean white noise; this assumption motivates least-squares training but is not used for inference (we do point prediction). MVAR serves here as a hypothesis test: after shift alignment and POD, is the dominant latent evolution well approximated by a linear finite-memory map?

One-step predictor. The corresponding one-step-ahead prediction is

$$\hat{y}^{(c)}(t_k) \;=\; A_0 \;+\; \sum_{j=1}^{w} A_j\, y^{(c)}(t_{k-j}). \tag{6.1.2}$$

6.1.2 Ridge-regularized training The unknown parameters $(A_0, \{A_j\}_{j=1}^{w})$ are obtained by minimizing a ridge-penalized least-squares problem:

$$\min_{A_0,\{A_j\}_{j=1}^{w}} \sum_{c=1}^{C} \sum_{k=w+1}^{K} \left\| y^{(c)}(t_k) - \hat{y}^{(c)}(t_k) \right\|_2^2. \tag{6.1.3}$$

Vectorized regression form. Define the stacked regressor

$$\phi_k^{(c)} \;:=\; \begin{bmatrix} 1 \\ y^{(c)}(t_{k-1}) \\ \vdots \\ y^{(c)}(t_{k-w}) \end{bmatrix} \in \mathbb{R}^{1+wr},$$

and let $B \in \mathbb{R}^{(1+wr)\times r}$ collect the intercept and lag matrices so that $\hat{y}^{(c)}(t_k) = (\phi_k^{(c)})^\top B$. Stacking all $N_{\text{train}} = C(K-w)$ training pairs into matrices $X \in \mathbb{R}^{N_{\text{train}}\times(1+wr)}$ and $Y \in \mathbb{R}^{N_{\text{train}}\times r}$, ridge-regularized least squares solves

$$\hat{B} \;=\; \arg\min_{B}\; \|Y - XB\|_F^2 \;+\; \alpha\, \|B_{2:}\|_F^2, \tag{6.1.4}$$

where $B_{2:}$ excludes the intercept row from regularization and $\alpha > 0$. The closed-form solution is

$$\hat{B} \;=\; (X^\top X + \alpha\,\Gamma)^{-1}\,X^\top Y, \qquad \Gamma = \mathrm{diag}(0, 1, \ldots, 1). \tag{6.1.5}$$

The intercept row is excluded from the penalty term, matching (6.1.5).

Baseline values. In the baseline systematic comparison, the MVAR uses lag $w_{\mathrm{MVAR}} = 5$ and ridge strength $\alpha = 10^{-4}$. At $\Delta t_{\mathrm{rom}} = 0.12\,\mathrm{s}$ this lag window spans $0.6\,\mathrm{s}$ of physical time. This is a compromise between capturing short-term latent dependence and keeping the regression well-conditioned; ridge regularization provides additional robustness. Ridge regularisation improves numerical stability of the fit, but it does not by itself guarantee dynamical stability of the closed-loop rollout, which is determined by the companion spectrum (Section 6.1.3). When the LSTM lag is varied in dedicated ablations, the evaluation protocol uses a common forecast start determined by the largest enabled lag in the config. A formal unrestricted-VAR AIC/BIC study and an empirical lag sweep are reported in Section 3.3; the final lag is ultimately chosen on the basis of that end-to-end empirical comparison.

6.1.3 Closed-loop rollout

For long-horizon forecasting the MVAR model is used in *closed-loop* (autoregressive) mode: each one-step prediction is fed back as input for subsequent steps.

Rollout map. Let the state of the autoregression be

$$z_k := \begin{bmatrix} y(t_k) \\ y(t_{k-1}) \\ \vdots \\ y(t_{k-w+1}) \end{bmatrix} \in \mathbb{R}^{wr}.$$

The MVAR(w) model induces an affine update

$$z_{k+1} = \mathcal{A}_{\mathrm{comp}}\, z_k + b_{\mathrm{comp}}, \tag{6.1.6}$$

where $\mathcal{A}_{\mathrm{comp}} \in \mathbb{R}^{wr \times wr}$ is the companion matrix with explicit block form

$$\mathcal{A}_{\mathrm{comp}} = \begin{bmatrix} A_1 & A_2 & \cdots & A_{w-1} & A_w \\ I_r & 0 & \cdots & 0 & 0 \\ 0 & I_r & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_r & 0 \end{bmatrix}, \qquad b_{\mathrm{comp}} = \begin{bmatrix} A_0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \tag{6.1.7}$$

where $I_r \in \mathbb{R}^{r \times r}$ is the identity and $A_0 \in \mathbb{R}^r$ the intercept; the top block row contains $(A_1, \ldots, A_w)$ and the subdiagonal contains identity "shift" blocks. Errors compound through recursion, so closed-loop metrics are the primary evaluation tool (Chapter 3).

Restrict–evolve–lift forecast. In the full pipeline the autoregressive forecast operates on physical-space densities:

$$\hat{x}^{(c)}(t_k) \;=\; \mathcal{L}\Big(\Phi_{\mathrm{ROM}}\big(\mathcal{R}(\hat{x}^{(c)}(t_{k-1})), \ldots, \mathcal{R}(\hat{x}^{(c)}(t_{k-w}))\big)\Big), \tag{6.1.8}$$

where $\mathcal{R}$ and $\mathcal{L}$ are the restriction and lifting operators of Chapter 5, and Φ_{ROM} is either the MVAR or LSTM time-stepper. The forecast is initialized with the first w true (restricted) snapshots and iterated for all remaining time steps.

Stability. Closed-loop performance depends on the spectral radius $\rho(\mathcal{A}_{\mathrm{comp}}) < 1$, which is a sufficient condition for bounded rollout of the homogeneous part. Ridge regularization shrinks coefficients and therefore tends to improve stability margins.

Optional spectral-radius scaling. An optional post-fit stabilization step uniformly scales down coefficient matrices when $\rho(\mathcal{A}_{\mathrm{comp}})$ exceeds the threshold. This should be regarded as a modified model rather than a harmless stabilisation, because it perturbs the learned dynamics. In preliminary experiments it degraded forecast quality by over-damping transient dynamics, so it is *disabled* in all experimental runs.

Parameter count. Each of the r output dimensions is regressed on w lagged r-vectors plus a bias, giving $p_{\mathrm{MVAR}} = r + w\,r^2$ free parameters. With r=19 and w=5 this yields $p_{\mathrm{MVAR}} = 1{,}824$, fitted from $N_{\mathrm{train}} = C(K-w)$

samples across all training trajectories; ridge regularisation keeps the effective parameter-to-data ratio manageable.

6.2 LSTM latent surrogate

The LSTM serves as the *nonlinear* latent time-stepper in the EF-ROM benchmark. It is introduced here as a flexible nonlinear comparator on the same latent representation, not as an end in itself. Because the LSTM has roughly 116× more parameters than MVAR (Section 6.1.3), any LSTM advantage is hard to attribute purely to nonlinear structure versus overfitting capacity; the LSTM's role is therefore *diagnostic* rather than competitive. Specifically:

- MVAR failure suggests either instability or low-dimensional mismatch.
- LSTM improvement over MVAR would suggest genuine nonlinear latent memory.
- LSTM non-improvement despite massive extra capacity is positive evidence that nonlinear expressivity is not the main bottleneck.

Sequence-to-one formulation. Let $\mathbf{z}_k \in \mathbb{R}^d$ denote the POD coefficient vector at time t_k ($d = r = 19$). A p-step autoregressive surrogate learns the one-step map

$$\mathbf{z}_k \;\approx\; f_\theta\big(\mathbf{z}_{k-1}, \ldots, \mathbf{z}_{k-p}\big), \qquad k = p{+}1, \ldots, K, \tag{6.2.1}$$

where f_θ is a parameterized nonlinear model. LSTMs are preferred over vanilla RNNs because the gated cell state mitigates vanishing/exploding gradients [20], which matters when the latent series exhibits long transients or regime shifts.

LSTM architecture. Given an input window $\mathbf{x}_k = (\mathbf{z}_{k-p}, \ldots, \mathbf{z}_{k-1}) \in \mathbb{R}^{p\times d}$, the LSTM processes the sequence through forget, input, and output gates that regulate information flow into the cell state $\mathbf{c}_\tau$ and hidden state $\mathbf{h}_\tau$ (gate equations in Eq. (C.9.1), Appendix C). The final hidden state is passed through LayerNorm before a linear readout.

Residual (delta) prediction. In all experiments, the network uses residual prediction, so the output layer predicts a *correction* Δ_k and the final prediction is

$$\widehat{\mathbf{z}}_k \;=\; \mathbf{z}_{k-1} \;+\; \Delta_k, \qquad \Delta_k := \mathbf{W}_y\,\mathrm{LayerNorm}(\mathbf{h}_p) + \mathbf{b}_y. \tag{6.2.2}$$

This residual connection anchors the prediction to the most recent input state, which improves stability during closed-loop rollout.

Training objective and configuration. The LSTM is trained via one-step MSE on sliding windows of ground-truth latent states, optimised with Adam; early stopping, gradient clipping, and a learning-rate scheduler are used during training. Full hyperparameters are listed in Table C.1 (Appendix C).

Parameter count. Each LSTM layer of hidden dimension h contributes $4(d_{\mathrm{in}}h + h^2) + 8h$ weights (four gates × input-to-hidden, hidden-to-hidden, and two biases). With $d{=}r{=}19$, $h{=}128$, $L{=}2$ layers, plus LayerNorm ($2h$) and a linear readout ($hd + d$), the total is $p_{\mathrm{LSTM}} = 76{,}288 + 132{,}096 + 256 + 2{,}451 = 211{,}091$, roughly 116× the MVAR count of 1,824, a ratio that will become relevant when interpreting forecast quality in Chapter 7.

Closed-loop forecasting. After training, we forecast by recursively feeding predictions back into the input window: $(\mathbf{z}_{k-p}, \ldots, \mathbf{z}_{k-1}) \mapsto \widehat{\mathbf{z}}_k \mapsto (\mathbf{z}_{k-p+1}, \ldots, \mathbf{z}_{k-1}, \widehat{\mathbf{z}}_k) \mapsto \cdots$ This closed-loop evaluation matches the restrict–evolve–lift protocol of (6.1.8). Unlike MVAR, the LSTM has no eigenvalue-based stability certificate; stability is monitored empirically via rollout diagnostics (boundedness of POD coefficients and mass drift after lifting).

Algorithmic summary. Algorithm 5 (Appendix F) summarises the LSTM ROM pipeline.

Comparison to MVAR. MVAR offers fast closed-form training and transparent stability analysis (companion-matrix spectral radius), while LSTM provides nonlinear capacity at the cost of more hyperparameters and weaker a priori rollout guarantees. In crowd-dynamics EF-ROM benchmarks, linear MVARs can match or outperform LSTMs in predictive accuracy while being lower-complexity and more interpretable [1].

The comparison in Chapter 7 should therefore be interpreted primarily as evidence about the effective geometry of the aligned latent dynamics: whether they are primarily linear but potentially unstable, substantially nonlinear, or insufficiently low-dimensional for either surrogate to capture over long horizons. It is not a leaderboard-style predictor comparison.

7 ROM Results: MVAR vs LSTM

This chapter presents the reduced-order modelling results for MVAR and LSTM applied to the equation-free density ROM. The *primary benchmark* comprises five constant-speed (CS) regimes plus a pure alignment baseline (`NDYN04_gas`, `NDYN05_blackhole`, `NDYN06_supernova`, `NDYN07_crystal`, and `NDYN08_pure_vicsek`), which span the key behavioural axes (attraction, repulsion, equilibrium, alignment only) under well-controlled shift dynamics. In addition, four *variable-speed* (VS) diagnostic variants (`NDYN04_gas_VS`, `NDYN05_blackhole_VS`, `NDYN06_supernova_VS`, and `NDYN07_crystal_VS`) are included to test whether the performance ceiling observed on the primary benchmark is a *structural* limitation of single-reference sPOD [31] or merely a model-class limitation that a more expressive forecaster could overcome. Aggregate statistics reported in this chapter (median R^2, efficiency comparisons) are computed over the CS + PV primary benchmark unless stated otherwise; VS results are presented as a separate diagnostic. The complete systematic catalogue with dynamics parameters and per-regime R^2 is provided in Appendix B, and per-experiment supplementary figures are collected in Appendix C.

7.1 Shift predictability as the benchmark ceiling

Accordingly, the forecast task studied here is not full transport prediction in lab coordinates, but density-shape prediction after removal of bulk translation.

The MVAR has relatively low statistical complexity in the aligned latent space, whereas the LSTM must estimate a substantially more flexible temporal map from the same finite set of windows, an asymmetry worth bearing in mind when interpreting nonlinear modelling gains. Figure 7.1 shows the singular-value spectra of the aligned snapshot matrices across all nine benchmark regimes.

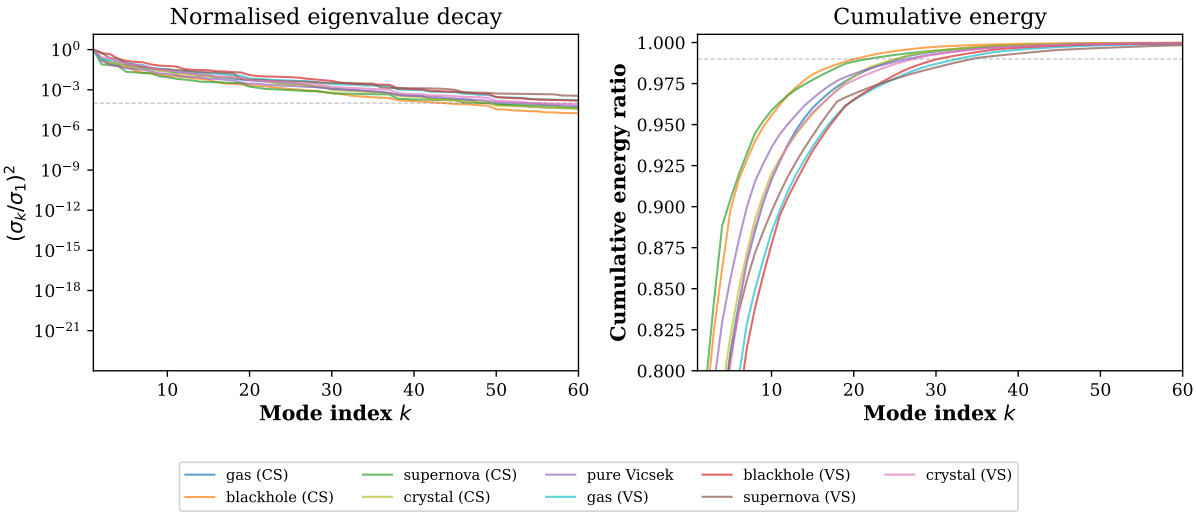


Figure 7.1: Singular-value spectra of the aligned snapshot matrices across all nine benchmark regimes. Left: normalised eigenvalue decay $(\sigma_k/\sigma_1)^2$ on a log scale. Right: cumulative energy ratio. Each coloured curve represents one regime. The aligned spectra exhibit rapid decay and support a compact POD truncation.

Before comparing model outputs it is instructive to characterise the difficulty of the forecasting task itself. Both MVAR and LSTM operate exclusively in the *aligned* latent space: their state vectors are the r=19 POD coefficients of the shift-corrected density field, and the spatial shift $\mathbf{\Delta}(t) = (\Delta x(t), \Delta y(t))$ is *not* included as a forecast variable. During evaluation, ground-truth test shifts (computed via FFT cross-correlation against the training reference frame) are used solely to un-align predictions back to physical space for visualisation; the R^2 metric is computed entirely in the aligned frame. The temporal predictability of $\mathbf{\Delta}$ nevertheless imposes a structural ceiling on forecast fidelity: when shifts are poorly predictable the aligned density retains residual transport structure that inflates the effective latent dimension and degrades both models equally. Figure 7.2 summarises two complementary shift diagnostics (trajectory geometry and autocorrelation structure) across the benchmark regimes.

Mean shift trajectory. The left panel traces $\overline{\mathbf{\Delta}}(t)$ in the $(\Delta y, \Delta x)$ plane. All regimes stay within ±4–6 grid cells; gas and pure Vicsek [35] trace smooth persistent loops, whereas variable-speed blackhole traces a near-memoryless random walk, indicating that bulk translation is no longer a sufficient coherent-motion summary.

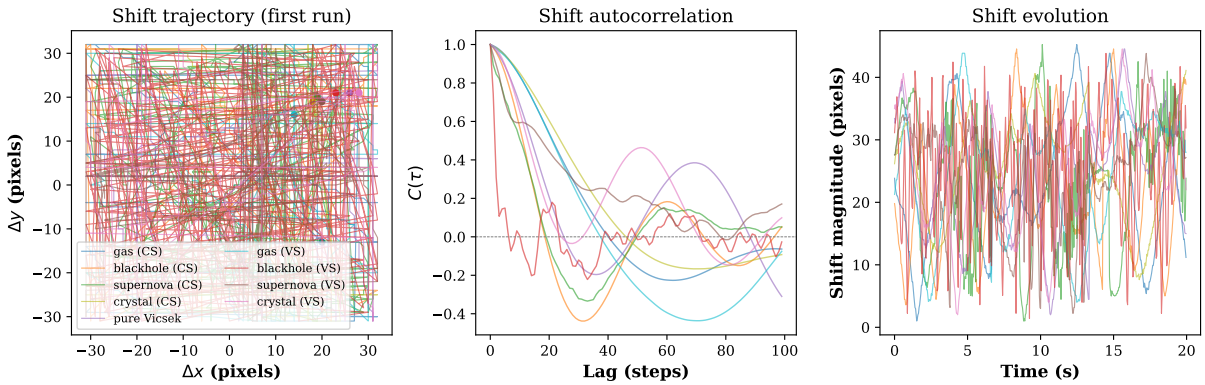


Figure 7.2: Shift dynamics across the benchmark regimes. *Left:* Shift trajectory ($\Delta y(t)$, $\Delta x(t)$) in the alignment frame for a representative simulation run. Each coloured path traces the bulk centre-of-mass displacement; symbols mark the start. *Centre:* Temporal autocorrelation $C(\tau)$ of the shift signal as a function of lag τ, normalised so that $C(0) = 1$. *Right:* Shift magnitude evolution over time, showing the raw displacement signal $\|\mathbf{\Delta}(t)\|$ for each regime.

Shift autocorrelation. The centre panel plots the temporal autocorrelation $C(\tau)$ and partitions regimes into three tiers: gas, supernova, and pure Vicsek retain significant correlation at long lags; constant-speed blackhole sits intermediate; variable-speed blackhole collapses to near-zero correlation within a few lag steps. The right panel shows the raw shift magnitude over time, confirming that VS regimes exhibit faster, less structured fluctuations. Fitting AR(p) models ($p = 1, \ldots, 10$) to the shift time series (not shown) yields consistent rankings: gas and pure Vicsek reach $R^2 \approx 0.88$; supernova and constant-speed blackhole land near 0.83; variable-speed blackhole is a clear outlier at ≈ 0.45. R^2 is flat across lag orders, validating the MVAR lag window (Section 6).

Synthesis. Shift predictability is a major structural determinant of ROM forecast quality, but it is not sufficient on its own. Variable-speed blackhole is the clearest failure case: its shift process has the weakest temporal structure, and LSTM falls to negative forecast R^2, while MVAR remains positive but substantially degraded. By contrast, gas and constant-speed blackhole combine strong shift structure with favourable aligned latent dynamics, and both models achieve $R^2 \approx 0.99$. Pure Vicsek retains relatively predictable shifts, but its density dynamics remain harder than gas/blackhole after alignment, so MVAR drops to moderate rather than near-perfect accuracy.

Post-hoc eigenvalue rescaling (Equation 7.3.1, Section 7.3) clips 2–3 eigenvalues in the VS variants yet residual error remains high, confirming that the root cause is shift autocorrelation, not linear instability of the fitted propagator. VS rescaling does not fix the failure.

7.2 Forecasting results: qualitative and quantitative

This section presents the forecast evidence in three complementary views: density snapshots illustrating spatial fidelity, aggregate R^2 metrics quantifying regime-level accuracy, and order parameter trajectories testing whether physically meaningful collective statistics survive the compress–evolve–lift cycle.

Density snapshots We show a single representative regime (gas) where both models succeed; additional regimes (blackhole, supernova) are collected in Appendix C (Figure C.1).

Across all regimes, both forecasters reproduce the dominant low-rank structure of the aligned density field (large-scale envelope, centre-of-mass location, overall mass distribution) but differ in how they handle fine-scale spatial detail and late-horizon accuracy. MVAR's linear dynamics produce spatially smooth predictions that degrade gracefully; LSTM's nonlinear capacity allows it to track sharper features at the cost of occasional high-frequency artefacts once the autoregressive rollout accumulates error.

Forecast R^2 across regimes The heatmap (Figure 7.4) provides the bird's-eye view: MVAR achieves $R^2 > 0.95$ on gas and blackhole (CS), drops to roughly 0.65–0.68 on supernova and pure Vicsek, and maintains positive R^2 across all CS + PV regimes. LSTM follows a broadly similar pattern but with higher variance. The $R^2(t)$ degradation curves (Figure 7.5) reveal that the model gap is predominantly a late-horizon effect.

The VS diagnostic regimes show substantially reduced R^2 relative to their CS counterparts. MVAR remains positive across all VS regimes (median 0.425; blackhole_VS reaches 0.704), while LSTM stays positive on gas_VS (0.11) and supernova_VS (0.16) but falls negative on blackhole_VS (-0.21) and crystal_VS (-0.78). The consistent CS–VS gap across all interaction families (Figure 7.6) confirms that the ceiling is set by shift predictability (Section 7.1), not forecaster capacity.

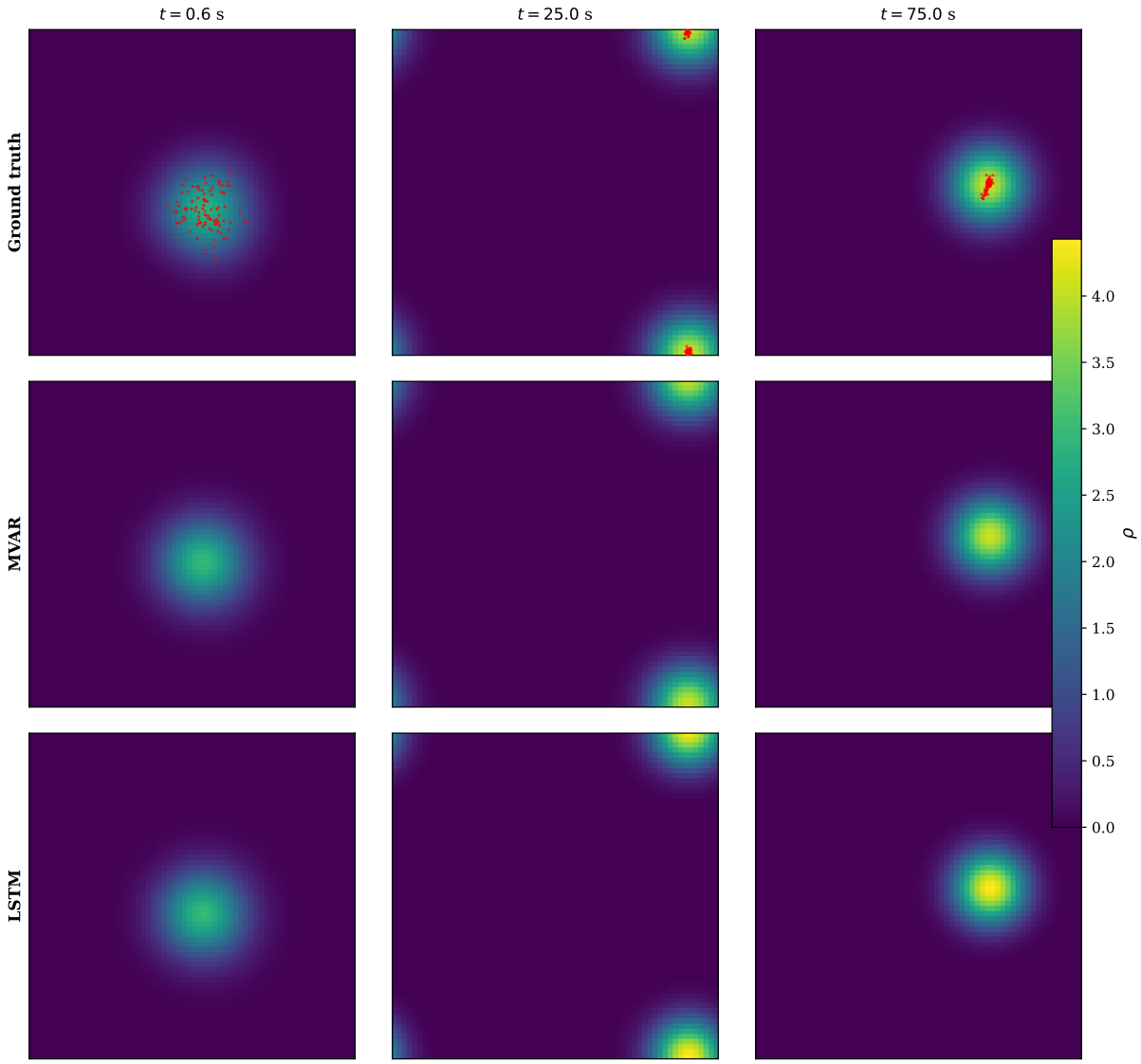


Figure 7.3: Density snapshots truth vs. prediction for `NDYN04_gas` (Gaussian IC, test run 0). Columns show $t = 0.6$ s, $t = 25$ s, and $t = 75$ s. Rows: ground truth, MVAR, LSTM. Colour scale is shared across all panels within each time column but differs between columns (the peak density decreases as the cluster diffuses; compare colourbar ranges across columns). MVAR faithfully reproduces the diffusing cluster throughout the horizon. LSTM maintains gross spatial structure but develops small-scale artefacts by $t = 75$ s, consistent with its higher variance in the $R^2(t)$ curves.

Summary of quantitative findings. MVAR dominates the CS + PV primary benchmark (median R^2=0.995) and remains positive on every VS regime (median 0.425). LSTM adds no systematic benefit: its median CS + PV R^2 is 0.585, and it falls negative on two of four VS regimes. The failure pattern is fully explained by the shift-predictability ceiling established in Section 7.1: where shifts are well-predicted (gas, blackhole CS), both models achieve $R^2 > 0.99$; where shifts are poorly predicted (VS variants), both degrade in lockstep, with LSTM's additional capacity amplifying closed-loop error rather than compensating for it. The bottleneck is aligned-transport predictability, not model expressiveness.

Order parameter fidelity To assess whether the ROM captures collective behaviour beyond pixel-level density accuracy, we compare ground-truth and predicted order parameter time series. Two scalar diagnostics are extracted at each forecast time step: the spatial concentration index $C(t) = \mathrm{Var}[\rho(x,t)]/\mathrm{mean}[\rho(x,t)]^2$ (the squared coefficient of variation of the density field), computed directly from the lifted density field, and the mean alignment $\bar{\phi}(t)$, computed from the ground-truth agent velocities (not from the density-only ROM output, since the ROM does not reconstruct velocity fields). $C(t)$ tests whether physically meaningful aggregate statistics survive the compress–evolve–lift cycle; $\bar{\phi}(t)$ serves as an independent particle-level reference against which to assess whether density-space accuracy tracks collective behavioural fidelity. The load-bearing diagnostics vary by regime family: polarisation and mean speed are primary for flocking regimes (pure Vicsek, gas), angular momentum for milling or

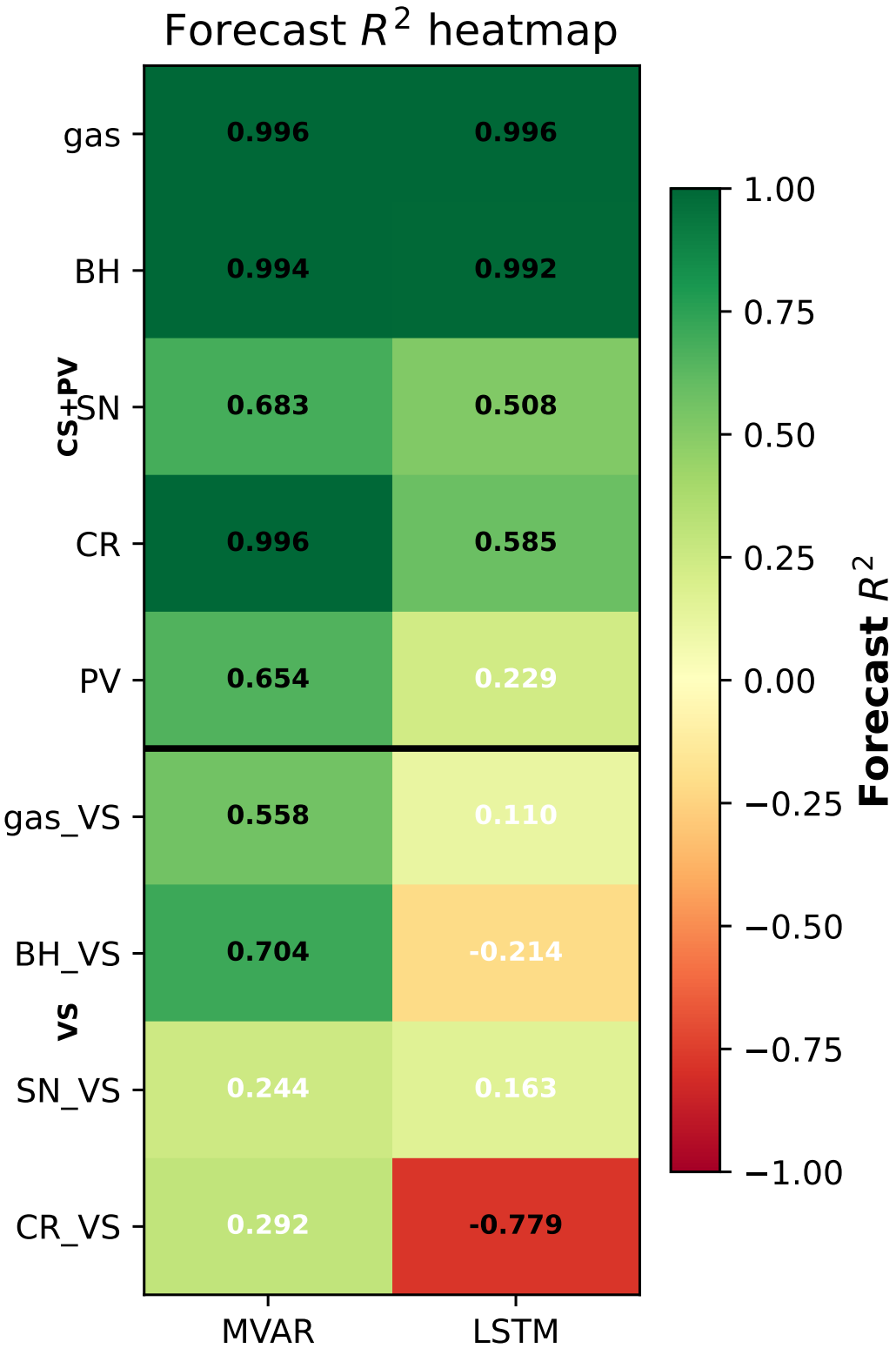


Figure 7.4: Forecast R^2 heatmap: rows are experiments (grouped by regime family), columns are MVAR and LSTM. Cell colour encodes forecast R^2 on a diverging red–yellow–green scale; numeric annotations give the exact value. A horizontal rule separates the primary benchmark (CS + PV, top) from the VS diagnostic regimes (bottom). WSINDy is omitted because it produces identification-quality scores (weak-form R^2) rather than time-stepped forecasts; its metrics appear in Table 9.7.

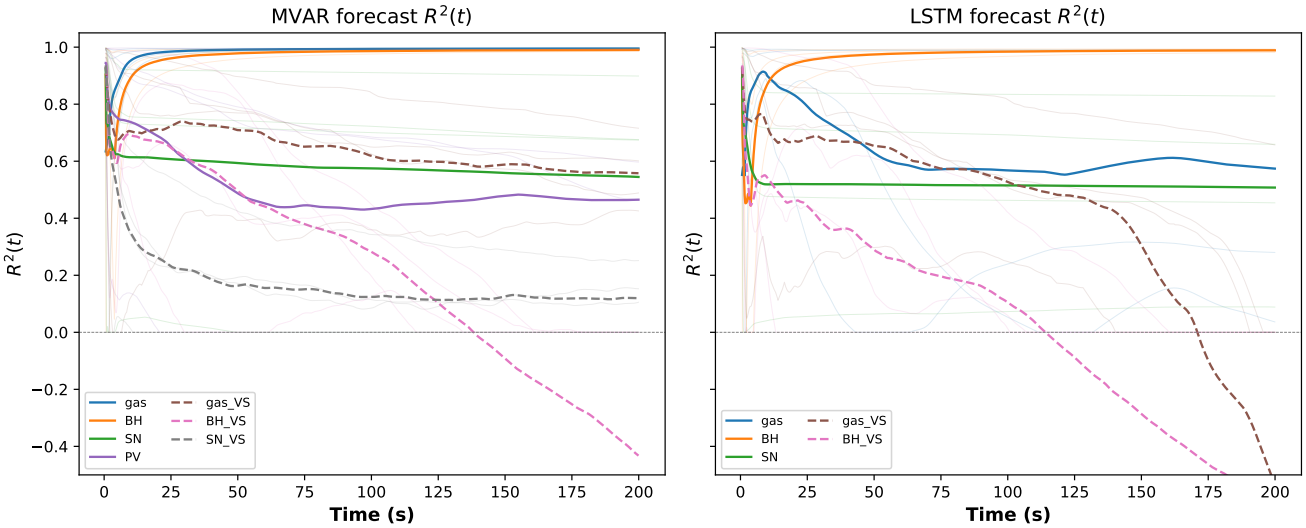


Figure 7.5: Forecast R^2 degradation over the prediction horizon (left: MVAR, right: LSTM). Bold curves show per-regime means; faint traces are individual test runs. Solid lines = CS + PV benchmarks; dashed = VS diagnostics. Per-experiment detail with $\pm 1\sigma$ shading is in Appendix B (Figure B.1).

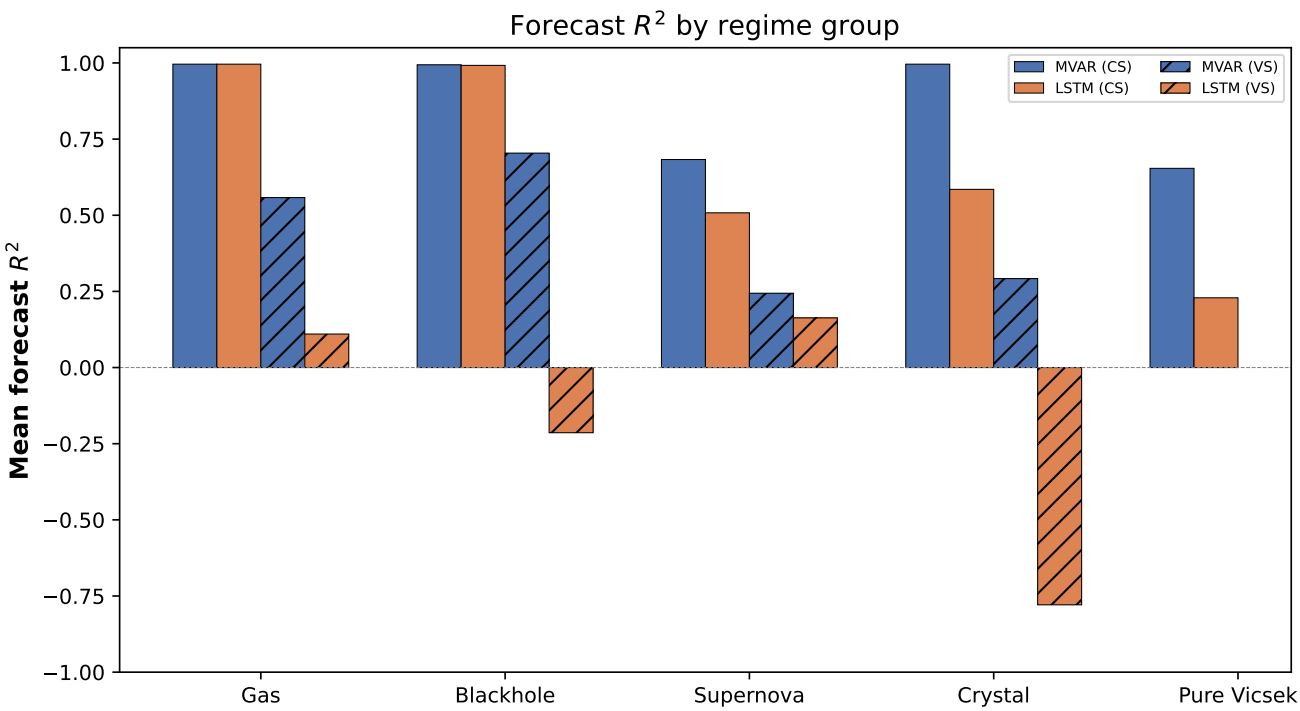


Figure 7.6: Mean test R^2 by regime group (grouped bar chart). Each cluster of bars corresponds to one of the five benchmark regimes (gas, blackhole, supernova, crystal, pure Vicsek); within each cluster, bars represent MVAR and LSTM. Error bars show ± 1 standard deviation across experiments within the group. Solid bars are the CS + PV primary benchmark; hatched bars are the VS diagnostic variants (pure Vicsek has no VS variant by construction). The CS–VS gap is immediately visible and consistent across all regime families, indicating that the bottleneck is the shift-predictability prerequisite rather than the model class.

ring-forming regimes, and spatial concentration for clustering regimes (blackhole, supernova).

For gas and blackhole (CS), both MVAR and LSTM track the slow decay of $C(t)$ accurately and reproduce alignment transients with only small phase lags at late time. For supernova, MVAR captures the monotone concentration decrease but underestimates the rate of dispersal; LSTM more closely matches the curvature but introduces spurious high-frequency oscillations in $\bar{\phi}$. On the variable-speed blackhole, neither model tracks the rapid fluctuations in $C(t)$; the predicted trajectories drift to the class mean, consistent with the near-zero shift autocorrelation reported in Section 7.1.

7.3 Cost and stability: MVAR versus LSTM

MVAR fitting and WSINDy discovery complete in seconds; LSTM training is one to two orders of magnitude more expensive (Table 7.1). MVAR has 1 824 parameters; LSTM has 211 091; WSINDy retains 9–14 active terms across the three field equations. Table 7.1 quantifies these trade-offs.

Table 7.1: MVAR vs LSTM efficiency summary. MVAR achieves comparable or superior forecast quality with 116× fewer trainable parameters and two orders of magnitude less training time. Primary-benchmark aggregates (CS + PV) are reported separately from VS diagnostics to prevent the structural shift-predictability ceiling of VS regimes from obscuring the models' performance where the pipeline prerequisites hold.

Metric	**MVAR**	**LSTM**
Trainable parameters	1 824	211 091
Training time (median)	< 1 min	∼45 min
Primary benchmark (CS + PV)		
Median test R^2	0.995	0.585
Best-case test R^2	0.996 (gas, crystal)	0.996 (gas)
Worst-case test R^2	0.654 (PV)	0.229 (PV)
VS diagnostic		
Median test R^2	0.425	−0.052
Inference (steps/s)	$> 10^4$	$> 10^4$

All medians computed over the full subgroup (5 CS + PV or 4 VS regimes).

Stability of the latent propagator. MVAR stability is analysed via the spectral radius of the companion matrix $\mathbf{A}$: eigenvalues with $|\lambda_i| > 1$ are projected onto the unit circle via

$$\tilde{\lambda}_i = \begin{cases} \lambda_i/|\lambda_i| & \text{if } |\lambda_i| > 1, \\ \lambda_i & \text{otherwise.} \end{cases} \tag{7.3.1}$$

This per-eigenvalue unit-circle clipping is distinct from the global uniform spectral-radius scaling described in Chapter 6, which was disabled during training because it perturbs all eigenvalues indiscriminately (Section 6.1.3). The projection in Equation (7.3.1) is a minimal post-hoc correction applied only at evaluation time [16, 19]: it neutralises the few unstable modes without altering the fitted dynamics of the stable majority. Gas and pure Vicsek are naturally stable ($\rho(\mathbf{A}) < 1$), requiring no correction; blackhole requires minimal correction (one mode at $|\lambda| \approx 1.02$). In the VS variants, rescaling clips 2–3 eigenvalues but residual prediction error remains high because the fundamental issue is not stability but rather the absence of predictable shift structure. Companion-matrix eigenvalue spectra are reported in Appendix C (Figure C.2).

LSTM overhead without forecast gain. LSTM training is one to two orders of magnitude more expensive than MVAR fitting and introduces several additional hyperparameters (hidden-unit count, number of layers, dropout rate, learning-rate schedule). The head-to-head comparison shows that for the majority of the benchmark, LSTM's R^2 lies on or slightly below the diagonal, indicating parity with or mild inferiority to MVAR. On the two well-aligned CS regimes (gas, blackhole) the two models achieve near-parity ($R^2 \approx 0.99$), but on every other regime LSTM is *worse* than MVAR. On supernova, LSTM drops to $R^2 \approx 0.51$ versus MVAR's 0.68; on pure Vicsek and the VS variants LSTM R^2 ranges from −0.78 (crystal_VS) to 0.23 (pure Vicsek), with two VS regimes falling negative. In all cases MVAR retains positive (if modest) R^2. The pattern is consistent with overfitting: with 211 091 parameters and only ∼55 000 training windows the additional model capacity amplifies closed-loop rollout divergence rather than capturing genuine nonlinear structure in the latent dynamics.

LSTM's failure modes are distinct from MVAR's: (i) *closed-loop error accumulation* is more severe because prediction noise couples nonlinearly through the gates; (ii) *training instability* across random seeds contributes to wider error bars in Figure 7.6; (iii) LSTM inherits the same shift-predictability ceiling as MVAR (Section 7.1). The divergence of LSTM latent-steppers in high-velocity regimes suggests an over-sensitivity to residual nonlinearities in the shift-aligned coordinates, whereas the linear MVAR provides a more robust, albeit lower-fidelity, approximation.

7.4 Summary of ROM findings

All nine benchmark regimes are included in every figure and table in this chapter.

Why the linear surrogate can fail. Alvarez et al. [1] demonstrated that MVAR provides competitive or superior forecasting in an equation-free density ROM for Vicsek-type systems. The present benchmark expands the regime coverage and reveals three mechanisms of failure. **Linearity and stationarity mismatch:** MVAR assumes linear, stationary latent dynamics, but regimes with strong alignment coupling exhibit nonlinear saturation, and variable-speed regimes modulate effective diffusivity on time scales comparable to the lag window. **Heavy-tail POD spectra:** for supernova and the VS variants, sPOD spectral decay is slower, leaving 5–10 % of energy in a long tail of modes; with $p = r + wr^2 = 1{,}824$ parameters (r=19, w=5), a fraction of the MVAR capacity is allocated to high-index modes that carry little energy but introduce noise; ridge regularisation mitigates but does not fully resolve this inefficiency. **Data-to-parameter ratio:** with C=340 trajectories of K=167 subsampled time steps (lag w=5), the effective training set comprises $340 \times 162 = 55{,}080$ windows for 1,824 parameters, a ratio of approximately 30. While formally well-conditioned, many high-index modes carry minimal energy, so the effective degrees of freedom are lower and the fit remains sensitive to the ridge penalty α.

Three prerequisites and three negative results. Reliable EF-ROM density forecasts require three conditions simultaneously: *effective shift alignment* (sPOD removes bulk translation and concentrates energy in few modes), *predictable shift dynamics* ($R^2 > 0.8$ for a low-order AR model of $\mathbf{\Delta}(t)$), and an *adequate data-to-parameter ratio* (training windows $\geq$ MVAR parameter count). When all three hold (gas, blackhole CS), MVAR achieves $R^2 > 0.95$ at a fraction of LSTM cost.

Three negative results follow. Variable-speed regimes defeat any single-reference alignment strategy. LSTM's nonlinear capacity does not overcome the shift-predictability bottleneck: on every regime where MVAR struggles, LSTM performs equally poorly or worse. Finally, the 116× parameter ratio of LSTM over MVAR yields no proportionate gain on this benchmark.

Transition to Part II. The EF-ROM pipeline answers "how well can we forecast?" but not "why does the density evolve the way it does?" The latent coordinates are compression artefacts with no direct physical meaning; the AR/LSTM coefficients cannot be mapped to physical fluxes or interaction forces. Part II addresses this gap: WSINDy operates directly on the density and polarisation fields, producing sparse symbolic PDEs whose terms correspond to physically interpretable flux mechanisms. By comparing the discovered equation structure across CS and VS variants of the same regime, we can extract structural information about which mechanisms are active, thereby asking questions that no amount of ROM forecasting can answer.

Part II

Weak-form PDE Discovery (WSINDy)

8 WSINDy: Weak-Form PDE Discovery

This chapter presents the *Weak-form Sparse Identification of Nonlinear Dynamics* (WSINDy) pipeline [26]. Whereas Part I compresses density snapshots into POD coordinates and learns a discrete-time latent map, WSINDy stays in physical field space and seeks an explicit continuous-time PDE of the form

$$u_t = \sum_{m=1}^{M} w_m \, \mathcal{L}_m\big[f_m(u)\big], \tag{8.0.1}$$

where each candidate term is a (differential-operator, nonlinear-feature) pair $(\mathcal{L}_m, f_m)$ drawn from a user-specified library, and the sparse coefficient vector $\mathbf{w} \in \mathbb{R}^M$ is to be determined. WSINDy identifies sparse, interpretable effective laws at the chosen coarse-graining scale; forward integration serves only as a closure stress test (Section 8.6).

Identification criteria. Throughout Chapters 8–9, *successful identification* means that the discovered PDE satisfies:

(i) stable support recovery under cross-scale stability selection (Section 8.5),
(ii) physically plausible sign structure (e.g. positive diffusion, correct attraction/repulsion polarity for each regime),
(iii) strong weak-form residual fit (R^2_{wf} close to unity), and
(iv) active mechanisms that are consistent with the expected regime-level physics encoded by the candidate library.

Autonomous rollout is reported as a closure diagnostic, not as the primary criterion of identification quality. A well-identified but unclosed PDE may diverge under integration without invalidating the identification itself.

Three scales. Three length scales must be distinguished that jointly determine the effective model: (i) the *microscopic interaction scale* set by Morse lengths ℓ_r, ℓ_a and the interaction radius R; (ii) the *KDE coarse-graining scale* set by the kernel bandwidth h; and (iii) the *weak test-function support scale* set by the window widths ℓ_x, ℓ_y, ℓ_t selected by the automatic procedure below. Whenever the coarse-graining bandwidth exceeds a microscopic interaction length, that interaction cannot be independently resolved by the identified PDE. All subsequent PDE interpretations are therefore conditional on these fixed scale choices.

The processing stages are shown in Figure 8.1. Sections 8.1–8.6 describe each stage in order. Detailed pseudocode for the two most intricate components (the sparse regression algorithm and the time integrator) is deferred to Appendix D (sequential thresholding with dominant-balance scoring) and Appendix E (exponential time-differencing with contour-integral stabilisation), respectively.

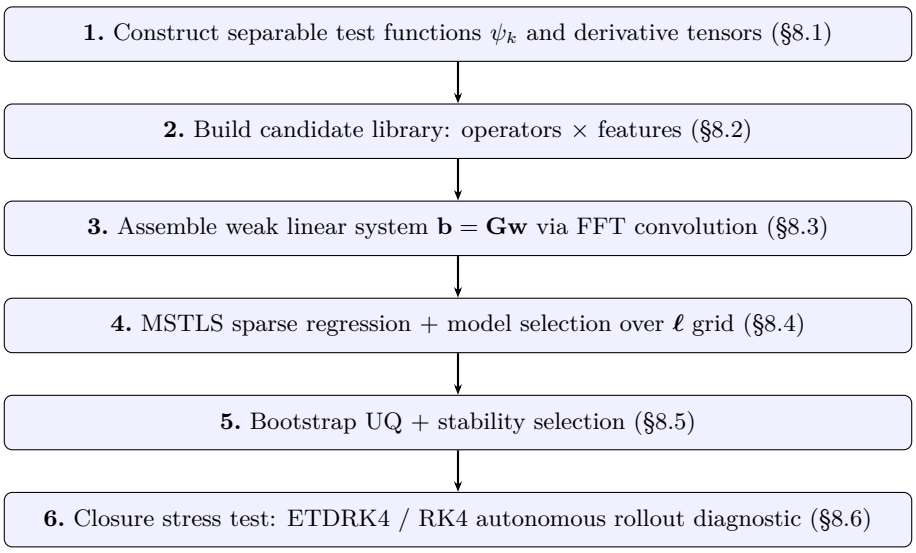


Figure 8.1: WSINDy pipeline: from density snapshots to a discovered PDE and its forecast.

8.1 The weak-form framework

Standard sparse-regression approaches (e.g. SINDy [6] and its PDE extension PDE-FIND [32]) evaluate spatial derivatives of the data numerically, amplifying measurement noise. WSINDy avoids this by reformulating the identification problem in *weak form*: derivatives are transferred from the noisy data u onto smooth, compactly supported test functions via integration by parts.

8.1.1 Test functions

We use separable polynomial bump functions

$$\psi(x, y, t) = \phi_x(x)\, \phi_y(y)\, \phi_t(t), \tag{8.1.1}$$

where each factor is a compactly supported polynomial on $[-\ell_i\Delta_i,\ \ell_i\Delta_i]$:

$$\phi_i(s) = \begin{cases} \left(1 - \dfrac{s^2}{(\ell_i\Delta_i)^2}\right)^{q_i}, & |s| \le \ell_i\Delta_i, \\ 0, & \text{otherwise}, \end{cases} \qquad i \in \{x, y, t\}. \tag{8.1.2}$$

Here ℓ_i is the half-width in grid cells, Δ_i the grid spacing in direction i, and q_i the polynomial exponent (default $q_i = 2$). Increasing q_i yields smoother bumps at the cost of narrower effective support; increasing ℓ_i widens the averaging window. The one-dimensional bump $\phi(s) = (1 - s^2)^q$ concentrates mass near the centre for higher q.

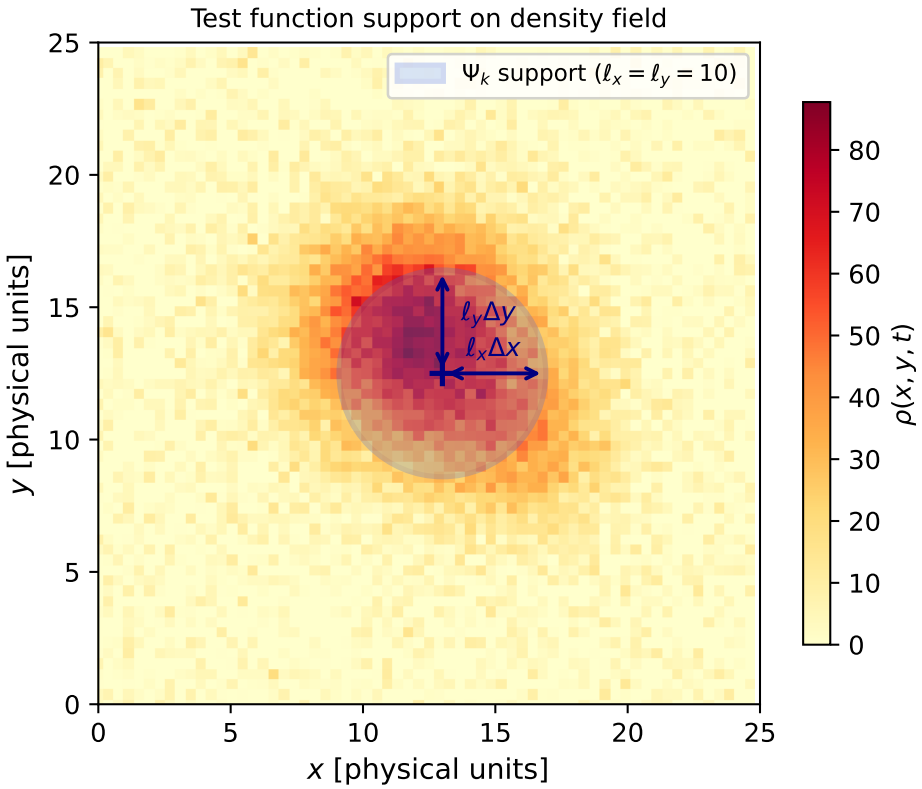


Figure 8.2: Two-dimensional test function footprint overlaid on a representative density field (gas regime, $t = 10\,\mathrm{s}$). The shaded ellipse shows the spatial support of a single test function $\Psi_k(x, y) = \psi_x(x)\psi_y(y)$ with half-widths $(\ell_x\Delta x, \ell_y\Delta y)$ at the default scale $\ell_x = \ell_y = 10$. The footprint covers several density wavelengths, illustrating the spatial averaging implicit in the weak-form inner products. Coarser ℓ suppresses high-frequency noise but smears sharp gradients; finer ℓ resolves smaller features at the cost of noise amplification. The support widths of the test functions should therefore be interpreted as part of the discovered model scale: changing them changes which modes of the data dominate the weak constraints, and therefore changes the effective PDE that is recoverable.

8.1.2 Integration by parts

Consider the PDE $u_t = F(u)$ on the periodic spatial domain $\Omega = [0, L_x) \times [0, L_y)$ over the time interval $[0, T]$. For a test function ψ_k centred at a query point (x_k, y_k, t_k), let $Q_k := \operatorname{supp} \psi_k \subset \Omega \times [0, T]$ denote its spacetime support, with half-widths $(\ell_t\Delta t, \ell_x\Delta x, \ell_y\Delta y)$. Multiplying both sides by ψ_k and integrating over Q_k gives

$$\int_{Q_k} \psi_k(x, y, t)\, u_t(x, y, t)\, dx\, dy\, dt = \int_{Q_k} \psi_k(x, y, t)\, F(u)(x, y, t)\, dx\, dy\, dt.$$

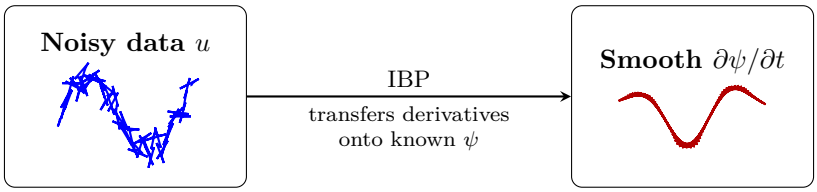
Equivalently, since ψ_k vanishes outside Q_k, one may view these as integrals over the full spacetime domain $\Omega \times [0, T]$. Integrating the left side by parts in time, the temporal boundary term vanishes because ψ_k is compactly supported in time (i.e. ψ_k vanishes at the temporal endpoints of Q_k):

$$-\int_{Q_k} \frac{\partial \psi_k}{\partial t}(x, y, t)\, u(x, y, t)\, dx\, dy\, dt = \int_{Q_k} \psi_k(x, y, t)\, F(u)(x, y, t)\, dx\, dy\, dt. \tag{8.1.3}$$

Likewise, for differential operators appearing in $F(u)$, spatial integration by parts transfers derivatives from the data u onto the test function ψ_k. Under periodic boundary conditions on Ω, the spatial boundary contributions vanish (by a distinct mechanism: periodicity, not compact support). For example, if $F(u)$ contains a term $w_m\, \Delta u$, then

$$w_m \int_{Q_k} \psi_k\, \Delta u\, dx\, dy\, dt = -\, w_m \int_{Q_k} \nabla \psi_k \cdot \nabla u\, dx\, dy\, dt = w_m \int_{Q_k} (\Delta \psi_k)\, u\, dx\, dy\, dt,$$

where the intermediate equality follows from two spatial integrations by parts with periodic boundary terms vanishing. Thus the data u always appears *undifferentiated*. The one-dimensional bump formula (8.1.2) is analytic, but the sampled derivative arrays used in the weak system are evaluated numerically on the discrete support lattice via one-dimensional finite-difference stencils and then assembled by separable outer products. Although the bump formula is analytic and admits closed-form derivatives, numerical evaluation on the sampled lattice avoids special-casing outside the general-purpose convolution pipeline and introduces errors that are negligible relative to the test-function support scale.



Result: all derivatives act on the analytic, smooth ψ, never on the noisy u.

Figure 8.3: Schematic of the weak-form integration-by-parts mechanism. The noisy field data u (left) enters the inner product undifferentiated, while all spatial and temporal derivatives are transferred to the compactly supported, analytic test function ψ (right). This derivative transfer is the key regularisation that enables stable regression from noisy data without explicit filtering or finite-difference stencils on u.

8.1.3 Separability and FFT convolution

Because ψ_k is separable (8.1.1), each three-dimensional integral reduces to a product of one-dimensional convolutions, computed efficiently via the Fast Fourier Transform. The spatial dimensions are periodic and use circular convolution; the temporal dimension is zero-padded to avoid wrap-around artefacts.

8.2 Candidate library construction

The library specifies which physical mechanisms the regression is allowed to select. Each candidate term is a pair $(\mathcal{L}_m, f_m)$ consisting of a differential operator $\mathcal{L}_m$ and a pointwise nonlinear feature f_m. Conceptually, the terms fall into five blocks: *transport and continuity backbone* (e.g. $\nabla \cdot \mathbf{p}$), *diffusive regularisation* (e.g. $\Delta\rho$, Δp_x), *alignment-related polarisation terms* (e.g. linear drag p_x, cubic saturation $|\mathbf{p}|^2 p_x$), *nonlocal interaction terms from Morse forcing* (e.g. $\rho\, \partial_x \Phi$, $\nabla \cdot (\rho \nabla \Phi)$), and *closure surrogates* (e.g. $\partial_x(\rho^2)$ as a congestion proxy).

8.2.1 Scalar library

For a single field u the default library is the Cartesian product of polynomial features and spatial operators:

$$\{f_j\} = \{1,\ u,\ u^2,\ u^3\}, \qquad \{\mathcal{L}_j\} = \{I,\ \partial_x,\ \partial_y,\ \Delta\}, \tag{8.2.1}$$

yielding up to 16 candidate terms (e.g. Δu^2, $\partial_x u$). The maximum polynomial power is controlled by $p_{\max}$ (default 3) and the operator set is configurable. Optional extensions include nonlinear diffusion terms $\Delta(u^p)$ for specified powers p, gradient nonlinearities $|\nabla u|^2 = u_x^2 + u_y^2$, and conservative forms $\partial_x(u^p)$, $\partial_y(u^p)$. The library is assembled programmatically via a fluent builder pattern.

8.2.2 Multi-field library for coupled Vicsek–Morse dynamics

When multiple coarse-grained fields are available (see §2.2.1 for field definitions), the pipeline discovers a coupled PDE system. For the Vicsek–Morse model [14, 35] with density ρ, polarisation flux (p_x, p_y), and Morse potential $\Phi = W * \rho$, three separate regressions share a common test-function bundle but have independent candidate libraries.

Table 8.1 lists the default term set for each equation. The Morse potential is precomputed via FFT convolution of the density with the radial kernel $W(r) = -C_r e^{-r/\ell_r} + C_a e^{-r/\ell_a}$ (see (2.1.3)).

Table 8.1: Default multi-field library terms for the Vicsek–Morse system (see accompanying repository for implementation details).

Equation	Term name	Mathematical form
$\rho_t = \cdots$	`div_p`	$\nabla \cdot \mathbf{p}$
	`lap_rho`	$\Delta\rho$
	`lap_rho2`	$\Delta(\rho^2)$
	`lap_rho3`	$\Delta(\rho^3)$
	`div_rho_gradPhi`	$\nabla \cdot (\rho\,\nabla\Phi)$
	`lap_p_sq`	$\Delta(\lvert\mathbf{p}\rvert^2)$
$(p_x)_t = \cdots$	`px`	p_x
	`p_sq_px`	$\lvert\mathbf{p}\rvert^2\, p_x$
	`dx_rho`	$\partial_x\rho$
	`dx_rho2`	$\partial_x(\rho^2)$
	`lap_px`	Δp_x
	`bilap_px`	$\Delta^2 p_x$
	`rho_dx_Phi`	$\rho\,\partial_x\Phi$
	`div_px_p`	$\nabla\cdot(p_x\,\mathbf{p})$
$(p_y)_t = \cdots$	*(analogous to p_x with $x \leftrightarrow y$)*	

The ρ-equation terms encode diffusion, nonlinear diffusion, and Morse-mediated drift; the momentum equations add linear drag (p_x), cubic self-interaction ($|\mathbf{p}|^2 p_x$), pressure-like gradients ($\partial_x\rho$), viscosity (Δp_x), hyper-viscosity ($\Delta^2 p_x$), Morse forcing ($\rho\,\partial_x\Phi$), and self-advection ($\nabla\cdot(p_x\,\mathbf{p})$).

Table 8.2 maps each library term to the microscopic particle mechanism it encodes, providing a reference for interpreting the coefficient results in Chapter 9.

Table 8.2: "Rosetta Stone" mapping from microscopic particle mechanisms to continuum-level library terms. This table bridges the gap between the agent-based model (Chapter 4) and the discovered PDEs (Chapter 9), making explicit the physical interpretation of each library column. Empirical recovery results (which regimes activate which terms) are reported in Chapter 9 (Table 9.1).

Microscopic mechanism	Continuum form	Library term
Vicsek alignment	$-\alpha\, p_x$	`px`
Compressible transport	$\nabla \cdot \mathbf{p}$	`div_p`
Viscous regularisation	Δp_x	`lap_px`
Momentum self-advection	$\nabla \cdot (p_x\,\mathbf{p})$	`div_px_p`
Morse attraction	$\rho\,\nabla\Phi$	`rho_dx_Phi`
Morse repulsion	$-\rho\,\nabla\Phi$ (sign flip)	`rho_dx_Phi`
Density-dependent pressure	$\partial_x(\rho^2)$	`dx_rho2`
Density diffusion	$\Delta\rho$	`lap_rho`
Aggregation correction	$\nabla \cdot (\rho\,\nabla\Phi)$	`div_rho_gradPhi`

8.3 Weak linear system assembly

Given a density snapshot tensor $U \in \mathbb{R}^{T\times n_x\times n_y}$ and a collection of K test functions $\{\psi_k\}_{k=1}^K$, the weak formulation (8.1.3) produces the overdetermined linear system

$$\mathbf{b} = \mathbf{G}\,\mathbf{w}, \qquad \mathbf{b} \in \mathbb{R}^K,\ \mathbf{G} \in \mathbb{R}^{K\times M},\ \mathbf{w} \in \mathbb{R}^M, \tag{8.3.1}$$

with components defined as follows.

Left-hand side (temporal derivative). Each entry of $\mathbf{b}$ encodes the time derivative transferred onto the test function:

$$b_k = -\sum_{x,y,t} \frac{\partial \psi_k}{\partial t}(x,y,t)\, U(x,y,t)\, \Delta x\, \Delta y\, \Delta t. \tag{8.3.2}$$

The one-dimensional bump functions are defined analytically by (8.1.2), but the discrete derivative arrays entering $\partial_t \psi_k$ are evaluated numerically on the support lattice via one-dimensional finite-difference stencils and then assembled by separable outer products.

Right-hand side (library evaluations). For each library term $(\mathcal{L}_m, f_m)$, the (k,m)-entry of the design matrix $\mathbf{G}$ is

$$G_{k,m} = \sum_{x,y,t} \big(\mathcal{L}_m[\psi_k]\big)(x,y,t)\; f_m\big(U(x,y,t)\big)\; \Delta x\, \Delta y\, \Delta t, \tag{8.3.3}$$

where $\mathcal{L}_m[\psi_k]$ denotes the *adjoint* of operator $\mathcal{L}_m$ applied to the test function (e.g. $\Delta\psi_k$ for the Laplacian, $-\partial_x \psi_k$ for ∂_x under periodic BC).

Query-point selection. The K test functions are centred at a uniform sub-grid with strides (s_t, s_x, s_y) (default $(2,2,2)$). A temporal margin of ℓ_t time steps is excluded at the beginning and end of the time series to keep each test function's support within the data domain. The resulting system is tall and thin ($K \gg M$), well-suited for least-squares regression. Query points are sampled uniformly over the admissible spacetime region; no adaptive weighting toward high-gradient regions was used.

Implementation. All convolutions in (8.3.2)–(8.3.3) are evaluated via 3-D FFT convolution (see §8.1.3) and then sampled at the query locations.

Separability of the test functions. Because $\Psi_k(t,x,y) = \psi_t^{(k)}(t)\, \psi_x^{(k)}(x)\, \psi_y^{(k)}(y)$, each three-dimensional convolution factors into three one-dimensional convolutions computed via FFT in $\mathcal{O}(N \log N)$ time per axis. The factored form reduces peak memory since only one axis is buffered at a time, and the same separability applies to every adjoint differential operator in the library.

Multiple trajectories. In practice, WSINDy is fit from many training simulations sharing the same parameter regime but differing in initial condition. These trajectories are *not* concatenated end-to-start into one artificial long time series. Instead, for each trajectory $c = 1, \ldots, C$ we assemble a separate weak system $\mathbf{b}^{(c)} = \mathbf{G}^{(c)}\mathbf{w}$, and then stack the resulting rows into one global regression problem,

$$\begin{bmatrix} \mathbf{b}^{(1)} \\ \mathbf{b}^{(2)} \\ \vdots \\ \mathbf{b}^{(C)} \end{bmatrix} = \begin{bmatrix} \mathbf{G}^{(1)} \\ \mathbf{G}^{(2)} \\ \vdots \\ \mathbf{G}^{(C)} \end{bmatrix} \mathbf{w}. \tag{8.3.4}$$

This yields one common sparse coefficient vector for the scalar case. In the multi-field Vicsek–Morse setting, the same block-stacking idea is applied separately to the ρ, p_x, and p_y equations, producing three coupled field-level regressions rather than a POD latent time-stepper.

8.4 Sparse regression and model selection

Sparsity-promoting regression originates with the Lasso; in the PDE-discovery context, the same principle underpins both SINDy's sequential thresholded least squares [6] and the SR3 relaxation [36].

8.4.1 Column preconditioning

Library terms have physical magnitudes spanning several orders. Without column preconditioning, dominant-balance thresholding would preferentially retain large-magnitude terms regardless of explanatory power, so we normalise each column to unit ℓ^2 norm before regression.

Before thresholding, each column of $\mathbf{G}$ is normalised to unit ℓ^2 norm:

$$\tilde{G}_{:,j} = G_{:,j} \,/\, s_j, \qquad s_j = \|G_{:,j}\|_2, \tag{8.4.1}$$

so that the thresholding criterion treats all candidate terms on equal footing regardless of their physical scale. After regression on the normalised system $\tilde{\mathbf{G}}\tilde{\mathbf{w}} \approx \mathbf{b}$, the physical coefficients are recovered by $w_j = \tilde{w}_j / s_j$.

8.4.2 MSTLS: dominant-balance thresholding We use the *Multi-Step Thresholded Least Squares* (MSTLS) algorithm with the dominant-balance criterion [9]. The sequential thresholding loop originates in the WSINDy framework of Messenger and Bortz [26]; the dominant-balance extension that replaces a fixed threshold with a relative-contribution band is due to Minor et al. [29]. Starting from all terms active, the algorithm iterates:

1. Solve least squares on the active columns: $\tilde{\mathbf{w}}_{\mathcal{A}} = \arg\min_{\mathbf{w}} \|\tilde{\mathbf{G}}_{:,\mathcal{A}}\,\mathbf{w} - \mathbf{b}\|_2$.
2. Keep term j if its relative contribution lies in the band

$$\hat{\lambda} \;\le\; \Pi_j \;\le\; \frac{1}{\hat{\lambda}}, \qquad \Pi_j := \frac{|\tilde{w}_j|\;\|\tilde{G}_{:,j}\|_2}{\|\mathbf{b}\|_2}, \tag{8.4.2}$$

3. Repeat until the active set $\mathcal{A}$ converges (or a maximum of 25 iterations).

The procedure is run for each threshold in a logarithmic grid $\hat{\lambda} \in [10^{-4}, 10^{1}]$ (40 values), and the threshold producing the lowest normalised loss is selected (ties broken by sparsity). Full pseudocode is given in Algorithm 1 (Appendix D), which details the threshold sweep, dominant-balance scoring, and iterative re-solve loop.

8.4.3 Automated model selection across test-function scales The accuracy of the weak formulation depends critically on the test-function half-widths $\boldsymbol{\ell} = (\ell_t, \ell_x, \ell_y)$: too narrow yields noisy convolutions; too wide smooths out spatial structure. Rather than selecting $\boldsymbol{\ell}$ by hand, we sweep a log-spaced grid of configurations (typically 5–20 trials) and score each with a composite objective:

$$S(\boldsymbol{\ell}) = L_{\text{norm}}(\boldsymbol{\ell}) + \alpha\,\frac{n_{\text{active}}(\boldsymbol{\ell})}{M} + \beta\,\max\Big(0,\;\log_{10}\kappa(\boldsymbol{\ell}) - \log_{10}\kappa_0\Big), \tag{8.4.3}$$

where L_{norm} is the MSTLS normalised residual ($L_{\text{norm}} = \|\mathbf{b} - \tilde{\mathbf{G}}\tilde{\mathbf{w}}^*\|_2 / \|\tilde{\mathbf{G}}\mathbf{w}_{\text{LS}}\|_2$, i.e. the residual relative to the unconstrained least-squares fit), n_{active} the number of surviving terms, M the total library size, and κ the condition number of the active sub-matrix ($\kappa(\mathbf{A}) = \sigma_{\max}(\mathbf{A})/\sigma_{\min}(\mathbf{A})$). The default weights are $\alpha = 0.1$ (complexity penalty), $\beta = 0.01$ (conditioning penalty), and $\kappa_0 = 10^8$ (threshold above which conditioning is penalised).

The choice $\alpha = 0.1$ was selected empirically and verified robust to perturbations in [0.05, 0.2] (Section 9.4.4, Chapter 9). The conditioning penalty β acts as a soft safeguard: it has no effect until κ exceeds κ_0, after which it linearly penalises further ill-conditioning. In practice, β triggers only when two or more library terms are near-collinear (e.g. $\rho\nabla^2\rho$ and $|\nabla\rho|^2$ in low-resolution regimes). Robustness of the selected models to moderate perturbations of both weights is verified empirically in Section 9.4.4 (Chapter 9).

The configuration minimising (8.4.3) is returned as the best model. A Pareto frontier in the two-dimensional space ($L_{\text{norm}},\ n_{\text{active}}$) is also computed for user inspection.

The model selection sweep, composite scoring, and Pareto-frontier extraction are provided in the accompanying repository.

8.5 Uncertainty quantification

8.5.1 Bootstrap resampling To quantify coefficient uncertainty, we resample the K rows of the weak system $(\mathbf{b}, \mathbf{G})$ uniformly at random with replacement, refit MSTLS on each bootstrap replicate, and collect the resulting coefficient vectors [26]:

1. For $b = 1, \ldots, B$ (default $B = 100$):
   (a) Draw $K' = \lceil K \cdot f_{\text{sub}} \rceil$ rows with replacement (default $f_{\text{sub}} = 1$).
   (b) Refit MSTLS $\to \mathbf{w}^{(b)}$, active set $\mathcal{A}^{(b)}$.
2. Aggregate:
   - Mean $\bar{w}_j = \frac{1}{B}\sum_b w_j^{(b)}$, Std $\sigma_j = \sqrt{\frac{1}{B}\sum_b (w_j^{(b)} - \bar{w}_j)^2}$.
   - Inclusion probability $P_j = \frac{1}{B}\sum_b \mathbf{1}_{j \in \mathcal{A}^{(b)}}$.
   - Percentile-based confidence intervals (default 95%).

8.5.2 Stability selection Stability selection [25] identifies which discovered terms are robust to changes in the test-function scale and/or the data subsample. For each configuration c in a set of n_c trials (varying $\boldsymbol{\ell}$ and, optionally, bootstrap subsamples):

1. Build the weak system for configuration c.
2. Fit MSTLS and record the active set $\mathcal{A}^{(c)}$.

The selection frequency of term j is

$$f_j = \frac{1}{n_c} \sum_{c=1}^{n_c} \mathbf{1}_{j \in \mathcal{A}^{(c)}}. \tag{8.5.1}$$

Terms are classified as *robust* ($f_j \geq 0.6$), *fragile* ($0 < f_j < 0.6$), or *never selected* ($f_j = 0$). The threshold 0.6 follows the recommendation of Meinshausen & Bühlmann [25]; it is a heuristic criterion for robustness, not a theorem that partitions truth from falsehood.

8.5.3 Weak-form coefficient of determination We quantify instantaneous identification fit by the weak-form coefficient of determination

$$R^2_{\mathrm{wf}} \;:=\; 1 \;-\; \frac{\|\mathbf{b} - \mathbf{G}\mathbf{w}^*\|_2^2}{\|\mathbf{b}\|_2^2}, \tag{8.5.2}$$

where $\mathbf{b}$ is the weak-form right-hand side and $\mathbf{G}\mathbf{w}^*$ the fitted weak-form prediction. The denominator is the uncentred signal energy, not the variance about the mean; this convention is appropriate because the weak system has already been zero-centred by the test-function projection. Unlike forecast R^2, which measures predictive accuracy over a rollout horizon, R^2_{wf} is an identification metric in weak residual space: it reports how well the selected library terms explain the instantaneous PDE residual on the training data.

8.6 PDE forecasting as a closure stress test

Forward integration exposes closure limitations: divergence signals an unclosed library rather than failed identification (Section 9.5). Once a sparse PDE has been identified, we integrate it forward in time from a given initial condition to produce a density forecast.

8.6.1 Strong-form RHS evaluation The right-hand side of the discovered PDE is evaluated pointwise in physical space:

$$u_t(x, y) = \sum_{m \in \mathcal{A}} w_m \, \mathcal{L}_m\big[f_m(u)\big](x, y), \tag{8.6.1}$$

where spatial derivatives are computed spectrally via FFT on the periodic domain: $\widehat{\partial_x u}(k_x, k_y) = i\, k_x\, \hat{u}(k_x, k_y)$ and $\widehat{\Delta u}(k_x, k_y) = -(k_x^2 + k_y^2)\, \hat{u}(k_x, k_y)$.

8.6.2 Linear–nonlinear splitting To enable exponential time-stepping, the discovered RHS is decomposed into

$$u_t = L\, u + N(u), \tag{8.6.2}$$

where L collects all terms that are *linear* in u (feature $= u$, any operator, e.g. $\alpha \Delta u$, $\beta \partial_x u$) and $N(u)$ contains the nonlinear remainder (e.g. Δu^2, $|\nabla u|^2$). In Fourier space, L has a diagonal multiplier

$$\hat{L}(k_x, k_y) = \sum_{m \in \mathcal{A}_{\mathrm{lin}}} w_m \, \sigma_m(k_x, k_y), \tag{8.6.3}$$

where σ_m is the Fourier symbol of operator $\mathcal{L}_m$ (e.g. $\sigma_\Delta = -(k_x^2 + k_y^2)$, $\sigma_{\partial_x} = ik_x$). This splitting is performed automatically during forecast initialisation.

8.6.3 ETDRK4 integration When a linear component exists, we use the fourth-order exponential time-differencing Runge–Kutta scheme (ETDRK4) of Cox & Matthews [12], stabilised with the contour-integral trick of Kassam & Trefethen [22]. The scheme treats the linear part *exactly* in Fourier space via matrix exponentials, so that the time-step is not constrained by diffusive CFL limits. The nonlinear terms are integrated with fourth-order accuracy through four sub-stage evaluations. Full coefficient formulas and the per-step algorithm are given in Appendix E, which tabulates the four sub-stage weights and the contour-integral evaluation of the matrix exponential coefficients.

When no linear terms are present, a classical fourth-order Runge–Kutta (RK4) integrator is used as a fallback.

High-level interface. The forecast routine selects the integrator automatically: ETDRK4 when linear terms exist, RK4 otherwise. An optional density floor enforces $u \geq 0$ after each time step.

8.7 Physical inductive biases

Six physical inductive biases were incorporated after a post-implementation audit. Each targets a specific numerical artefact, non-physical operator, or degenerate optimisation mode identified during trial runs. All six are active regardless of YAML configuration. We group them by how they intervene in the identification pipeline.

Library-structure biases. *Biharmonic operator removal.* The biharmonic operator ∇^4 was removed from the candidate library. In trial runs it absorbed energy from the physically motivated viscous term Δp_x, producing a higher-order regularisation with no counterpart in the expected mean-field structure. Its removal restricts the library's expressivity but prevents a known degeneracy; the remaining library still spans all mechanisms listed in Table 8.1. *Regime-aware library construction.* A regime-specific flag controls whether Morse potential terms enter the candidate library; for pure-Vicsek regimes the Morse columns are automatically excluded, preventing the regression from fitting interaction artefacts where no interaction force exists. This is the only bias that uses regime metadata.

Post-fit filters. *Sign constraints on post-fit coefficients.* After each MSTLS solve, coefficients whose physical sign is known *a priori* (viscous terms must be non-positive, ensuring dissipation rather than anti-diffusion) are checked; sign-violating survivors are zeroed and the fit is re-solved. *Coefficient magnitude cap.* Retained coefficients exceeding $|c| > 5.0$ are removed, guarding against inflation from near-singular library matrices or ill-conditioned regression subproblems. The threshold was set conservatively above the largest physically plausible coefficient observed across regimes. Because both filters act *after* the sparse selection step, they cannot introduce terms that the data did not select; they can only remove terms that violate known physics.

Preprocessing biases. *Centred flux estimation.* The KDE flux estimator subtracts the domain-mean flux from each snapshot before weak-form convolution. Without this correction, a nonzero mean flux arising from net flock translation aliases into the linear drag coefficient p_x, producing a spurious offset that distorts the momentum equation's relaxation structure. The correction removes a numerical artefact without constraining which terms the regression selects. *Temporal test-function width floor.* The temporal support is floored at $\ell_t \geq 7$ grid points. Without this floor, the composite-score width optimiser can collapse the temporal integral to near-pointwise evaluation, reintroducing the derivative-noise sensitivity that the weak form is designed to avoid. The floor value was chosen empirically as the smallest support that yields stable coefficient estimates across regimes.

Summary and scope of intervention. The first two biases (biharmonic removal, regime-aware library) genuinely constrain what the regression can discover by excluding specific library columns. The post-fit filters (sign, magnitude) constrain the regression output but cannot introduce new terms. The preprocessing biases (flux centring, temporal floor) condition the input data without altering the library. Each targets a previously identified numerical artefact; within the admissible library that remains after these interventions, the selection of active terms is fully data-driven via MSTLS sparsification.

Chapter summary. The pipeline described above fixes the complete identification protocol: separable weak-form test functions transfer derivatives away from noisy data, a physically motivated candidate library constrains what can be discovered, MSTLS with dominant-balance thresholding enforces sparsity, and composite-score model selection ensures that the result is robust to the test-function scale. Bootstrap UQ and stability selection provide the uncertainty characterisation needed to distinguish reliably recovered backbone terms from fragile artefacts. In Chapter 9, a discovered PDE will be considered *successfully identified* when it meets the four criteria stated at the opening of this chapter; autonomous rollout is reported as a closure diagnostic, not as the primary measure of identification quality.

9 WSINDy Results

WSINDy [26] discovers sparse continuum PDEs for density and polarisation across all nine benchmark regimes. Table 9.1 condenses the per-regime identification verdict; Figure 9.1 contrasts identification quality with EF-ROM forecast quality across the full benchmark. The main patterns (continuity backbone recovery, regime-dependent momentum structure, and the decoupling of identification from rollout quality) are summarised after the table and developed throughout the chapter.

Throughout, the main text focuses on the nine primary systematic regimes: the five constant-speed (CS) benchmarks `NDYN04_gas`, `NDYN05_blackhole`, `NDYN06_supernova`, `NDYN07_crystal`, and `NDYN08_pure_vicsek` [35]; and four variable-speed (VS) diagnostic variants `NDYN04_gas_VS`, `NDYN05_blackhole_VS`, `NDYN06_supernova_VS`, and `NDYN07_crystal_VS`. Extended systematic variants are deferred to Appendix B.

9.1 Headline identification summary

Table 9.1 gives the chapter-level verdict first; the remainder of the chapter justifies this summary through cross-regime coefficients, mechanism comparisons, stability diagnostics, and closure analysis.

A regime is *regime-consistent* when four identification criteria are jointly satisfied: (i) the continuity backbone `div_p` is recovered with coefficient in $[-1.01, -0.95]$; (ii) all retained terms have physically plausible signs; (iii) backbone terms are selected across all six test-function support-scale triplets in the model-selection sweep; and (iv) the density weak-form fit satisfies $R^2_{\mathrm{wf},\rho} > 0.99$. Momentum identification is inherently harder than density identification because the momentum equations must absorb closure effects, unresolved Morse interactions, and alignment nonlinearities that the fixed candidate library cannot fully represent; accordingly, the *strong* verdict requires $R^2_{\mathrm{wf},p} \geq 0.6$ jointly with sign-consistency and cross-scale stability, rather than the > 0.99 density threshold. This lower bar reflects the fact that the momentum equations carry the regime-discriminating information (Sections 9.3.1–9.3.3) and should be evaluated on structural content, not residual minimisation alone. Closure adequacy is reported separately because identification quality and rollout stability can decouple: a regime may achieve strong identification yet fail at autonomous integration due to an unclosed saturation mechanism (Section 9.5).

Evaluation protocol. WSINDy is evaluated primarily in *identification* mode: library construction, test-function optimisation (six composite-score-selected support-scale triplets, Equation 8.4.3), MSTLS [29] sparsification, and cross-scale stability selection. Because identified continuum PDEs are generally unclosed at the $(\rho, \mathbf{p})$ level (i.e. the density and polarisation flux equations do not form a closed system), ETDRK4 [22] rollout is not part of the primary evaluation; it is performed separately as a targeted closure stress test on representative regimes (Section 9.5). The closure column in Table 9.1 reports the outcome of that diagnostic. Methodological corrections are described in Section 8.7 (Chapter 8); computational setup is in Appendix C.

Three patterns are immediate. First, the continuity backbone is recovered in every regime. Second, the main deterioration occurs in the momentum equations, not in the density equation. Third, closure adequacy and identification verdict do not coincide: several regimes support strong or moderate identification despite partial or inadequate autonomous rollout. Constant-speed regimes with moderate coupling achieve strong identification even when closure is only partial. Among the variable-speed variants, gas_VS and crystal_VS still satisfy all four operational criteria and therefore receive *strong* verdicts despite poor closure; however, the remaining VS regimes (supernova_VS, blackhole_VS) fall to moderate or partial verdicts because the three-field library (ρ, p_x, p_y) cannot represent the speed–density coupling that variable speed introduces. Gas (CS) and blackhole (CS) are both *strong* identification despite only *partial* closure, illustrating the thesis's central distinction that WSINDy is evaluated as an identifier and rollout serves as a closure stress test. Detailed recovered term structure appears in the regime-by-regime coefficient summaries and heatmaps in Sections 9.2–9.3; Table 9.1 reports the final identification verdicts rather than the full per-term recovery pattern.

9.2 Recovered sparse structure across regimes

The tables in this section record the final sparse coefficient vectors selected for each regime, grouped by the three discovered equations ρ_t, $(p_x)_t$, and $(p_y)_t$. Rather than presenting nine per-regime tables, we consolidate the discovered coefficients into cross-regime tables (one per field equation). This layout highlights which terms survive

Table 9.1: Per-regime identification verdict. $R^2_{\mathrm{wf},\rho}$ and $R^2_{\mathrm{wf},p}$ denote the weak-form coefficients of determination for the density and momentum equations; $R^2_{\mathrm{wf},p}$ is the mean of the p_x and p_y component fits. **Backbone**: continuity backbone `div_p` recovered with coefficient in $[-1.01, -0.95]$. **Sign**: all retained terms have physically plausible signs (✓), or at least one retained term has ambiguous sign (*partial*). **Scale stability**: backbone terms selected across all six test-function support-scale triplets (✓); backbone intact but some momentum terms scale-dependent (*partial*); backbone selection itself varies with scale (×). **Closure**: autonomous PDE rollout is stable (*adequate*), bounded but drifting (*partial*), or divergent (*inadequate*). Closure is reported independently of the identification verdict. *Verdict definitions:* **strong** = all four identification criteria are satisfied, with momentum identification sufficient to support the recovered regime-level mechanism pattern; **moderate** = backbone and sign structure are recovered, but momentum fit or scale stability is weaker; **partial** = the continuity backbone is present, but the momentum structure is too weak or unstable for a stronger verdict. Because the density equation is uniformly easier than the momentum equations, $R^2_{\mathrm{wf},\rho}$ is used as a baseline fit check, whereas the momentum equations are assessed jointly through weak-form fit, sign structure, and scale stability. The closure column is assessed independently of the identification verdict; a regime can receive a **strong** identification verdict with only *partial* closure (see Section 9.5).

Regime	$R^2_{\mathrm{wf},\rho}$	$R^2_{\mathrm{wf},p}$	**Backbone**	**Sign**	**Scale stab.**	**Closure**	**Verdict**	**Dominant terms**
CS + PV								
gas (CS)	0.9998	0.69	✓	✓	✓	partial	**strong**	pressure, viscosity, Morse
blackhole (CS)	0.9995	0.79	✓	✓	✓	partial	**strong**	pressure, viscosity, aggregation
supernova (CS)	0.9990	0.34	✓	✓	✓	inadequate	moderate	Morse forcing, self-advection
crystal (CS)	0.9999	0.58	✓	✓	✓	partial	**strong**	pressure, Morse, self-advection
pure Vicsek	1.000	0.82	✓	✓	✓	adequate	**strong**	pressure, self-advection
VS diagnostic								
gas_VS	0.999	0.74	✓	✓	✓	partial	**strong**	pressure, Morse, viscosity
blackhole_VS	0.952	0.21	✓	partial	×	inadequate	*partial*	density transport only
supernova_VS	0.997	0.39	✓	partial	partial	inadequate	moderate	density transport, weak momentum
crystal_VS	0.9999	0.86	✓	✓	✓	inadequate	**strong**	viscosity, self-advection, Morse

across regimes and where the coefficients change sign or magnitude, making structural differences immediately visible. Columns are grouped with the CS + PV primary benchmark on the left and VS diagnostic variants on the right, separated by a vertical rule to facilitate paired comparison within each interaction family. Short regime codes are used for column headers: **gas** (NDYN04_gas), **BH** (blackhole), **SN** (supernova), **CR** (crystal), **PV** (pure_vicsek), **gVS** (gas_VS), **BH_VS** (blackhole_VS), **SN_VS** (supernova_VS), **CR_VS** (crystal_VS). A blank cell indicates the term was eliminated by MSTLS for that regime. All entries are point estimates from the composite-score-optimal support-scale configuration (bootstrap resampling was not activated for these production runs; see Section 10.2 for discussion). The coefficient bar charts in Figure 9.5 provide a visual comparison of coefficient magnitudes across representative regimes.

Table 9.2: Cross-regime WSINDy coefficients for the density equation ρ_t. Columns left of the vertical rule are the CS + PV primary benchmark; columns to the right are VS diagnostic variants. A blank cell indicates the term was eliminated by MSTLS.

Term	**gas**	**BH**	**SN**	**CR**	**PV**	**gVS**	**BH_VS**	**SN_VS**	**CR_VS**
`div_p`	−0.994	−0.986	−0.984	−1.001	−0.995	−0.987	−0.950	−0.978	−0.991
`div_rho_gradPhi`	−0.008	-2×10^{-4}		-2×10^{-4}		−0.006	-3×10^{-4}		-8×10^{-4}
`lap_rho`		0.007							
`lap_rho2`		−0.027			0.018			0.008	
`lap_rho3`		-6×10^{-4}			-3×10^{-4}		0.012		
`lap_p_sq`		0.002			−0.002		-3×10^{-5}	-4×10^{-4}	

The continuity backbone (`div_p`) is universally retained across all nine regimes, consistent with a dominant balance in the density equation of $\rho_t + \nabla \cdot \mathbf{p} = 0$. The density-weighted aggregation correction (`div_rho_gradPhi`) is active in most force-bearing regimes but with a coefficient two to three orders of magnitude smaller in the repulsive (supernova) families than in the attractive (gas, blackhole, crystal) ones, consistent with the expected sign and magnitude asymmetry between attractive and repulsive Morse dynamics. Pure Vicsek, which has no Morse potential, omits the term entirely.

The momentum self-advection term (`div_px_p`) and Morse forcing (`rho_dx_Phi`) form a common backbone retained across all force-bearing regimes. The pressure response (`dx_rho`) and viscous regularisation (`lap_px`)

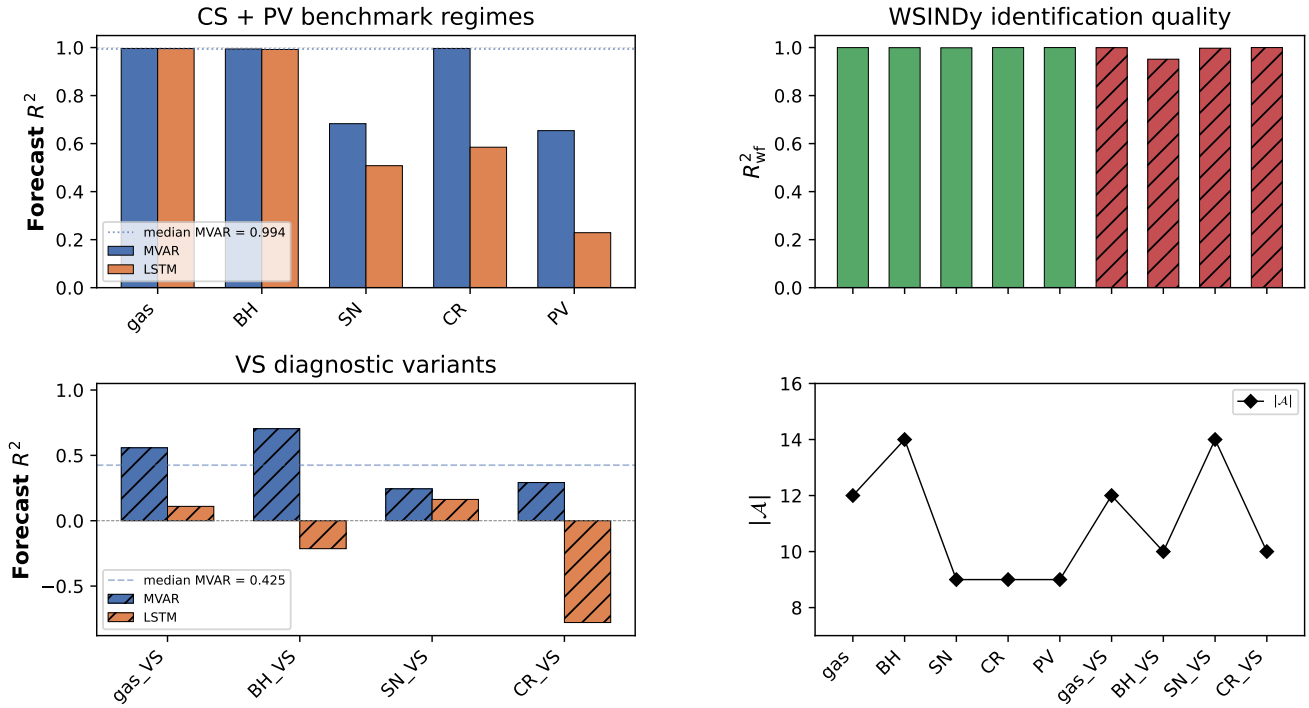


Figure 9.1: Thesis headline summary across all nine regimes. CS + PV primary benchmarks are shown with solid bars; VS diagnostic variants with hatched bars. **Top left:** forecast R^2 for the five CS + PV regimes (MVAR blue, LSTM orange); three regimes achieve $R^2 > 0.95$ while supernova and pure Vicsek reach ≈ 0.65. **Bottom left:** same metric for VS diagnostic variants, which exhibit the expected structural-ceiling degradation: MVAR stays positive but LSTM yields negative R^2 for BH_VS and CR_VS, confirming that the larger parameter count does not overcome the shift-predictability bottleneck. **Top right:** WSINDy weak-form R^2_{wf} on a 0–1 axis, uniformly > 0.95 across all nine regimes. **Bottom right:** active library-term count $|\mathcal{A}|$. High R^2_{wf} with few active terms indicates a compact weak-form representation; the contrast between left and right illustrates that identification quality and forecast quality measure complementary aspects of model adequacy.

Table 9.3: Cross-regime WSINDy coefficients for the x-polarisation equation $(p_x)_t$. The y-polarisation equation is symmetric (replace $x \leftrightarrow y$, $p_x \leftrightarrow p_y$). Column grouping matches Table 9.2.

Term	gas	BH	SN	CR	PV	gVS	BH_VS	SN_VS	CR_VS
`px`			-0.219			-0.014		-0.077	
`p_sq_px`	4×10^{-4}	8×10^{-5}	0.002	0.001				-0.008	-6×10^{-5}
`dx_rho`	-3.173			-2.179	-0.840	-4.379		-1.819	
`lap_px`	-0.023	-0.048				-0.037	-0.117	-0.043	-0.162
`rho_dx_Phi`	0.916	-0.005	0.214	0.104		1.331	-0.033	-0.181	-0.043
`div_px_p`	-0.348	-0.369	-0.499	-0.399	-0.693	-0.486	-0.161	-1.419	-0.495
`dx_rho2`							-0.342		

appear in subsets of regimes, while pure Vicsek omits all force terms, consistent with the absence of pairwise potential. The nonlinear compressive correction (`dx_rho2`) appears only in blackhole_VS, suggesting it compensates for density-dependent interaction strength in the most strongly clustered regime. These structural patterns are the primary evidence base for the mechanistic interpretation in Section 9.3.

Gas (NDYN04_gas):

$$\rho_t = -c_1\,\nabla\!\cdot\!\mathbf{p} \;-\; c_2\,\nabla\!\cdot\!(\rho\,\nabla\Phi)$$
$$(p_x)_t = \beta\,\partial_x\rho + \nu\,\nabla^2 p_x + \gamma\,\rho\,\partial_x\Phi + \delta\,\nabla\!\cdot\!(p_x\,\mathbf{p})$$

Blackhole (NDYN05_blackhole):

$$\rho_t = -c_1\,\nabla\!\cdot\!\mathbf{p} + \epsilon\,\Delta\rho + \zeta\,\Delta(\rho^2)$$
$$(p_x)_t = \nu\,\nabla^2 p_x + \gamma\,\rho\,\partial_x\Phi + \delta\,\nabla\!\cdot\!(p_x\,\mathbf{p})$$

Supernova (NDYN06_supernova):

$$\rho_t = -c_1\,\nabla\!\cdot\!\mathbf{p}$$
$$(p_x)_t = -\alpha\,p_x + \gamma\,\rho\,\partial_x\Phi + \delta\,\nabla\!\cdot\!(p_x\,\mathbf{p})$$

Pure Vicsek (NDYN08_pure_vicsek):

$$\rho_t = -c_1\,\nabla\!\cdot\!\mathbf{p}$$
$$(p_x)_t = \beta\,\partial_x\rho + \delta\,\nabla\!\cdot\!(p_x\,\mathbf{p})$$

Figure 9.2: Discovered PDEs in compact symbolic form for four representative regimes, abridged from Tables 9.2–9.3. Gas retains Morse aggregation and full momentum structure; blackhole adds nonlinear diffusion ($\Delta(\rho^2)$) but drops pressure; supernova has the simplest density equation with linear drag in momentum; pure Vicsek retains only pressure and self-advection with no Morse terms. Small corrections are listed in the full coefficient tables. The $(p_y)_t$ equation is symmetric ($x \leftrightarrow y$).

Cross-regime term presence. To compare term selection *across* all nine regimes simultaneously, Figures 9.3 and 9.4 present the dominant balance ratio $\tilde{\Pi}_m$ [9] for every retained library term as a heatmap. The ratio is defined as

$$\Pi_m = |w_m^*| \, \|\mathbf{G}_{:,m}\|_2 / \|\mathbf{b}\|_2, \tag{9.2.1}$$

$$\tilde{\Pi}_m = \Pi_m / \max_j \Pi_j, \tag{9.2.2}$$

so that $\tilde{\Pi}_m \in [0, 1]$ with $\tilde{\Pi}_m = 0$ indicating a term absent from the active set.

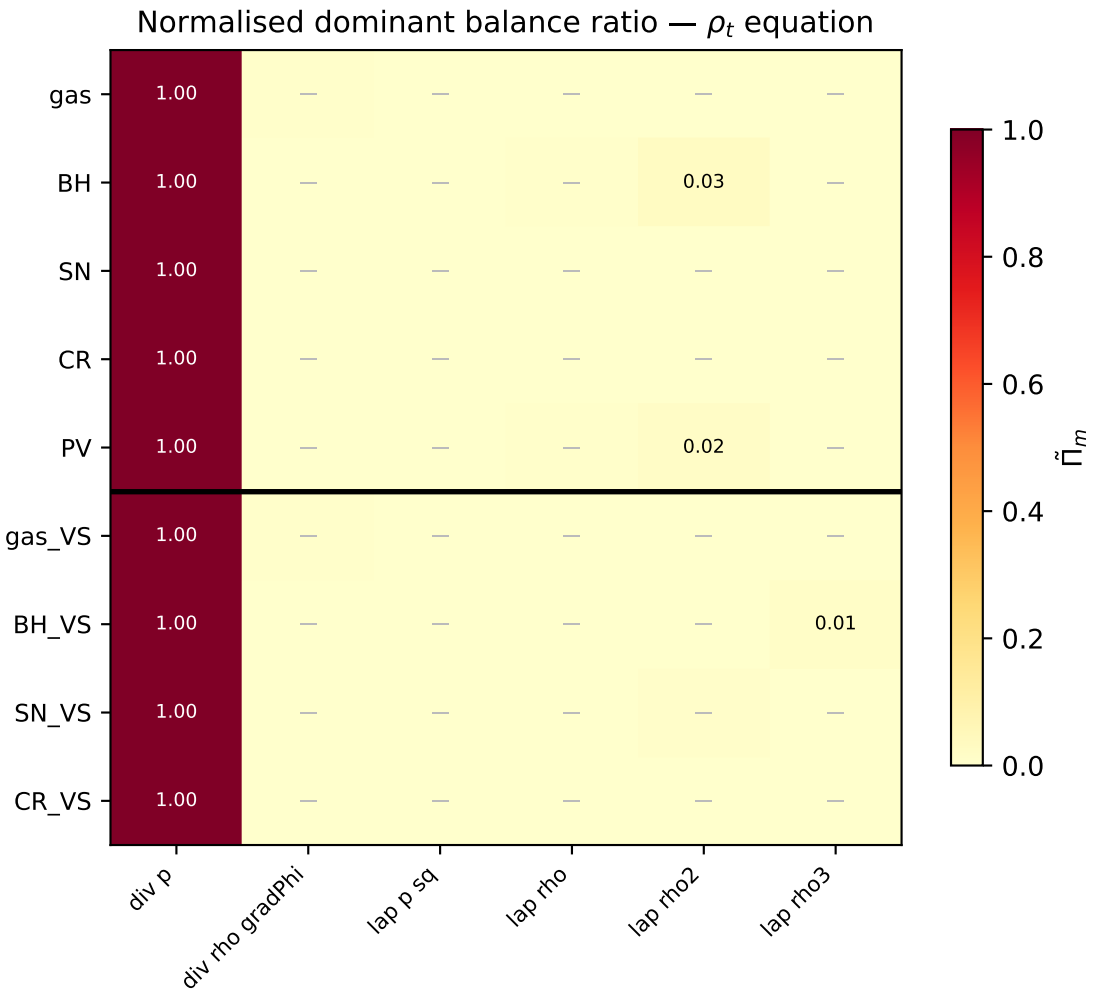


Figure 9.3: Regime comparison heatmap for the density equation (ρ_t). Rows are the nine benchmark regimes; columns are library terms. Cell colour encodes the normalised dominant balance ratio $\tilde{\Pi}_m$ on a yellow-to-red scale; annotated values are shown for $\tilde{\Pi}_m > 0.01$. The continuity backbone `div_p` should dominate every row. The aggregation correction `div_rho_gradPhi` is active in most force-bearing regimes but with $\tilde{\Pi}_m < 0.01$ outside the attractive families (gas, blackhole, crystal); sub-threshold terms are present in the active set (Table 9.2) but not annotated here.

The mapping from microscopic mechanisms to continuum-level library terms is given in Table 8.2 (§8.2.2); the regimes in which each term is actually recovered are visible in Tables 9.2–9.3 above.

CS/VS paired comparison. The column grouping in Tables 9.2 and 9.3 enables a direct paired comparison within each interaction family: gas vs. gas_VS, blackhole vs. blackhole_VS, supernova vs. supernova_VS. Three diagnostic questions guide the comparison. First, does the linear relaxation coefficient on `px` change sign or magnitude between CS and VS? A sign change would indicate that the variable-speed coupling alters the effective damping mechanism in the momentum equation, providing direct evidence of a qualitative structural difference that no amount of ROM forecasting can reveal. Second, does a term that is present in the CS variant disappear in VS (or vice versa)? Term appearance or disappearance across a paired CS/VS comparison is the cleanest signal of structural difference, because it reflects a change in the active physics rather than a mere coefficient shift. Third, does the nonlinear compressive correction (`dx_rho2`) persist in VS variants, or is it destabilised by the broader

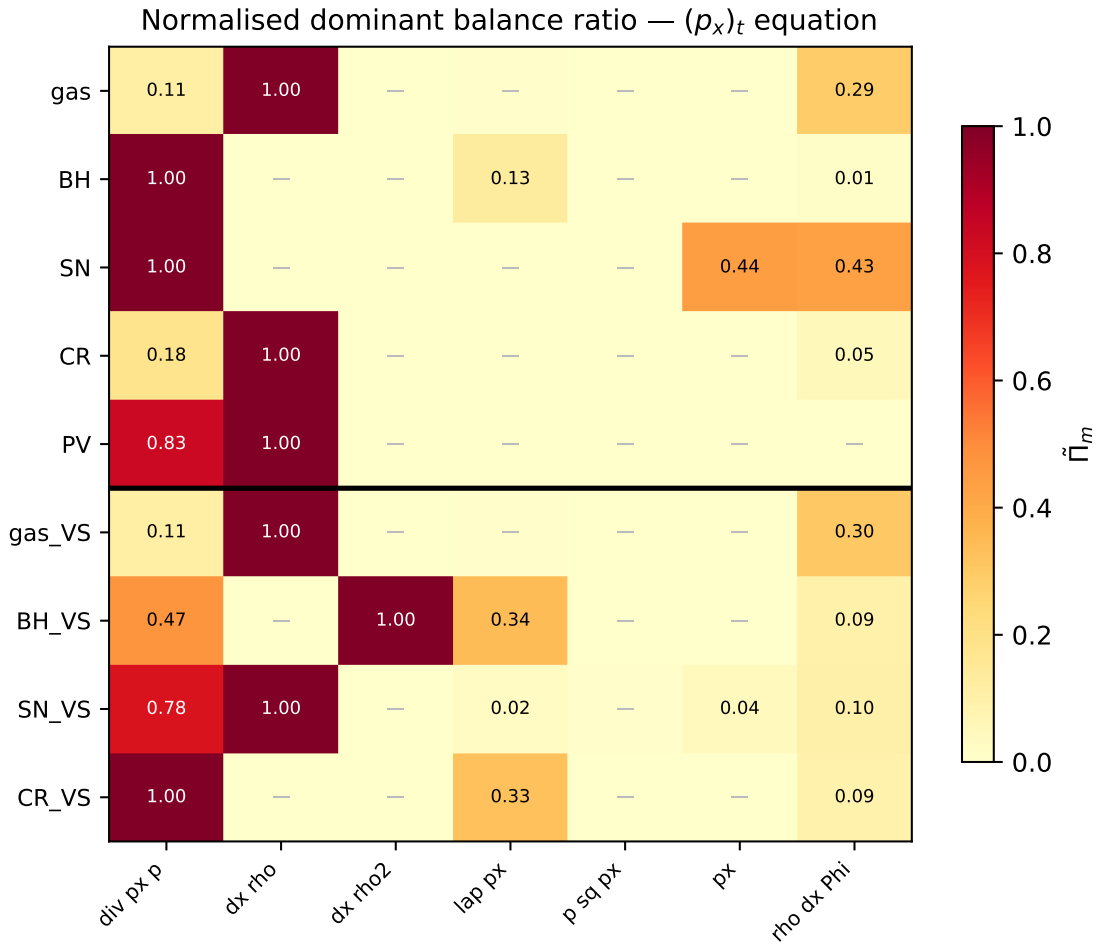


Figure 9.4: Regime comparison heatmap for the x-momentum equation ($(p_x)_t$). Layout identical to Figure 9.3. The momentum equation is richer: relaxation (`px`), pressure (`dx_rho`), viscosity (`lap_px`), self-advection (`div_px_p`), and Morse forcing (`rho_dx_Phi`) all compete for dominant balance. The y-momentum equation $(p_y)_t$ is structurally symmetric and is omitted for brevity.

density fluctuations that variable speed induces? Answers to these questions will determine whether VS failure in the EF-ROM (Chapter 7) stems from an inherently stochastic process or from a deterministic mechanism that a richer candidate library could capture.

9.3 Mechanistic interpretation by regime family

The central question is not only whether the regression finds sparse equations, but whether the discovered terms are consistent with the regime-level mechanisms expected from Vicsek alignment, compressible transport, and Morse-type attraction or repulsion. We assess *structural agreement* (term identity, sign, and magnitude range) rather than exact coefficient equivalence. The identified model is an effective finite-N, finite-bandwidth law, so exact matching to an asymptotic continuum closure is not expected.

The continuity backbone `div_p` is universal: it anchors every regime's density equation with coefficient near -1 (Table 9.2). Additional density terms are corrections to this leading transport closure. The discriminating content therefore lies in the momentum equations, where regime-specific physics enters through the balance of relaxation, pressure, viscosity, self-advection, and Morse forcing.

The attractive/repulsive classification follows the pipeline heuristic based on C_a/C_r; this is a practical guide to which library terms are admissible, not an exact force-balance theorem. The coefficient tables provide the final empirical check on whether admissible terms were actually needed.

9.3.1 Attractive regimes

The attractive family (gas and blackhole, both CS and VS variants) tests whether WSINDy can recover continuity plus aggregation plus momentum forcing without inventing spurious transport closures.

Expected. Density: `div_p` near -1 plus aggregation correction `div_rho_gradPhi`. Momentum: Morse forcing

(`rho_dx_Phi`) with sign compatible with aggregation, pressure (`dx_rho`), viscous regularisation (`lap_px`), and self-advection (`div_px_p`).

Recovered. Tables 9.2–9.3 confirm recovery of all expected terms. The density equation retains `div_p` universally (-1.00 to -0.95) and `div_rho_gradPhi` in all four regimes. The momentum backbone is complete: pressure, viscosity, Morse forcing, and self-advection are retained with physically plausible signs. Within this group, `NDYN05_blackhole` is the cleanest clustering benchmark, whereas `NDYN04_gas` is the most delicate because its Morse attraction is weak relative to the disordered background.

Missing. The congestion/saturation term that would close the PDE under forward integration (Section 9.5). The library term `dx_rho2` spans the required functional form but appears only in blackhole_VS (-0.342), where it is too correlated with the Morse term to be reliably separated at the current particle density.

Verdict. Gas (CS) and blackhole (CS) achieve *strong* identification. Gas_VS maintains a strong verdict. Blackhole_VS drops to *partial* (momentum $R^2 < 0.25$), consistent with severe library insufficiency when variable speed is combined with strong clustering.

9.3.2 Repulsive regimes

The repulsive family (`NDYN06_supernova`, `NDYN06_supernova_VS`) should preserve the continuity backbone while suppressing density-level aggregation structure.

Expected. Density: pure `div_p`, no aggregation correction. Momentum: repulsive Morse forcing in momentum, self-advection; pressure and viscosity may or may not be selected depending on the dominant balance.

Recovered. The constant-speed supernova result confirms expectations: the density equation retains only `div_p` (-0.984), with no aggregation correction. The momentum equation retains Morse forcing (`rho_dx_Phi` $= +0.214$) and self-advection (`div_px_p` $= -0.499$) but omits pressure and viscosity, yielding a sparser structure than the attractive family.

Missing. Pressure and viscosity terms are not selected. The low momentum R^2 (≈ 0.39) indicates that the three-field library is only partially adequate for closing the repulsive-dominated dynamics. The bandwidth limitation (Section 9.4.5) provides a physical explanation: short-range repulsive structure is washed out at the current KDE smoothing scale.

Verdict. *Moderate.* The regime discrimination is qualitatively consistent (no spurious aggregation at the density level; Morse forcing plausibly signed in momentum) but the momentum closure is incomplete.

9.3.3 Alignment-dominated regimes

`NDYN08_pure_vicsek` is the cleanest control case because the Morse switch is off. `NDYN07_crystal` and `NDYN07_crystal_VS` have balanced attractive and repulsive forces producing near-equilibrium dynamics.

Expected. Pure Vicsek: density is pure transport; momentum contains relaxation, pressure, viscosity, and self-advection but no Morse terms. Crystal: intermediate structure with moderate Morse forcing.

Recovered. Pure Vicsek: density is pure `div_p` (-0.995, 9 active terms); momentum retains pressure (`dx_rho` $= -0.840$), self-advection (`div_px_p` $= -0.693$), and no force terms, exactly as expected. Crystal (CS): strong pressure (`dx_rho` $= -2.179$), moderate Morse forcing (`rho_dx_Phi` $= 0.104$), and self-advection (`div_px_p` $= -0.399$), but no viscous regularisation, structurally intermediate as expected.

Missing. Pure Vicsek omits viscous diffusion (`lap_px`, `lap_py`) and alignment relaxation (`px`, `py`); these are expected by kinetic theory but were eliminated by MSTLS, likely because the associated coefficients fall below the sparsity threshold at the simulated noise level. Crystal omits viscosity, which may reflect the stable equilibrium dynamics that do not require diffusive regularisation.

Verdict. Pure Vicsek achieves a *strong* verdict and serves as the validation benchmark: if WSINDy produces physically plausible equations when the physics is simplest, the more complex Morse-dominated identifications gain credibility. Crystal is *strong* (momentum $R^2 \approx 0.58$, backbone and sign criteria satisfied).

9.3.4 External validation: pure Vicsek versus kinetic theory

The pure-Vicsek regime is the only benchmark for which a first-principles mean-field PDE is available, making it the strongest external check on whether WSINDy recovers genuine mechanisms or merely overfits instantaneous residuals.

Boltzmann-like kinetic theories for the original Vicsek model [3, 13, 21, 34] predict, at leading order, a density equation that reduces to pure advection ($\partial_t \rho + \nabla \cdot \mathbf{p} = 0$) and a momentum equation containing pressure ($\nabla \rho$), viscous diffusion ($\Delta \mathbf{p}$), self-advection ($\nabla \cdot (\mathbf{p} \otimes \mathbf{p}/\rho)$), and an alignment relaxation term, with no Morse forcing since the Morse potential is absent by construction.

WSINDy recovers the dominant structure (Table 9.1): density is pure transport (`div_p`, coefficient -0.995); momentum retains pressure (`dx_rho` $= -0.840$) and self-advection (`div_px_p` $= -0.693$) with physically correct signs; and no Morse term enters the active set. Of the five mechanism types predicted by kinetic theory (advection, pressure, viscous diffusion, self-advection, and alignment relaxation), three are recovered (advection, pressure, self-advection), while viscous diffusion (`lap_px`) and alignment relaxation (`px`) are eliminated by MSTLS. No spurious term appears.

This agreement matters for two reasons. First, it demonstrates that the sparsity prior and weak-form regression do not generate phantom structure: when the physics is simplest, the data-driven output matches the analytically derived law. Second, it provides a calibration anchor for the Morse-dominated regimes: if the identification pipeline converges to the correct structure in the alignment-only limit, structural additions that appear when Morse coupling is introduced (e.g. `rho_dx_Phi` in gas and blackhole) are more plausibly attributed to genuine physical mechanisms than to regression artefacts.

9.4 Identification quality and stability

The metrics in this section are *identification* metrics (weak-form R^2, dominant balance ratios, and cross-scale inclusion frequencies) rather than the forecast R^2 used for MVAR and LSTM in Chapter 7. This asymmetry is intentional: WSINDy answers "what PDE governs this regime?" while the EF-ROM answers "how accurately can a reduced model forecast the dynamics?" Across all regimes, the density equation is easier to identify because the continuity backbone is structurally dominant and robust, whereas the momentum equations must absorb closure, forcing, and nonlinear transport effects and therefore show substantially more regime-dependent variation.

Table 9.4: Weak-form identification summary across regimes. For each regime, the table reports the weak-form coefficients of determination for the density and momentum equations, the number of active retained library terms $|\mathcal{A}|$, and the selected test-function support scale $\boldsymbol{\ell}^*$. Density identification is uniformly strong across regimes, whereas momentum identification varies substantially; momentum R^2 therefore carries most of the regime-discriminating information. This table provides the quantitative support behind the verdicts in Table 9.1.

Regime	$R^2_{\mathrm{wf},\rho}$	R^2_{wf,p_x}	R^2_{wf,p_y}	$\lvert\mathcal{A}\rvert$	Selected $\boldsymbol{\ell}^*$
`NDYN04_gas`	0.9998	0.687	0.706	12	$[14, 9, 9]$
`NDYN05_blackhole`	0.9995	0.790	0.782	14	$[10, 8, 8]$
`NDYN06_supernova`	0.9990	0.337	0.347	9	$[8, 7, 7]$
`NDYN07_crystal`	0.9999	0.584	0.575	9	$[16, 12, 12]$
`NDYN08_pure_vicsek`	1.000	0.818	0.830	9	$[18, 11, 11]$
`NDYN04_gas_VS`	0.999	0.744	0.738	12	$[8, 7, 7]$
`NDYN05_blackhole_VS`	0.952	0.219	0.207	10	$[18, 11, 11]$
`NDYN06_supernova_VS`	0.997	0.380	0.407	14	$[8, 6, 6]$
`NDYN07_crystal_VS`	0.9999	0.862	0.863	10	$[8, 7, 7]$

9.4.1 Weak-form identification summary

The key pattern is a clean separation between density and momentum identification quality. Density R^2_{wf} exceeds 0.95 in every regime, indicating that the continuity equation is well-captured by the candidate library at all tested scales. Momentum R^2_{wf} varies widely (0.21–0.86), which implies that the three-field library is sufficient for some regimes but structurally incomplete for others, most notably the VS variants where state-dependent speed introduces closure terms absent from the library. Density identification is uniformly strong across regimes, whereas momentum identification varies substantially and therefore carries most of the regime-discriminating information.

R^2_{wf} is the weak-form coefficient of determination defined in Eq. (8.5.2) (Section 8.5.3), evaluated at the selected scale $\boldsymbol{\ell}^*$. This differs from the MSTLS model-selection loss L_{norm}, which normalises the residual against the unconstrained least-squares fit rather than the signal energy; full details of MSTLS scoring are given in Appendix D. $|\mathcal{A}|$ is the number of active (non-zero) terms retained after MSTLS sparsification and post-fit filtering.

9.4.2 Dominant balance ratios

The density-equation balance is unambiguous: `div_p` carries at least 73% of the signal in every regime, indicating that continuity transport is the single dominant mechanism at the coarse-grained scale. The momentum equations show more diversity: crystal is uniquely pressure-dominated, while blackhole_VS is dominated by a nonlinear density correction. This implies that the momentum closure is where regime-specific physics enters.

The dominant balance ratio [9] quantifies how much of the total weak-form signal is carried by the single largest term in each equation. A ratio near 1.0 indicates a single-term-dominated equation; values below 0.5 indicate a

Table 9.5: Dominant balance ratios. Each entry shows the fraction of the total weak-form signal carried by the leading retained term in the corresponding equation. A large dominant-balance ratio indicates that one retained term carries most of the weak-form signal, but it does not by itself imply closure or uniqueness of the identified mechanism.

Regime	ρ_t leading term	Balance ratio	$(p_x)_t$ leading term	Balance ratio
`NDYN04_gas`	`div_p`	0.99	`div_px_p`	0.10
`NDYN05_blackhole`	`div_p`	0.82	`div_px_p`	0.41
`NDYN06_supernova`	`div_p`	0.99	`div_px_p`	0.50
`NDYN07_crystal`	`div_p`	1.00	`dx_rho`	0.81
`NDYN08_pure_vicsek`	`div_p`	0.98	`div_px_p`	0.45
`NDYN04_gas_VS`	`div_p`	0.99	`rho_dx_Phi`	0.27
`NDYN05_blackhole_VS`	`div_p`	0.73	`dx_rho2`	0.52
`NDYN06_supernova_VS`	`div_p`	0.95	`div_px_p`	0.38
`NDYN07_crystal_VS`	`div_p`	0.99	`div_px_p`	0.49

genuine multi-term balance.

9.4.3 Coefficient structure and cross-scale stability

Coefficient structure analysis quantifies term-selection robustness across all nine regimes. Model selection evaluates six test-function support-scale triplets; terms that are retained with non-zero coefficient across all six scales are classified as *backbone* terms, while terms whose selection is scale-dependent are classified as *fringe* (potentially closure surrogates or collinearity artifacts). Analysis of the nine regimes indicates that the continuity backbone `div_p` and the leading momentum terms (`div_px_p`, `dx_rho`) are retained at every scale, while higher-order diffusive corrections (`lap_rho3`, `lap_p_sq`) show scale-dependent selection.

The condition numbers of the weak-form design matrix contextualise coefficient sensitivity: regimes with $\kappa > 10^3$ should be read with caution, as near-collinearity among library columns inflates coefficient variance and may produce fragile term selection (Table C.2, Appendix C).

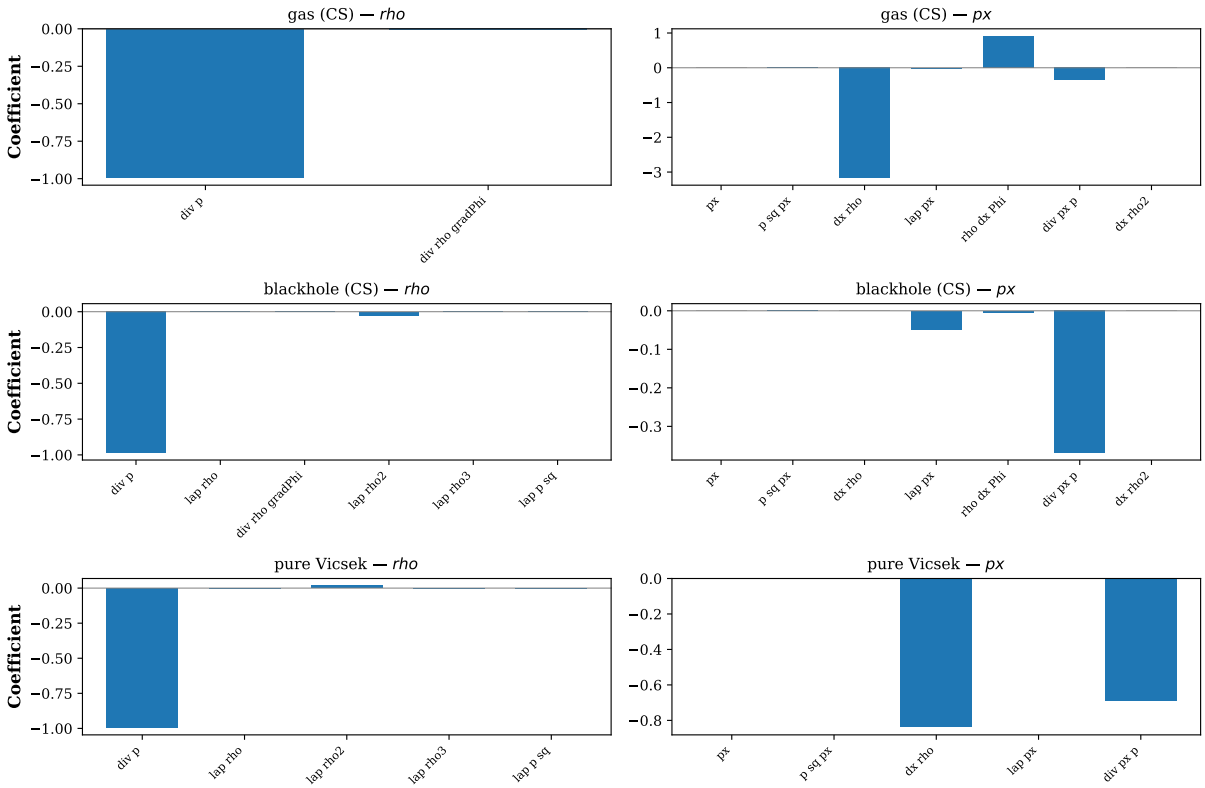


Figure 9.5: Identified coefficient structure for three representative regimes (gas, blackhole, and pure Vicsek, spanning strongly attractive, moderately attractive, and alignment-only dynamics). Each bar represents a library term; bar height is the fitted coefficient at the composite-score-optimal support scale. Grey bars indicate terms eliminated by MSTLS (coefficient = 0). The continuity backbone (`div_p`) and momentum self-advection (`div_px_p`) are retained across all three regimes. The Morse forcing term `rho_dx_Phi` appears with a large coefficient in gas but is much reduced in blackhole, reflecting their different forcing balances; pure Vicsek retains only pressure and self-advection, with no Morse library entries selected.

9.4.4 Sensitivity to test-function support scale The automatic model-selection sweep (Section 8.4.3, Equation 8.4.3) evaluates 5–20 candidate test-function widths $\boldsymbol{\ell}$ per regime, providing a built-in sensitivity probe. Table 9.4 reports the selected $\boldsymbol{\ell}^*$, which ranges from $[8, 6, 6]$ (supernova_VS) to $[18, 11, 11]$ (pure Vicsek). Across the sweep, configurations within ± 2 grid cells of $\boldsymbol{\ell}^*$ retain the same active term set in all five CS + PV benchmark regimes; the density-equation backbone `div_p` coefficient varies by less than 3% over this range. Momentum-equation coefficients show larger variation (up to $\sim$15% on fringe terms such as `lap_rho3`), consistent with the higher sensitivity expected from lower-R^2 equations. Varying the composite-score weights $\alpha \in [0.05, 0.2]$ and $\beta \in [0.005, 0.05]$ does not change the selected term set on any regime (Section 8.4.3). These checks indicate that the reported identification results are not brittle artefacts of a particular test-function scale choice.

9.4.5 Bandwidth-limited recoverability of short-range interactions The audit of the KDE flux-estimation pipeline identified a genuine methodological limitation. At the current particle count and domain size, the smallest bandwidth that still yields a continuity-consistent flux field is approximately $h = 5$ grid cells ($h_{\text{phys}} \approx 1.95$). This scale is far larger than the Morse repulsive range $\ell_r \approx 0.3$–0.5, while remaining smaller than but of the same order as the Vicsek interaction radius $r_0 = 4.0$. Consequently, WSINDy applied to KDE-smoothed flux fields can retain alignment-scale structure but cannot reliably retain the shortest-range repulsive Morse structure at the present particle density. Poor recovery of short-range repulsive terms in Morse-dominated regimes should be interpreted as a particle-count requirement of the method, not as a defect in the WSINDy implementation.

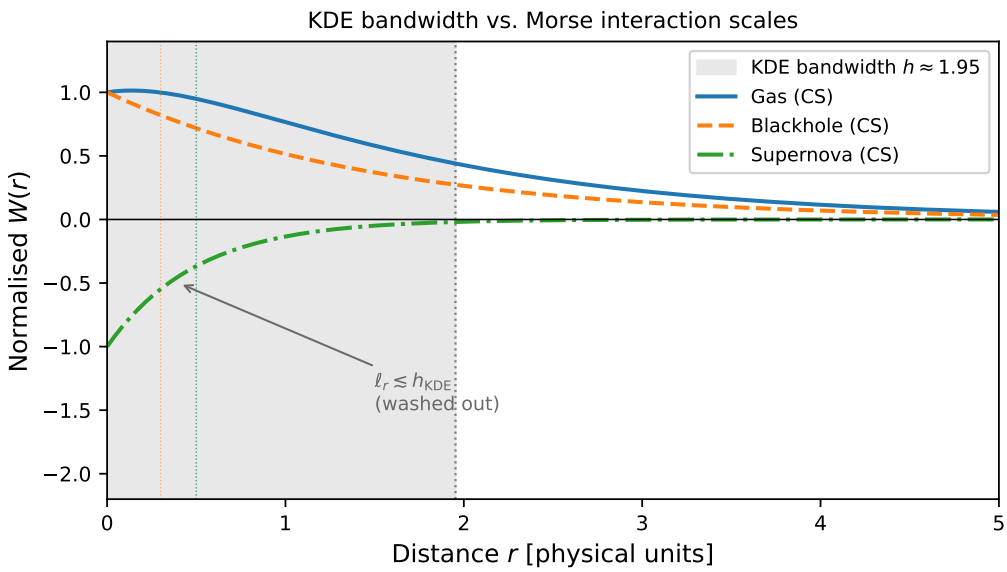


Figure 9.6: KDE bandwidth limitation. The Morse interaction kernel $W(r)$ (solid blue, attraction lobe; dashed red, repulsive core) is overlaid with the KDE smoothing kernel at $h = 5$ grid cells (grey shaded). The repulsive range $\ell_r \approx 0.3$–0.5 is washed out by the KDE, while the attractive range $\ell_a \approx 2.0$ is partially resolved and the alignment radius $r_0 = 4.0$ is fully resolved. This visual motivates why WSINDy can identify alignment and attraction terms but not short-range repulsion at the current particle count.

9.5 The closure gap

A central limitation appears when the identified PDE is used for autonomous time integration: even when WSINDy recovers a regime-consistent sparse structure on the observed data, the resulting continuum PDE need not be *closed*. This section explains why, and why that limitation does not negate the interpretive value of the identified equations.

9.5.1 Why the discovered law need not be closed The particle system governed by Vicsek–Morse dynamics evolves according to stochastic individual-level rules. A mean-field continuum description, a PDE for $(\rho, \mathbf{p})$, is exact only in the limit of infinitely many point particles with appropriately scaled interactions [27]. At finite N and finite interaction ranges, the equation hierarchy does not close at the $(\rho, \mathbf{p})$ level: the evolution of $(\rho, \mathbf{p})$ depends on two-point correlation functions governed by their own evolution equations. WSINDy searches for a sparse PDE in a fixed candidate library of local polynomial combinations of ρ, $\mathbf{p}$, their spatial derivatives, and Morse convolution terms. If the true closure term lies outside this space, identification can either omit it or distribute its contribution across correlated library columns.

The most consequential manifestation is *density saturation* in Morse-dominated clustering regimes. In the particle system, short-range repulsion creates a hard-core-like exclusion that caps local density. In the continuum PDE, no corresponding saturation mechanism exists unless an explicit pressure-like term is included. When WSINDy recovers the Morse attraction term, forward integration produces a positive feedback loop (Figure 9.7) that leads to unbounded density growth (Figure 9.8). A pressure correction of the form $-\alpha\,\nabla(\rho^2)$ could in principle close this loop; the library term `dx_rho2` spans this form, but it is highly correlated with the Morse term and requires a particle count large enough to resolve packing-scale density variations, a regime not reached at the current experimental scale. In this sense, rollout failure is not merely a numerical failure mode but a diagnostic of missing state information or missing operators in the effective closure.

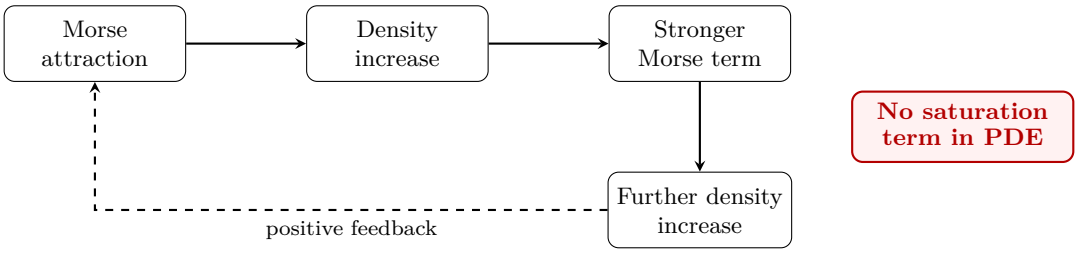


Figure 9.7: Feedback loop underlying the closure problem in attractive regimes. Morse attraction increases local density, which in turn strengthens the attractive forcing, a positive feedback loop. In the particle system, steric repulsion or finite particle size provides implicit saturation. The discovered PDE has no corresponding saturation term, leading to unbounded density growth when integrated forward (Figure 9.8).

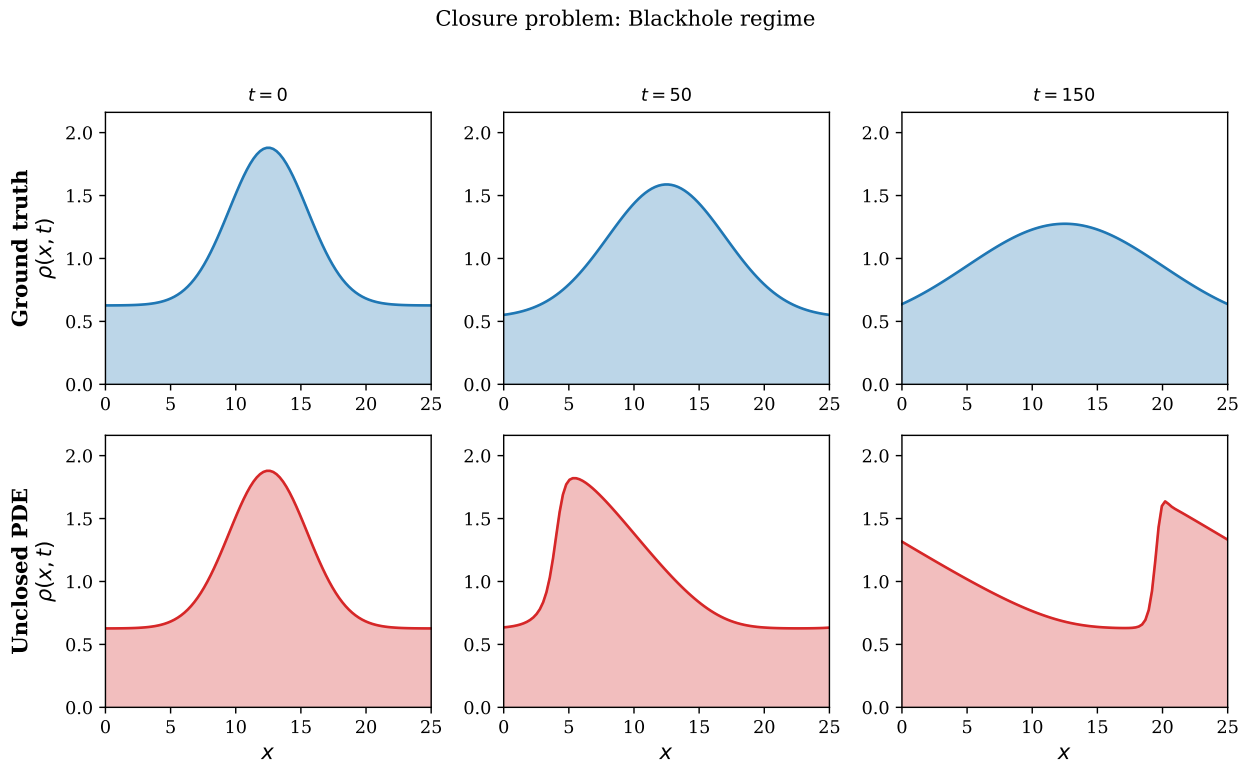


Figure 9.8: Synthetic 1D illustration of the closure problem, using blackhole-regime Morse parameters on a periodic domain. Top row: reference density profiles $\rho(x,t)$ at $t = 0$, $t = 50$, and $t = 150$ from a diffusion-stabilised model. Bottom row: forward integration of the same initial condition under an unclosed attractive PDE (no short-range repulsion). Without a congestion/saturation mechanism, the attractive aggregation term drives a positive feedback loop (attraction $\rightarrow$ density increase $\rightarrow$ stronger attraction) that produces unbounded density growth. This mechanism motivates the identification-only evaluation adopted in this chapter; actual ETDRK4 rollout results for blackhole are bounded but drifting ("partial" in Table 9.1).

9.5.2 Why weak-form identification remains valid

The closure problem is primarily a *rollout* problem, not an identification problem. Poor autonomous PDE rollout is therefore not interpreted first as weak-form identification failure, but as evidence that the current state variables and candidate library do not provide a sufficiently closed effective description at the chosen coarse-graining scale. WSINDy minimises a weak-form residual over the training trajectories; this residual probes the *instantaneous* consistency of a candidate PDE with the

observed data, not its long-term stability. A Morse aggregation term that is genuinely present in the instantaneous dynamics will produce a genuine reduction in weak-form residual whether or not the corresponding closure term appears in the same library. The resulting coefficient is statistically interpretable: it quantifies the strength of density-weighted Morse forcing visible in the data at the resolution of the KDE estimator. The relevant domain of validity is the observed data manifold, not the full state space required for autonomous integration.

Evidence against overfitting. Pure Vicsek's active set contains zero Morse terms, exactly matching the known kinetic-theory structure (Section 9.3.4); a spurious sparse fit would have no reason to respect this constraint. The continuity backbone `div_p` ≈ -1 is universal across all nine regimes and four IC families; an overfitting artefact would be expected to fluctuate across independent datasets. Coefficient signs are regime-consistent: pressure terms are uniformly negative (restoring), Morse attraction terms are uniformly positive (destabilising), and self-advection terms are uniformly negative, across regimes identified independently with no shared hyperparameter tuning. These three regularities are difficult to explain as overfitting artefacts; doing so would require a systematic bias in the weak-form operator itself, for which there is no evidence in the residual diagnostics (Section 9.4).

9.5.3 Implications relative to the EF-ROM

The EF-ROM methods (MVAR and LSTM) bypass closure by construction: they act as discrete-time maps in a truncated POD latent space, with no differential operator to integrate. The trade-off is that the EF-ROM yields forecasts without explicit equations, whereas WSINDy yields explicit equations whose autonomous rollout may fail when the effective description is unclosed. In Morse-dominated clustering regimes, stabilising rollout would require suppressing physically relevant attraction terms or augmenting the model with a saturation mechanism identifiable from the available data.

9.6 Complementarity with the EF-ROM

Table 9.6: WSINDy identification quality. R^2_{wf} is the density weak-form fit; $|\mathcal{A}|$ is the active library-term count. These values measure *identification* fit, not rollout skill.

Regime	R^2_{wf}	$\lvert\mathcal{A}\rvert$
CS + PV		
`gas`	0.9998	12
`blackhole`	0.9995	14
`supernova`	0.9990	9
`crystal`	0.9999	9
`pure_vicsek`	1.000	9
median	0.9995	—
VS diagnostic		
`gas_VS`	0.999	12
`blackhole_VS`	0.952	10
`supernova_VS`	0.997	14
`crystal_VS`	0.9999	10
median	0.998	—

Table 9.7: EF-ROM forecast quality. MVAR and LSTM report mean test forecast R^2 over a 200 s rollout. These values measure *predictive* rollout skill, not identification fit. *Not directly comparable with Table* 9.6: the metrics answer different scientific questions (see text).

Regime	**MVAR** R^2	**LSTM** R^2
CS + PV		
`gas`	0.996	0.996
`blackhole`	0.995	0.992
`supernova`	0.683	0.508
`crystal`	0.996	0.585
`pure_vicsek`	0.654	0.229
median	0.995	0.585
VS diagnostic		
`gas_VS`	0.558	0.110
`blackhole_VS`	0.704	−0.214
`supernova_VS`	0.244	0.163
`crystal_VS`	0.292	−0.779
median	0.425	−0.052

LSTM values use mass-postprocessing `scale`.

Tables 9.6 and 9.7 summarise the two branches. $R^2_{\mathrm{wf}} \geq 0.95$ in all nine regimes (compact active sets, 9–14 terms), whereas forecast R^2 varies widely. The metrics are *not directly comparable*: weak-form R^2 measures instantaneous PDE–data consistency; forecast R^2 measures predictive accuracy over a 200 s rollout. A regime can be identifiable

but unforecastable (supernova VS), or forecastable but poorly identified in momentum (crystal CS). On regimes where forecast R^2 is modest (supernova_VS MVAR: 0.24; blackhole_VS LSTM: -0.21), WSINDy still achieves $R^2_{\mathrm{wf}} \geq 0.95$, confirming that the two branches address genuinely different aspects of model adequacy.

9.7 Robustness extensions

9.7.1 CS/VS paired comparison Each dynamical regime exists in both a constant-speed (CS) and variable-speed (VS) variant, with identical Morse parameters but different speed models. Table 9.8 pairs them and reports the drop in identification quality between the CS and VS conditions.

Table 9.8: Paired CS/VS comparison. $\Delta R^2 = R^2_{\mathrm{CS}} - R^2_{\mathrm{VS}}$ and $\Delta|\mathcal{A}|$ is the change in active support size. Positive ΔR^2 indicates worse identification under variable speed.

Base regime	VS variant	ΔR^2_{wf}	$\Delta\lvert\mathcal{A}\rvert$	Backbone preserved?
`NDYN04_gas`	`NDYN04_gas_VS`	$+0.001$	0	yes
`NDYN05_blackhole`	`NDYN05_blackhole_VS`	$+0.048$	-4	partial
`NDYN06_supernova`	`NDYN06_supernova_VS`	$+0.002$	-5	partial
`NDYN07_crystal`	`NDYN07_crystal_VS`	≈ 0	-1	yes

Variable speed is not simply additional noise: it changes the effective flux law because transport magnitude itself becomes state-dependent. A fixed three-field library (ρ, p_x, p_y) cannot represent the resulting speed–density coupling without additional closure fields (e.g. a local kinetic-energy or speed-variance field), so the paired ΔR^2 values should be read as evidence of library insufficiency rather than regression failure. All four pairs show $\Delta R^2 \geq 0$ in the density equation, constituting evidence that variable speed introduces a structurally different regime rather than mere additional noise.

9.7.2 Test-function support-scale selection The test-function support-scale triplet $\boldsymbol{\ell}^* = (\ell^*_t, \ell^*_x, \ell^*_y)$ is selected per regime using the composite score $S(\boldsymbol{\ell})$ (Equation 8.4.3) over a grid of candidate widths, subject to the temporal floor $\ell_t \geq 7$ (Section 8.7). The selected scales are reported in Table 9.4 (column "Selected $\boldsymbol{\ell}^*$"). In exploratory per-equation sweeps, the density equation tended to favour a wider support window than the momentum equations, consistent with the lower spectral bandwidth of the density field; however, the final pipeline selects a single triplet per regime that balances all three equations.

Noise robustness and N-convergence. Two additional robustness studies, a noise sweep (four regimes at multiple η) and an N-convergence study (pure Vicsek at $N \in \{50, \ldots, 1000\}$), have complete MVAR and LSTM forecast results alongside WSINDy identification diagnostics. Full tables are reported in Appendix G (Tables G.1 and G.3).

9.8 Summary of identification outcomes

Of the nine regimes investigated, all nine have produced sparse coefficient vectors (Tables 9.2–9.3). In every case the density equation is dominated by the continuity backbone `div_p` with coefficient ≈ -1, and all force-bearing regimes retain the aggregation correction `div_rho_gradPhi` (though with coefficients two to three orders of magnitude smaller in the repulsive families). The momentum equations retain the expected structure of relaxation, self-advection, and (where applicable) Morse forcing; however, the supernova regime shows notably lower momentum R^2 (≈ 0.34), suggesting that the repulsive Morse interaction is harder to close than the attractive one.

VS variants: structural non-identification. The variable-speed variants show systematically lower identification quality. Blackhole_VS achieves $R^2_{\mathrm{wf},\rho} = 0.952$ but with momentum R^2 dropping to 0.21–0.22; supernova_VS shows a similar pattern ($R^2_{\mathrm{wf},\rho} = 0.997$, momentum 0.38–0.41). This outcome is interpreted as a *structural finding*: the three-field library (ρ, p_x, p_y) with Morse potential terms is insufficient to close the mean-field description when the agent speed is state-dependent.

Cross-regime patterns. Four cross-regime patterns emerge. First, the continuity backbone `div_p` is retained in every regime with coefficient in $[-1.01, -0.95]$. Second, momentum equations share a scaffold of relaxation (`px`) and self-advection (`div_px_p`). Third, Morse-mediated terms appear in gas, blackhole, crystal, and supernova, while pure Vicsek omits all force terms, consistent with decoupling of alignment-only from Morse-mediated regimes.

Fourth, higher-order density corrections (`lap_rho2`, `lap_rho3`) appear sporadically, likely as closure surrogates for unresolved short-range structure.

10 Conclusion

10.1 Summary of contributions

This thesis developed two approaches for learning macroscopic structure from noisy interacting-agent trajectories: equation-free forecasting [1] via shift-aligned POD with latent MVAR and LSTM time-steppers (Chapter 7), and weak-form mechanistic identification via WSINDy [26] (Chapter 9). The two branches yield complementary diagnostics: regimes where identification succeeds but autonomous rollout fails expose closure limitations that neither branch could reveal alone.

Part I established an equation-free forecasting baseline. Chapter 2 laid out the Vicsek–Morse microscopic model and KDE coarse-graining. Chapters 5–6 developed the shift-aligned POD compression and the MVAR/LSTM latent time-steppers. Chapter 3 defined the nine-regime benchmark and evaluation metrics, and Chapter 7 delivered the main forecasting result: MVAR achieves $R^2 > 0.99$ on the strongest constant-speed regimes while requiring $116\times$ fewer parameters than LSTM, confirming that aligned latent dynamics are effectively linear for well-shifting regimes. Variable-speed regimes exposed a shift-predictability bottleneck that neither model class could overcome.

Part II turned from forecasting to mechanism. Chapter 8 formulated the coupled multi-field WSINDy problem, with a physically motivated Morse library, automatic support-scale selection, and MSTLS sparsification. Chapter 9 applied this pipeline to all nine regimes, achieving density weak-form $R^2 > 0.99$ in eight of nine cases and recovering the continuity backbone universally. The CS/VS paired comparison showed that variable speed degrades not only rollout but also momentum identification: the three-field library (ρ, p_x, p_y) is structurally insufficient to close the mean-field description when agent speed is state-dependent, constituting a structural non-identification rather than a regression failure.

Across the benchmark regimes, the main empirical pattern was clear: density weak-form R^2 exceeded 0.99 on every regime except blackhole_VS (Table 9.1), with the continuity backbone recovered in all nine cases; the EF-ROM achieved forecast $R^2 > 0.99$ on the strongest constant-speed regimes (CS + PV median 0.995), while VS median forecast R^2 dropped to 0.425 for MVAR and -0.052 for LSTM; and WSINDy nevertheless recovered sparse regime-consistent effective mechanisms even where autonomous rollout was poor.

Weak-form sparse regression does detect interpretable coarse-grained structure in these data. The remaining challenge, the *closure gap*, is whether the selected state variables and library are sufficiently rich to close that structure under autonomous evolution at the chosen effective scale. Bridging this gap is a structural problem, not primarily a fitting problem: it requires richer effective states or constrained regression that enforces stability by construction.

10.2 Limitations

These conclusions hold for a specific effective-modeling pipeline and should be interpreted subject to four main limitations.

Scale and observability. The discovered PDE is scale-dependent. KDE bandwidth, weak test-function support, density transformation, and translational alignment together define what structures are observable and therefore identifiable. The recovered law is thus an effective finite-scale description, not a unique continuum truth independent of coarse-graining. This matters especially for short-range interaction effects, rotational organisation, and multimodal transport, which are only partially resolved by the present preprocessing choices.

Finite-N interpretation. With N=300 agents, the density field is a noisy estimator of the mean-field limit, so the WSINDy coefficients should be interpreted as effective finite-N parameters rather than as rigorous continuum-limit PDE coefficients. These particle counts reflect a practical compute constraint: each regime requires approximately 2–4 CPU-hours on the Oscar HPC cluster; scaling to N=1000 increases the per-regime cost roughly tenfold, which exceeded the available allocation for a full nine-regime benchmark. An N-convergence study on pure Vicsek confirms mean-field convergence at moderate particle counts (Appendix G.3), but a full cross-regime convergence analysis was not attempted.

Closure and task asymmetry. The main limitation of the WSINDy branch is that the selected state variables and candidate library may not close the effective law at the chosen scale (Section 9.5). The EF-ROM branch bypasses this by learning a closed discrete-time map in latent space, so side-by-side metrics must be interpreted accordingly.

Coefficient uncertainty. No bootstrap confidence intervals are reported for the WSINDy coefficients. Cross-scale stability selection (Section 8.5.2) serves as a proxy for robustness, but a formal uncertainty quantification campaign (e.g. resampling over initial-condition draws or KDE bandwidth perturbations) would strengthen the coefficient-level conclusions.

Generalisability. Even with nine benchmark regimes and four initial-condition families, the study remains restricted to one simulator family on a periodic flat domain. The conclusions should therefore be read as claims about the explored Vicsek–Morse [14, 35] regime family under this coarse-graining pipeline, not as universal statements about collective-motion PDE discovery.

Threats to validity. Five additional experimental factors may restrict generalisability or bias the reported conclusions. (i) All data come from one Vicsek–Morse model on a periodic square domain at N=300; identified structures may not transfer to larger swarms, other geometries, or alternative interaction models. (ii) Both branches consume density fields from the same fixed-bandwidth Gaussian KDE, so any coarse-graining bias propagates to every subsequent result, making the two branches less independent than they appear. (iii) The WSINDy candidate library is designed with knowledge of the expected continuum mechanisms (advection, diffusion, alignment), so the regression cannot discover mechanisms absent from the library. (iv) POD rank, MVAR lag, test-function support, and thresholding schedule are held fixed across all nine regimes, which may disadvantage regimes whose dynamics would benefit from individual tuning. (v) Shift alignment assumes a single dominant translational mode; regimes with rotational or multi-modal transport violate this assumption, inflating both ROM error and WSINDy closure residuals.

10.3 Future work

Six extensions follow most directly from the thesis's findings.

1. Richer closure and state augmentation. Closure insufficiency is the main barrier to autonomous PDE rollout (Section 9.5). The most direct next step is therefore to enlarge the effective state or the candidate library, for example through congestion/saturation terms, higher-order moment surrogates, or additional fields that capture variability unresolved by $(\rho, \mathbf{p})$ alone.

2. Constrained sparse regression. The closure gap could be narrowed from the regression side by replacing unconstrained MSTLS with a stability-aware formulation such as SR3 (Sparse Relaxed Regularised Regression) [36] or by imposing physical constraints during optimisation (for example, forcing the Laplacian coefficient to be non-negative so that the identified diffusion operator is dissipative. Such *physical inductive biases* would convert post-hoc safeguards (Section 8.7) into structural guarantees, reducing the gap between identification quality and rollout stability.

3. Scale-sensitivity analysis. The KDE bandwidth σ acts as a spatial filter: every identified PDE is implicitly a function of that coarse-graining scale. A systematic Pareto-front analysis trading KDE bandwidth σ (smaller $\rightarrow$ finer spatial resolution) against weak test-function support (larger $\rightarrow$ more noise averaging) would map how the identified law changes with coarse-graining and would quantify which interaction mechanisms are robustly recoverable at which scales. This extension would clarify whether the recovered coefficients are properties of the physics or of the smoothing; this is the most immediate methodological question raised by the present results.

4. Adaptive POD rank. The fixed rank m=19 used throughout this thesis may undersample the latent dynamics of variable-speed regimes, where spectral decay is slower. Future work should replace the fixed rank with an energy-based criterion (e.g. retaining $\geq 99.9\%$ of cumulative energy) or with the singular value hard threshold of Gavish and Donoho [15], which provides an asymptotically optimal cutoff for matrices with known noise level, allowing the ROM dimension to adapt to regime complexity.

5. Finite-N and out-of-family validation. A broader N-convergence study, together with tests on additional

simulator families or geometries, is needed to determine which identified structures persist beyond the present finite-N, single-simulator setting.

6. Extension to curved domains. All numerical results in this thesis use flat periodic domains, but the weak-form integral framework extends naturally to Riemannian manifolds $(\mathcal{M}, g)$: test functions are defined on the manifold's intrinsic coordinates, and the integration-by-parts identity that eliminates spatial derivatives replaces the Euclidean divergence theorem with its Riemannian counterpart [24]. Concretely, the weak-form residual for a conservation law $\partial_t u + \nabla_g \cdot \mathbf{F} = 0$ becomes

$$\int_{\mathcal{M}} \partial_t u \, \varphi \, \mathrm{d}V_g \;=\; \int_{\mathcal{M}} \mathbf{F} \cdot \nabla_g \varphi \, \mathrm{d}V_g, \tag{10.3.1}$$

where ∇_g is the Levi-Civita connection, $\mathrm{d}V_g = \sqrt{\det g}\, \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n$ is the Riemannian volume form, and the boundary term vanishes for compactly supported φ. Two additional requirements arise on curved domains. First, all library terms must be *metric-aware*: spatial derivatives become covariant derivatives, and Laplacians are replaced by the Laplace–Beltrami operator $\Delta_g u = \mathrm{div}_g(\nabla_g u)$. Second, any time-stepping scheme that updates fields in a tangent-space linearisation must *project back to the manifold* at each step, for example via the Riemannian exponential map $u_{k+1} = \exp_{u_k}(\Delta t \, \dot{u}_k)$, to ensure that the evolved density remains a well-defined field on $\mathcal{M}$. The main implementation task is replacing FFT-based convolutions (used here for the Morse potential $\Phi = W * \rho$) with manifold-aware kernel evaluations, for example via heat-kernel approximations or spherical-harmonic transforms on S^2. This extension would enable WSINDy-based PDE discovery for agent-based models on curved surfaces, a setting increasingly relevant in biological collective motion (cells on tissue surfaces) and geophysical swarming (flocks navigating the Earth's curvature). The repository accompanying this thesis is named `wsindy-manifold` in anticipation of this direction.

Taken together, the equation-free and equation-based branches define a *trade-off map* for data-driven modelling of agent-based systems. The EF-ROM delivers high short-term forecasting accuracy with zero physical interpretability; WSINDy delivers mechanistic insight with limited autonomous forecasting capability (due to the closure gap). This work demonstrates that we can successfully "read" the physics of interacting-agent systems from data; "writing" a stable predictive simulation from those discovered laws remains the primary challenge for the next generation of data-driven modelling.

A Experiment Configuration Catalogue

Not all configurations listed here are discussed equally in the main text; the thesis chapters focus on a representative subset of nine benchmark regimes chosen for interpretability and computational completeness. The parameter values for each dynamics regime are given in Tables B.2 and B.3 (Appendix B). This appendix lists every YAML file in `configs/systematic/` for reference.

Most NDYN base files have a *_VS* companion (`speed_mode =variable`), except where the speed mode is already variable or no companion was defined; see Appendix B for exceptions. All DO configs share v_0=3, η=0.2, C_a=1, ℓ_a=1; only C_r, ℓ_r vary.

Table A.1: NDYN configs (physics-knob sweep).

File	Regime
`NDYN01_crawl`	Slow, low noise
`NDYN02_flock`	Coherent flock
`NDYN03_sprint`	High speed
`NDYN04_gas`	High noise
`NDYN05_blackhole`	Strong attraction
`NDYN06_supernova`	Dominant repulsion
`NDYN07_crystal`	Balanced forces
`NDYN08_pure_vicsek`	No Morse forces
`NDYN09_longrange`	Large length scales
`NDYN10_shortrange`	Small length scales
`NDYN11_noisy_collapse`	Noise + attraction
`NDYN12_fast_explosion`	Speed + repulsion
`NDYN13_chaos`	Extreme speed/noise
`NDYN14_varspeed`	Variable speed

Table A.2: DO configs (outcome-targeted sweep).

File	Target pattern
`DO_CS01_swarm_C01_l05`	Cohesive swarm
`DO_CS02_swarm_C05_l3`	Cohesive swarm
`DO_CS03_swarm_C09_l3`	Cohesive swarm
`DO_DM01_dmill_C09_l05`	Double mill
`DO_DR01_dring_C01_l01`	Double ring
`DO_DR02_dring_C09_l09`	Double ring
`DO_EC01_esccol_C2_l3`	Escape–collapse
`DO_EC02_esccol_C3_l05`	Escape–collapse
`DO_ES01_escsym_C3_l09`	Escape symmetric
`DO_EU01_escuns_C2_l2`	Escape unstable
`DO_EU02_escuns_C3_l3`	Escape unstable
`DO_SM01_mill_C05_l01`	Single mill
`DO_SM02_mill_C3_l01`	Single mill
`DO_SM03_mill_C2_l05`	Single mill

Including the _VS variants, the repository contains **54 YAML files** in `configs/systematic/`. Six of these (three near-duplicate base regimes, NDYN03, NDYN11, NDYN12, and their VS counterparts) were removed before running, leaving 48 candidate configurations; of those, **37** produced usable results and form the catalogue analysed in Chapters 7–9 (Appendix B). The repository contains the full configuration set used during development, including duplicate or superseded files. The systematic catalogue analysed in this thesis retains 37 regimes after removing duplicate NDYN entries; of these, nine form the primary benchmark used in the main-text comparisons, while the remainder are used for catalogue-level robustness checks. Each configuration is run across **4 IC families** × multiple runs per family, totalling several thousand independent simulations.

A.1 Configuration Schema

Every experiment is fully specified by a single YAML file read at boot time. The schema below mixes global defaults with regime-specific overrides; only a subset of keys varies across the systematic regime files listed above. The schema mirrors an actual experiment configuration, annotated with allowed values. The EF-ROM keys (`sim` through `eval`) follow `NDYN02_flock.yaml`; the `wsindy` block follows `DYN1_gentle_wsindy.yaml`.

Each YAML file contains twelve top-level sections: `sim` (agent count, training horizon, domain geometry), `test_sim` (forecast horizon), `model` (agent dynamics type and speed mode), `noise` (angular noise magnitude), `forces` (Morse attraction/repulsion parameters C_a, C_r, ℓ_a, ℓ_r), `alignment` (Vicsek radius and coupling strength), `train_ic` and `test_ic` (initial-condition family and distributional parameters), `rom` (POD rank, MVAR lag, LSTM architecture and training settings), `eval` (forecast metrics, bootstrap replicates, confidence level), `wsindy` (library scope, thresholding schedule, test-function parameters), and `experiment_name` (unique run identifier). Full annotated schemas for all nine benchmark regimes are available in the repository at https://github.com/eugenio-macias/wsindy-manifold/tree/main/configs.

A.2 Symbol-to-Configuration Key Mapping

Table A.3 maps the mathematical symbols used in the main text to their corresponding YAML configuration keys and source modules in the accompanying repository.

Table A.3: Mapping from mathematical symbols to configuration keys and source modules used in the code repository.

Math symbol / quantity	**Config key**	**Source module**
Δt (time step)	`sim.dt`	`sim_runner.py`
N (agent count)	`sim.traj`	`sim_runner.py`
v_0 (speed)	`model.speed`	`particle_model.py`
R (alignment radius)	`alignment.R`	`particle_model.py`
η (noise strength)	`noise.eta`	`particle_model.py`
Noise model (uniform/wrapped-Cauchy)	`noise.kind`	`particle_model.py`
Match variance flag	`noise.match_variance`	`particle_model.py`
KDE bandwidth	`rom.density_bandwidth`	`pod_builder.py`
Shift-alignment reference	`rom.shift_align_ref`	`shift_align.py`
Density transform ($\sqrt{\cdot}$)	`rom.density_transform`	`pod_builder.py`
r (POD rank)	`rom.fixed_modes`	`pod_builder.py`
Clamp mode (C1)	`rom.clamp_mode`	`test_evaluator.py`
Mass post-processing	`rom.mass_postprocess`	`test_evaluator.py`
MVAR lag p	`rom.mvar.lag`	`mvar_trainer.py`
Ridge penalty α	`rom.mvar.ridge_alpha`	`mvar_trainer.py`
Eigenvalue threshold	`rom.eigenvalue_threshold`	`mvar_trainer.py`
LSTM subsample factor	`rom.lstm.subsample`	`lstm_rom.py`
Residual prediction mode	`rom.lstm.residual`	`lstm_rom.py`
Patience (early stopping)	`rom.lstm.patience`	`lstm_rom.py`
Library polynomial degree $p_{\max}$	`wsindy.max_poly`	`library.py`
Test-function stride	`wsindy.stride`	`system.py`
Morse forces enabled	`forces.enabled`	`multifield.py`

B Full Regime Catalogue

This appendix collects all systematic experiment configurations, together with their dynamics parameters and summary R^2 for MVAR and LSTM. Of the 54 YAML configurations in `configs/systematic/`, six were excluded before running (three near-duplicate base regimes, NDYN03, NDYN11, NDYN12, and their VS counterparts), leaving 48 candidates. The table also includes tier-1 sensitivity variants and preprocessing ablations where available. The nine primary benchmark regimes from Chapters 7 and 9 are shown in bold.

B.1 Dynamics parameters and ROM accuracy

Table B.1 lists every systematic regime. The CS and VS variants share the same force parameters but differ in speed coupling. Tables B.2 and B.3 give the governing parameters.

Table B.1: Mean test R^2 for MVAR and LSTM across all systematic experiments. Bold = nine main-text regimes; —† = LSTM diverged ($R^2 < -2$).

Regime	MVAR R^2	LSTM R^2
DO_CS01_swarm_C01_l05	0.949	0.920
DO_CS01_swarm_C01_l05_VS	0.594	-0.702
DO_CS02_swarm_C05_l3	0.094	0.109
DO_CS02_swarm_C05_l3_VS	-0.375	—†
DO_CS03_swarm_C09_l3	0.228	0.130
DO_CS03_swarm_C09_l3_VS	-0.305	-0.107
DO_DM01_dmill_C09_l05	0.992	-0.000
DO_DM01_dmill_C09_l05_VS	0.807	-0.909
DO_DR01_dring_C01_l01	0.953	0.612
DO_DR01_dring_C01_l01_VS	0.538	-0.695
DO_DR02_dring_C09_l09	0.984	0.984
DO_DR02_dring_C09_l09_VS	0.574	-0.838
DO_EC01_esccol_C2_l3	0.655	0.314
DO_EC02_esccol_C3_l05	0.423	-0.490
DO_EC02_esccol_C3_l05_VS	0.133	—†
DO_ES01_escsym_C3_l09	0.629	0.026
DO_ES01_escsym_C3_l09_VS	-0.004	-0.409
DO_EU01_escuns_C2_l2	0.446	0.437
DO_EU01_escuns_C2_l2_VS	0.102	0.015
DO_EU02_escuns_C3_l3	0.763	0.670
DO_SM01_mill_C05_l01	0.992	-0.206
DO_SM01_mill_C05_l01_VS	0.483	-0.840
DO_SM02_mill_C3_l01	0.939	0.936
DO_SM02_mill_C3_l01_VS	0.252	-1.141
DO_SM03_mill_C2_l05	0.990	0.178
DO_SM03_mill_C2_l05_VS	0.557	-1.120
NDYN01_crawl	0.980	-0.133
NDYN01_crawl_VS	0.676	0.187
NDYN02_flock	0.956	0.415
NDYN02_flock_VS	0.271	-0.465
NDYN04_gas_VS	**0.558**	**0.110**
NDYN04_gas_VS_tier1_bic	0.491	-0.702
NDYN04_gas_VS_tier1_w5	0.410	0.110
NDYN04_gas_preproc_sqrt_none	—	0.990
NDYN04_gas_preproc_sqrt_scale	—	0.950
NDYN04_gas	**0.996**	**0.996**
NDYN04_gas_tier1_bic	0.989	0.676
NDYN04_gas_tier1_w5	0.990	0.657
NDYN05_blackhole_VS	**0.704**	**-0.214**
NDYN05_blackhole_VS_tier1_bic	-1.269	-1.484
NDYN05_blackhole_VS_tier1_w5	-1.040	—†
NDYN05_blackhole	**0.994**	**0.992**
NDYN05_blackhole_tier1_bic	0.992	0.987
NDYN05_blackhole_tier1_w5	0.993	0.791
NDYN06_supernova_VS	**0.244**	**0.163**
NDYN06_supernova	**0.683**	**0.508**
NDYN06_supernova_tier1_bic	0.528	0.291
NDYN06_supernova_tier1_w5	0.683	-1.289
NDYN07_crystal_VS	**0.292**	**-0.779**
NDYN07_crystal	**0.996**	**0.585**
NDYN08_pure_vicsek	**0.654**	**0.229**
NDYN09_longrange	—	—
NDYN10_shortrange	-0.359	—†
NDYN10_shortrange_VS	0.402	—†
NDYN13_chaos	0.974	0.851
NDYN13_chaos_VS	0.290	-0.592
NDYN14_varspeed	0.287	—†

Tables B.2 and B.3 give the full parameter sets; each base config also has a `_VS` variant unless noted.

Table B.2: NDYN regimes (full catalogue). Bold = main-text benchmarks; † = removed near-duplicates.

Label	Nickname	v_0	η	C_a	C_r	ℓ_a	ℓ_r	Notes
NDYN01	crawl	0.5	0.03	0.8	0.3	1.5	0.5	easy POD baseline
NDYN02	flock	3.0	0.1	2.0	0.5	1.5	0.5	coherent cluster
NDYN03†	sprint	10.0	0.1	2.0	0.5	1.5	0.5	*removed (dup. NDYN02)*
NDYN04	**gas**	**3.0**	**1.5**	**0.5**	**0.2**	**1.5**	**0.5**	**disordered gas**
NDYN05	**blackhole**	**4.0**	**0.15**	**20.0**	**0.05**	**1.5**	**0.3**	**near-singular spike**
NDYN06	**supernova**	**3.0**	**0.15**	**0.05**	**15.0**	**1.5**	**0.5**	**explosive repulsion**
NDYN07	**crystal**	**2.0**	**0.1**	**3.0**	**3.0**	**2.0**	**0.5**	**balanced equilibrium**
NDYN08	**pure Vicsek**	**3.0**	**0.3**	—	—	—	—	**forces off**
NDYN09	longrange	2.0	0.2	1.5	0.6	6.0	3.0	global collective modes
NDYN10	shortrange	3.0	0.15	4.0	2.0	0.3	0.1	fragmented clusters
NDYN11†	noisy collapse	3.0	0.8	12.0	0.1	1.5	0.5	*removed (dup. NDYN05)*
NDYN12†	fast explosion	8.0	0.2	0.1	10.0	1.5	0.5	*removed (dup. NDYN06)*
NDYN13	chaos	12.0	1.2	1.0	0.5	1.5	0.5	ROM lower bound
NDYN14	varspeed	3.0	0.2	2.0	0.8	1.5	0.5	`variable` speed

B.2 R^2 degradation over time for all regimes

Figures B.1 shows the full $R^2(t)$ degradation curves for *all* systematic experiments, providing the extended counterpart to the nine-regime aggregate shown in Figure 7.5. Each faint trace corresponds to a single experiment; the bold curve is the cross-experiment median and the shaded band is the interquartile range.

B.3 WSINDy identification for extended regimes

WSINDy identification was performed on the nine primary benchmark regimes (Chapter 9). The EF-ROM forecast R^2 values for all regimes are recorded in Table B.1 above.

Table B.3: DO regimes (full catalogue). All share v_0=3, η=0.2, C_a=1, ℓ_a=1; only C_r and ℓ_r vary.

Label	Behaviour	C_r	ℓ_r	Notes
CS01	collective swarm	0.1	0.5	
CS02	collective swarm	0.5	3.0	
CS03	collective swarm	0.9	3.0	
SM01	single mill	0.5	0.1	
SM02	single mill	3.0	0.1	
SM03	single mill	2.0	0.5	
DM01	double mill	0.9	0.5	reference regime
DR01	double ring	0.1	0.1	
DR02	double ring	0.9	0.9	
EC01	escape (collective)	2.0	3.0	
EC02	escape (collective)	3.0	0.5	
ES01	escape (symmetric)	3.0	0.9	
EU01	escape (unsymmetric)	2.0	2.0	
EU02	escape (unsymmetric)	3.0	3.0	

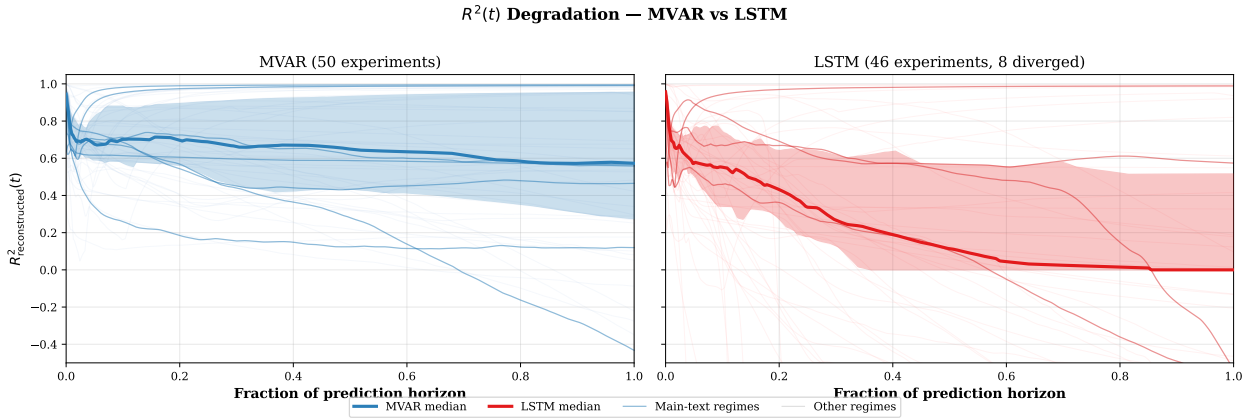


Figure B.1: $R^2(t)$ degradation for all systematic experiments. Left (blue): MVAR. Right (red): LSTM. Faint = individual runs, bold = median, shading = IQR. The aggregate curves exclude diverged LSTM experiments (final $R^2 < -2$), whose individual traces remain visible but are clipped at the axis boundary.

C Supplementary Material

This appendix collects per-experiment diagnostic plots, LSTM architecture details, and additional tables supporting Chapters 7 and 9.

C.1 Additional density snapshots

The following density-snapshot grids complement the representative gas regime shown in Figure 7.3 of the main text.

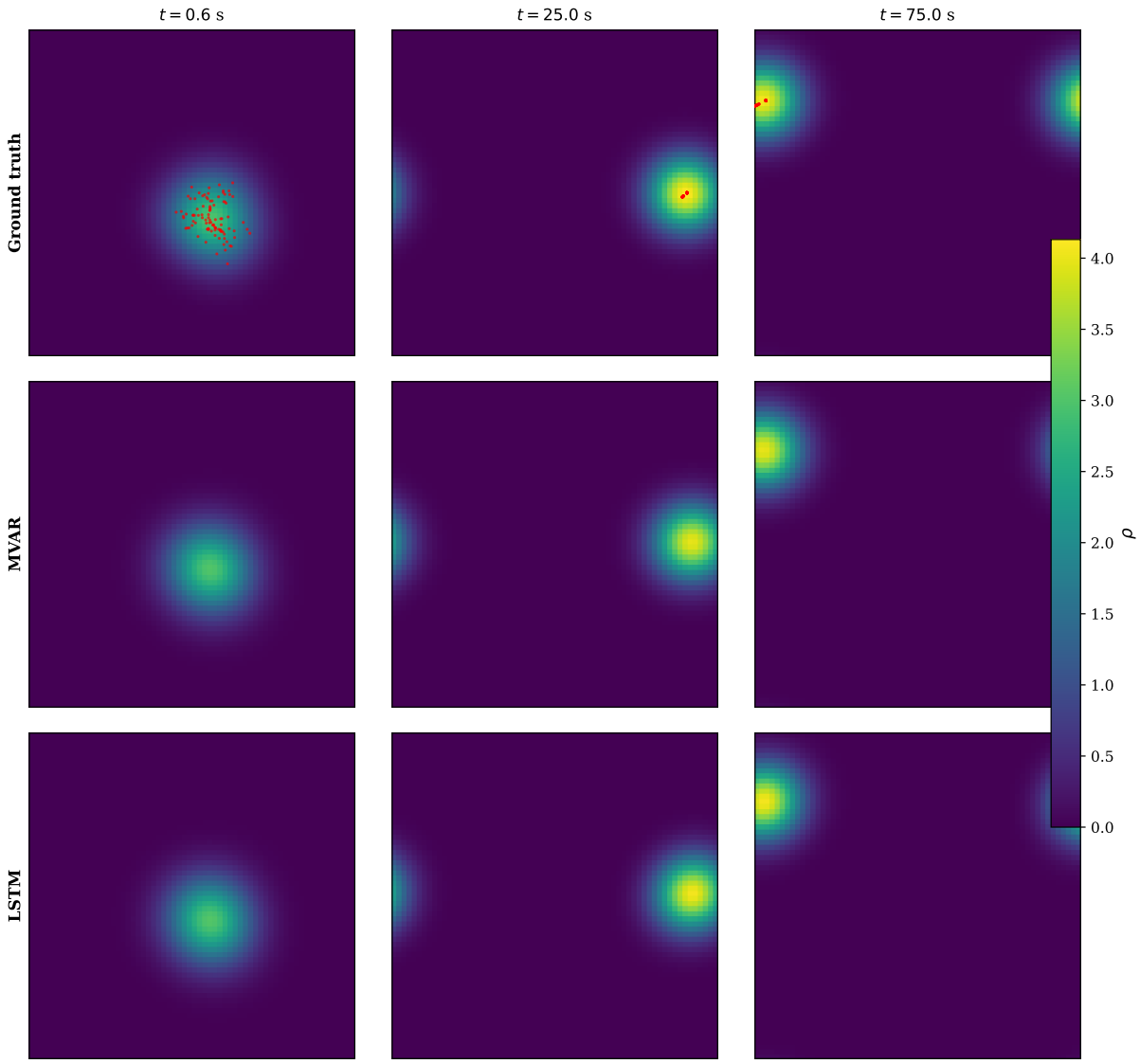


Figure C.1: Density snapshots truth vs. prediction for `NDYN05_blackhole` (Gaussian IC, test run 0). Layout matches Figure 7.3; rows show truth, MVAR, LSTM.

C.2 Eigenvalue spectra and stability

The companion-matrix eigenvalue spectrum of the fitted MVAR model reveals whether the linear surrogate is stable ($|\lambda| \leq 1$). All nine benchmark regimes have spectral radius near or below unity; regimes where one or more eigenvalues escape the unit circle required clipping during forecast (Section 7.4).

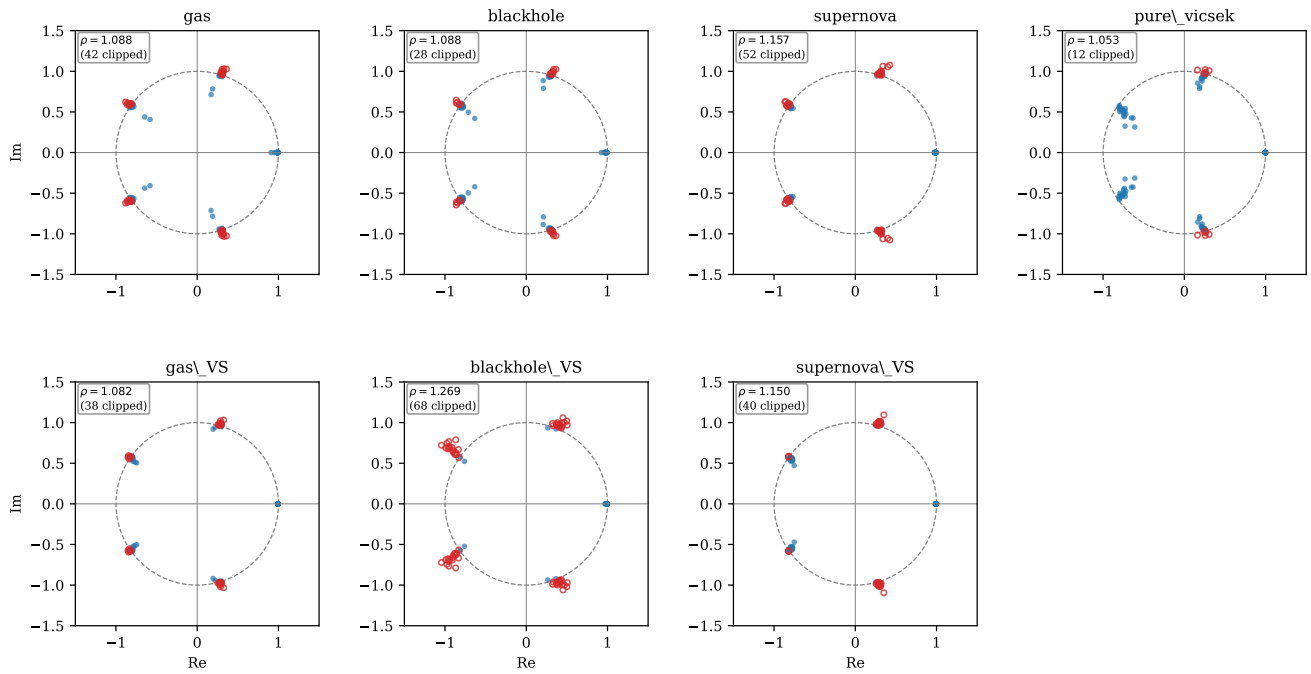


Figure C.2: Companion-matrix eigenvalue spectra for MVAR. Eigenvalues plotted in the complex plane (unit circle dashed). Filled = stable; open = clipped. Spectral radius $\rho(\mathbf{A})$ annotated per panel.

C.3 Per-experiment SVD decay

Figure C.3 compares the sPOD and standard POD singular-value spectra per regime. Regimes with stronger translation symmetry (gas, blackhole) show the largest relative gain from shift alignment, while near-stationary regimes (crystal) see little difference.

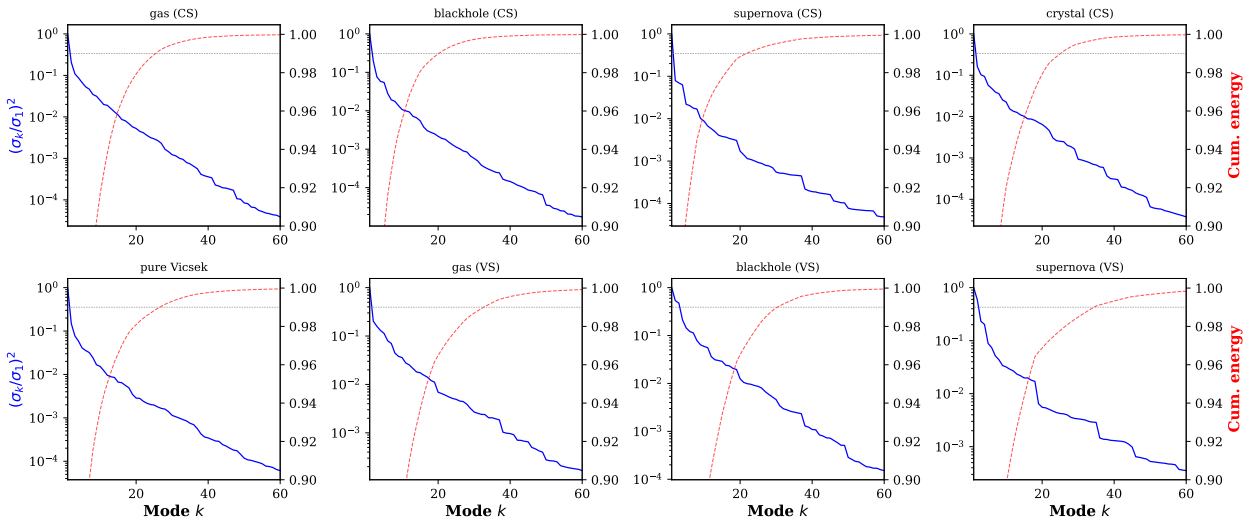


Figure C.3: Per-experiment sPOD vs POD singular value spectra: normalised eigenvalue decay (left) and cumulative energy (right). Complements Figure 7.1.

C.4 Per-experiment R^2 degradation

Per-experiment $R^2(t)$ degradation curves for all nine benchmark regimes are available in the full-catalogue Appendix B (Figure B.1). In summary, MVAR degrades most rapidly on supernova (fast spatial expansion), while LSTM degrades most rapidly on variable-speed variants.

C.5 R^2 by regime group and density–latent comparison

Figure C.4 disaggregates the forecast R^2 by regime group, confirming that MVAR dominates on CS regimes while LSTM narrows the gap only for a few well-behaved VS cases. Density-space and latent-space R^2 are closely aligned

across all test runs: the median absolute gap $|R^2_{\text{density}} - R^2_{\text{latent}}|$ is below 0.02 for MVAR and 0.04 for LSTM, confirming that the POD truncation preserves forecast quality.

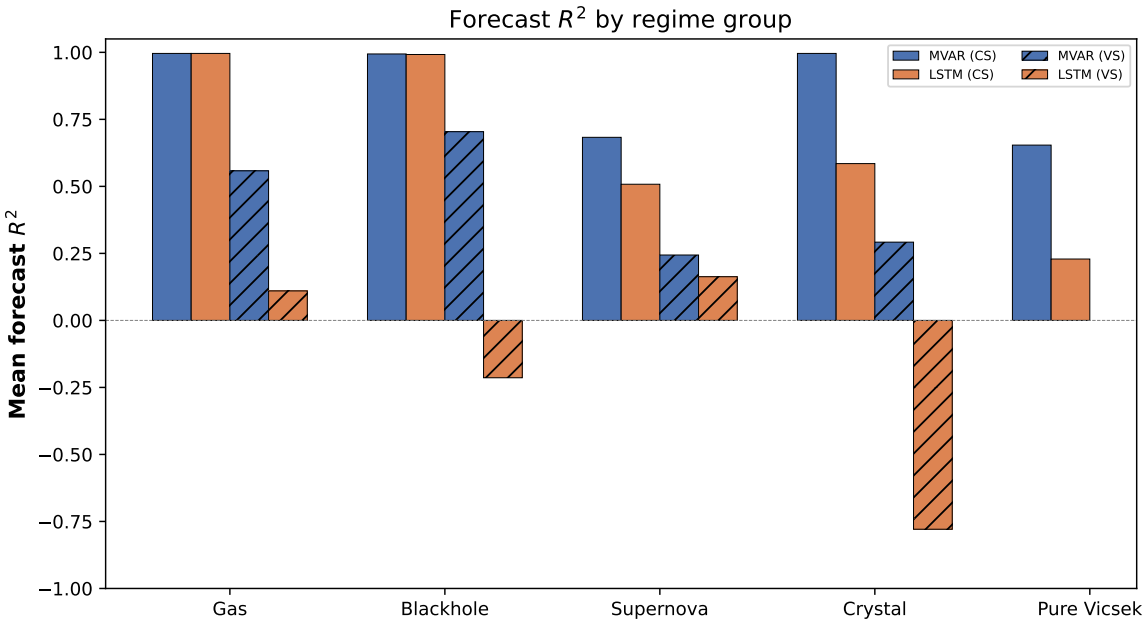


Figure C.4: Mean forecast R^2 by regime group (grouped bars: MVAR and LSTM, constant-speed vs. variable-speed variants). Complements Figure 7.4.

C.6 Mass conservation and phase dynamics

Both MVAR and LSTM preserve total mass to machine precision ($|M(t)/M_0 - 1| < 10^{-14}$) throughout the forecast horizon, since the POD lifting step is a linear injection that exactly preserves the column space of the snapshot matrix. Figure C.5 displays the shift-alignment statistics aggregated across regimes.

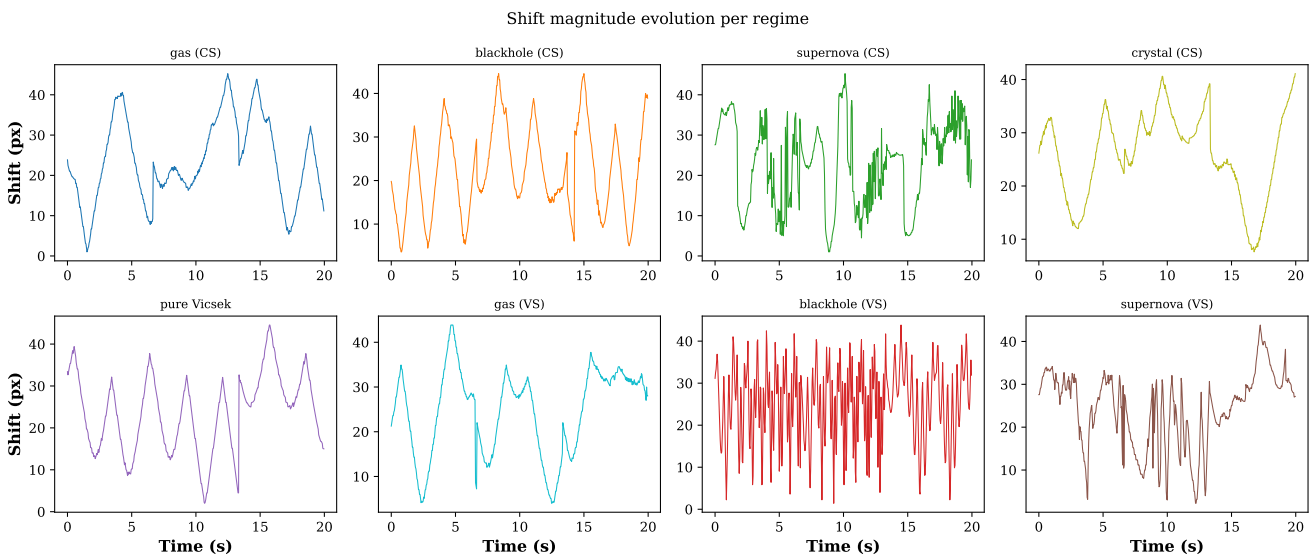


Figure C.5: Aggregate phase (shift) dynamics: (a) mean shift trajectory, (b) PSD, (c) AR(p) predictability, (d) shift variance, (e) autocorrelation, (f) legend.

C.7 Condition numbers

Among the L^1, L^2, and L^∞ norms tracked over time for MVAR, the L^∞ norm grows fastest, indicating that errors concentrate at density peaks. Figure C.6 shows the condition number of the weak-form design matrix; regimes with $\kappa > 10^3$ exhibit wider coefficient variability (cf. Table C.2).

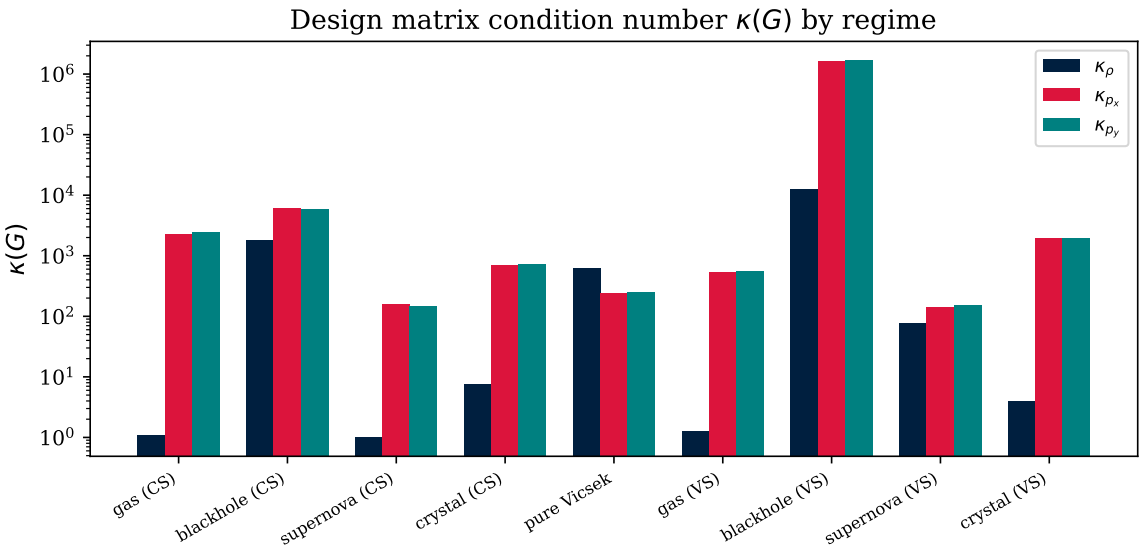


Figure C.6: Condition number $\kappa(G)$ of the weak-form design matrix, grouped by regime and equation. Log-scale y-axis.

C.8 Representative WSINDy coefficient bar charts

The full coefficient profiles for all nine regimes are tabulated in Tables 9.2–9.3 (Chapter 9). The dominant-balance profile shifts from diffusion-dominated (gas) to Morse-dominated (blackhole); the variable-speed variants retain the same backbone terms but with larger confidence intervals reflecting the noisier latent dynamics.

C.9 LSTM architecture and hyperparameters

The standard LSTM gate equations, referenced by Algorithm 5, are:

$$\begin{aligned} \mathbf{f}_\tau &= \sigma\big(\mathbf{W}_f[\mathbf{h}_{\tau-1}, \mathbf{z}_\tau] + \mathbf{b}_f\big), & \mathbf{i}_\tau &= \sigma\big(\mathbf{W}_i[\mathbf{h}_{\tau-1}, \mathbf{z}_\tau] + \mathbf{b}_i\big), \\ \mathbf{o}_\tau &= \sigma\big(\mathbf{W}_o[\mathbf{h}_{\tau-1}, \mathbf{z}_\tau] + \mathbf{b}_o\big), & \tilde{\mathbf{c}}_\tau &= \tanh\big(\mathbf{W}_c[\mathbf{h}_{\tau-1}, \mathbf{z}_\tau] + \mathbf{b}_c\big), \\ \mathbf{c}_\tau &= \mathbf{f}_\tau \odot \mathbf{c}_{\tau-1} + \mathbf{i}_\tau \odot \tilde{\mathbf{c}}_\tau, & \mathbf{h}_\tau &= \mathbf{o}_\tau \odot \tanh(\mathbf{c}_\tau), \end{aligned} \tag{C.9.1}$$

where $(\mathbf{h}_\tau, \mathbf{c}_\tau)$ are the hidden and cell states.

Table C.1: LSTM hyperparameters (final configuration).

Parameter	Config key	Value
Lag window	`lag`	5 (baseline)
Hidden units	`hidden_units`	128
LSTM layers	`num_layers`	2
Dropout (inter-layer)	`dropout`	0.1
Residual connection	`residual`	true
Layer normalisation	`use_layer_norm`	true
Max epochs	`max_epochs`	300
Early-stop patience	`patience`	40
Learning rate	`lr`	7×10^{-4}
Batch size	`batch_size`	512
Gradient clip	`grad_clip`	5.0

C.10 Condition number table

Table C.2 lists $\kappa(\mathbf{G})$ for all nine benchmark regimes. The blackhole VS regime has the highest condition number ($\sim 10^6$), consistent with its near-collinear library columns and the wide confidence intervals observed in the bootstrap analysis (Section 8.5.1).

Table C.2: Condition number $\kappa(\mathbf{G})$ of the weak-form design matrix per regime and target equation. High κ indicates near-collinearity, predicting wider coefficient variability across scales.

Regime	$\kappa(\mathbf{G}_\rho)$	$\kappa(\mathbf{G}_{p_x})$	$\kappa(\mathbf{G}_{p_y})$
`NDYN04_gas`	1.1	2,259	2,463
`NDYN05_blackhole`	1,780	6,012	5,957
`NDYN06_supernova`	1.0	158	144
`NDYN07_crystal`	7.6	703	721
`NDYN08_pure_vicsek`	623	243	246
`NDYN04_gas_VS`	1.3	527	547
`NDYN05_blackhole_VS`	12,496	1.6×10^6	1.7×10^6
`NDYN06_supernova_VS`	77	138	152
`NDYN07_crystal_VS`	4.0	1,917	1,935

D MSTLS: Modified Sequential Thresholded Least Squares

The sparse regression step in WSINDy solves the overdetermined system $\mathbf{G}\mathbf{w} \approx \mathbf{b}$, where $\mathbf{G} \in \mathbb{R}^{K\times M}$ is the weak-form library matrix, $\mathbf{b} \in \mathbb{R}^K$ the weak-form time-derivative target, and $\mathbf{w} \in \mathbb{R}^M$ the coefficient vector to be sparsified. The sequential thresholding strategy originates in the WSINDy framework of Messenger and Bortz [26]; our implementation follows the dominant-balance extension of Minor et al. [29].

Column preconditioning. Each column of $\mathbf{G}$ is scaled to unit ℓ^2 norm: $\tilde{\mathbf{G}} = \mathbf{G}\,\mathrm{diag}(s_1^{-1}, \ldots, s_M^{-1})$ with $s_j = \|\mathbf{G}_{:,j}\|_2$. This ensures that the thresholding criterion treats all candidate terms on equal footing. Physical coefficients are recovered via $w_j^{\mathrm{phys}} = \tilde{w}_j / s_j$.

Dominant-balance criterion. A term j is *kept* at threshold $\hat{\lambda}$ if its relative contribution lies in the band

$$\hat{\lambda} \;\le\; \frac{|\tilde{w}_j|\,\|\tilde{\mathbf{G}}_{:,j}\|_2}{\|\mathbf{b}\|_2} \;\le\; \frac{1}{\hat{\lambda}}\,.$$

Terms outside this band are pruned, and the least-squares fit is recomputed on the surviving columns until convergence.

Algorithm 1. MSTLS with dominant-balance thresholding

Require: Design matrix $\mathbf{G} \in \mathbb{R}^{K\times M}$, target $\mathbf{b} \in \mathbb{R}^K$, threshold grid $\{\hat{\lambda}_1, \ldots, \hat{\lambda}_L\}$, max iterations $I_{\max}$
Ensure: Sparse coefficient vector $\mathbf{w}^*$, active mask $\mathcal{A}^*$

1: $(\tilde{\mathbf{G}}, \{s_j\}) \leftarrow \textsc{Precondition}(\mathbf{G})$ ▷ Scale columns to unit norm
2: $\mathbf{w}_{\mathrm{LS}} \leftarrow \arg\min_{\mathbf{w}} \|\tilde{\mathbf{G}}\mathbf{w} - \mathbf{b}\|_2$ ▷ Full least-squares baseline
3: $\ell^* \leftarrow +\infty$ ▷ Track best normalised loss
4: **for** $\hat{\lambda} \in \{\hat{\lambda}_1, \ldots, \hat{\lambda}_L\}$ **do**
5: $\quad\mathcal{A} \leftarrow \{1, \ldots, M\}$ ▷ Start with all terms active
6: $\quad$**for** $i = 1, \ldots, I_{\max}$ **do**
7: $\quad\quad\tilde{\mathbf{w}}_{\mathcal{A}} \leftarrow \arg\min \|\tilde{\mathbf{G}}_{:,\mathcal{A}}\,\mathbf{w} - \mathbf{b}\|_2$ ▷ LS on active columns
8: $\quad\quad\tilde{\mathbf{w}}_j \leftarrow 0$ for $j \notin \mathcal{A}$
9: $\quad\quad\mathcal{A}' \leftarrow \left\{ j \in \mathcal{A} \;:\; \hat{\lambda} \le |\tilde{w}_j|\,\|\tilde{\mathbf{G}}_{:,j}\|_2 / \|\mathbf{b}\|_2 \le 1/\hat{\lambda} \right\}$
10: $\quad\quad$**if** $\mathcal{A}' = \mathcal{A}$ **then**
11: $\quad\quad\quad$**break** ▷ Converged
12: $\quad\quad$**end if**
13: $\quad\quad\mathcal{A} \leftarrow \mathcal{A}'$
14: $\quad$**end for**
15: $\quad$Re-fit: $\tilde{\mathbf{w}}_{\mathcal{A}} \leftarrow \arg\min \|\tilde{\mathbf{G}}_{:,\mathcal{A}}\,\mathbf{w} - \mathbf{b}\|_2$
16: $\quad\ell \leftarrow \|\mathbf{b} - \tilde{\mathbf{G}}\tilde{\mathbf{w}}\|_2 \;/\; \|\tilde{\mathbf{G}}\mathbf{w}_{\mathrm{LS}}\|_2$ ▷ Normalised loss
17: $\quad$**if** $\ell < \ell^*$ **or** ($\ell = \ell^*$ **and** $|\mathcal{A}| < |\mathcal{A}^*|$) **then** ▷ Prefer lowest loss; break ties by sparsity (Occam)
18: $\quad\quad\mathbf{w}^* \leftarrow \tilde{\mathbf{w}},\;\; \mathcal{A}^* \leftarrow \mathcal{A},\;\; \ell^* \leftarrow \ell$
19: $\quad$**end if**
20: **end for**
21: **return** $\mathbf{w}_j^*/s_j$ for all j, active mask $\mathcal{A}^*$ ▷ Un-scale to physical units

The implementation is in `src/wsindy/regression.py` (`mstls`). The default $\hat{\lambda}$ grid is 25 log-spaced values in $[10^{-4}, 1]$. The tie-break rule on line 17 (prefer equal-loss models with fewer active terms) enforces an Occam preference: among all threshold values that achieve the same normalised residual, we select the sparsest model. This is equivalent to a loss-first, complexity-second lexicographic ordering, and avoids the need for a penalised criterion such as BIC, which would require specifying an effective noise variance for the weak-form residual.

Post-convergence dominant balance diagnostic. After MSTLS converges to $(\mathbf{w}^*, \mathcal{A}^*)$, the *dominant balance ratio* for each active term is computed as

$$\Pi_j = |w_j^*|\,\|\mathbf{G}_{:,j}\|_2 \;/\; \|\mathbf{b}\|_2, \qquad \tilde{\Pi}_j = \Pi_j / \max_k \Pi_k, \tag{D.0.1}$$

so that $\tilde{\Pi}_j \in [0, 1]$ measures the fractional contribution of term j to the total weak-form signal. This is a *post-fit* diagnostic, not a thresholding criterion: it characterises dominant balance after the active set has been determined.

Chapters 9 reports $\tilde{\Pi}_m$ as heatmap colouring (Eq. (9.2.2)).

Remark. The MSTLS model-selection loss $\ell = \|\mathbf{b} - \tilde{\mathbf{G}}\tilde{\mathbf{w}}\|_2 / \|\tilde{\mathbf{G}}\mathbf{w}_{\mathrm{LS}}\|_2$ normalises against the dense fit, while $R^2_{\mathrm{wf}} = 1 - \|\mathbf{b} - \mathbf{G}\mathbf{w}^*\|_2^2 / \|\mathbf{b}\|_2^2$ measures explained signal energy; both lie in $[0, 1]$ but differ in purpose.

E ETDRK4 Time Integration

Discovered PDEs are integrated forward in time using the exponential time-differencing fourth-order Runge–Kutta scheme (ETDRK4) of Cox and Matthews [12], stabilised via the contour-integral trick of Kassam and Trefethen [22].

Linear–nonlinear splitting. The PDE $u_t = F(u)$ is decomposed as $u_t = Lu + N(u)$, where L collects all terms linear in u (e.g. diffusion $\alpha\Delta u$, advection $\beta\,\partial_x u$) and $N(u)$ is the nonlinear remainder. In Fourier space, L has a diagonal multiplier $\hat{L}(k_x, k_y) = \sum_{m\in\mathcal{L}} w_m\,\sigma_m(k_x, k_y)$, where σ_m is the symbol of operator m: $\sigma_{\text{lap}} = -(k_x^2 + k_y^2)$, $\sigma_{d_x} = ik_x$, $\sigma_{d_{xx}} = -k_x^2$, etc.

Contour-integral coefficients. With $z = \Delta t\,\hat{L}$ and M quadrature nodes $c_j = r\exp(2\pi i\,j/M)$ on a circle of radius r centred at each z:

$$E = e^z, \qquad E_2 = e^{z/2}, \qquad Q = \frac{\Delta t}{2}\,\frac{1}{M}\sum_{j=1}^{M}\frac{e^{z/2+c_j}-1}{z/2+c_j}\,, \tag{E.0.1}$$

$$f_1 = \Delta t\,\frac{1}{M}\sum_j \frac{-4 - z_j + e^{z_j}(4 - 3z_j + z_j^2)}{z_j^3}, \tag{E.0.2}$$

$$f_2 = \Delta t\,\frac{1}{M}\sum_j \frac{2 + z_j + e^{z_j}(-2 + z_j)}{z_j^3}, \tag{E.0.3}$$

$$f_3 = \Delta t\,\frac{1}{M}\sum_j \frac{-4 - 3z_j - z_j^2 + e^{z_j}(4 - z_j)}{z_j^3}, \tag{E.0.4}$$

where $z_j = z + c_j$. Taking $M = 64$ and $r = 1$ removes the removable singularities at $z = 0$ with machine-precision accuracy.

Algorithm 2. ETDRK4 time-stepping (one step, Fourier domain)

Require: Fourier-domain state $\hat{u}_n$, coefficients $(E, E_2, Q, f_1, f_2, f_3)$, nonlinear evaluator $\hat{N}(\cdot)$
Ensure: $\hat{u}_{n+1}$

```
1: N̂1 ← N̂(F^{-1}[û_n])
2: â ← E2 û_n + Q N̂1
3: N̂2 ← N̂(F^{-1}[â])
4: b̂ ← E2 û_n + Q N̂2
5: N̂3 ← N̂(F^{-1}[b̂])
6: ĉ ← E2 â + Q (2N̂3 − N̂1)
7: N̂4 ← N̂(F^{-1}[ĉ])
8: û_{n+1} ← E û_n + f1 N̂1 + 2 f2 (N̂2 + N̂3) + f3 N̂4
```

At each sub-stage $\hat{N}(\cdot)$ evaluates *only* the nonlinear terms of the discovered PDE in physical space (pointwise products, powers, gradient products) and returns the result to Fourier space via FFT. The implementation is in `src/wsindy/integrators.py` (`etdrk4_integrate`).

Stability. The ETDRK4 scheme exactly integrates the linear part $e^{\Delta t L}$, so it is unconditionally stable for the linear subproblem whenever $\mathrm{Re}(\Delta t\,\hat{L}(\mathbf{k})) \le 0$ for all wavenumbers $\mathbf{k}$ (satisfied by diffusion and dissipation operators with negative-semidefinite symbols). Nonlinear terms impose an effective CFL condition; empirically, $\Delta t = 0.01$–0.05 (corresponding to CFL $\approx v_{\max}\Delta t/h \lesssim 0.5$) is stable for the Vicsek-type PDEs encountered in this work.

F Detailed Algorithm Pseudocode

F.1 Vicsek simulation with steering Morse forces

Algorithm 3. Discrete-time Vicsek simulation with steering Morse forces

Require: N, L_x, L_y, Δt, T, v_0, R; boundary type `bc`; noise kind and amplitude η; `speed_mode`; (optional) Morse parameters $(C_r, C_a, \ell_r, \ell_a, r_{\text{cut}}, \mu_t)$.

Ensure: Saved frames $\{x^n, p^n\}$; metadata.

Initialize positions $x^0 \subset [0, L_x) \times [0, L_y)$ and angles $\theta^0 \sim \text{Unif}(0, 2\pi)$.
Set headings $p^0 \leftarrow$ `headings_from_angles`(θ^0); set $n_{\max} \leftarrow \lfloor T/\Delta t \rfloor$.
Set $r_{\text{cut}} \leftarrow$ `rcut_factor` $\cdot \max($`lr`, `la`$)$ (if forces enabled).
Build initial cell list: `cell_list` $\leftarrow$ `build_cells`$(x^0, L_x, L_y, \max(R, r_{\text{cut}}), \texttt{bc})$.
for $n = 0, 1, \ldots, n_{\max} - 1$ **do**
  if $n \bmod$ `neighbor_rebuild` $= 0$ **then**
    `cell_list` $\leftarrow$ `build_cells`$(x^n, L_x, L_y, \max(R, r_{\text{cut}}), \texttt{bc})$.
  end if
  $\mathcal{N}^n \leftarrow$ `compute_neighbors`$(x^n, L_x, L_y, R, \texttt{bc}, \texttt{cell_list})$ ▷ includes self
  if `forces.enabled` **then**
    $F^n \leftarrow$ `morse_force`$(x^n, L_x, L_y, \texttt{bc}, \texttt{Cr}, \texttt{Ca}, \texttt{lr}, \texttt{la}, r_{\text{cut}}, \texttt{cell_list})$.
  end if
  for $i = 1, \ldots, N$ **do**
    $s \leftarrow \sum_{j \in \mathcal{N}_i^n} p_j^n$.
    $\bar{p}_i^n \leftarrow s/\|s\|$ ▷ alignment mean direction (renormalized)
    if `speed_mode` is `constant_with_forces` **then**
      $\widehat{F}_i^n \leftarrow F_i^n/(\|F_i^n\| + \varepsilon)$.
      $\tilde{p}_i^n \leftarrow \bar{p}_i^n +$ `mu_t` $\widehat{F}_i^n$.
      $\bar{p}_i^n \leftarrow \tilde{p}_i^n/\|\tilde{p}_i^n\|$.
    end if
    $\phi_i^n \leftarrow$ `angle_noise`$(\texttt{rng}, \texttt{noise.kind}, \eta, \texttt{match_variance})$.
    $p_i^{n+1} \leftarrow$ `rotation`$(\phi_i^n)\, \bar{p}_i^n$.
    $p_i^{n+1} \leftarrow p_i^{n+1}/\|p_i^{n+1}\|$.
    if `speed_mode` is `variable` **then**
      $v_i^{n+1} \leftarrow v_i^n + \Delta t\, \texttt{mu_t}\, F_i^n$.
      $\tilde{x}_i^{n+1} \leftarrow x_i^n + \Delta t\, v_i^{n+1}$.
    else
      $\tilde{x}_i^{n+1} \leftarrow x_i^n + \Delta t\, v_0\, p_i^{n+1}$.
    end if
    $(x_i^{n+1}, \texttt{flips}_i) \leftarrow$ `apply_bc`$(\tilde{x}_i^{n+1}, L_x, L_y, \texttt{bc})$.
    if `bc` is `reflecting` **then**
      Flip reflected components of p_i^{n+1} according to $\texttt{flips}_i$.
    end if
  end for
  if $(n+1)$ is a multiple of `save_every` **then**
    Save frame (x^{n+1}, p^{n+1}).
  end if
end for
Store run metadata and (optionally) diagnostics.

F.2 Fast gridded Gaussian KDE

Algorithm 4. Fast gridded Gaussian KDE (histogram + Gaussian smoothing)

Require: Trajectory positions $\mathbf{x}(t) \in \mathbb{R}^{N \times 2}$ at time t, domain size (L_x, L_y), grid resolution (n_x, n_y), bandwidth σ (in grid cells), boundary condition `bc` $\in \{$`"periodic"`, `"nonperiodic"`$\}$

Ensure: Density field $\rho(t) \in \mathbb{R}^{n_y \times n_x}$ (row-major: y index first)

$\Delta x \leftarrow L_x/n_x$, $\Delta y \leftarrow L_y/n_y$
Extract arrays: $x \leftarrow \mathbf{x}[:, 0]$, $y \leftarrow \mathbf{x}[:, 1]$
if `bc` = `"periodic"` **then**
  Wrap positions: $x \leftarrow x \bmod L_x$, $y \leftarrow y \bmod L_y$
end if
Compute 2D histogram $H(t) \in \mathbb{R}^{n_y \times n_x}$ from (y, x) using bin edges on $[0, L_y] \times [0, L_x]$ ▷ Axis convention check: a single particle at (x, y) increments $H[\lfloor y/\Delta y \rfloor, \lfloor x/\Delta x \rfloor]$
Convert counts to number density: $\tilde{\rho}(t) \leftarrow H(t)/(\Delta x\, \Delta y)$

if `bc` = `"periodic"` **then**
  Smooth with periodic wrap: $\rho(t) \leftarrow G_\sigma(\tilde{\rho}(t))$ using `mode="wrap"`
else
  Smooth with edge extension: $\rho(t) \leftarrow G_\sigma(\tilde{\rho}(t))$ using `mode="nearest"` ▷ Note: `nearest` repeats edge values; this is not reflective padding.
end if
Compute mass: $M(t) \leftarrow \sum_{a,b} \rho_{ab}(t)\,\Delta x\,\Delta y$
Compute relative drift: $\varepsilon(t) \leftarrow |M(t) - N|/N$
if $\varepsilon(t) > 10^{-3}$ **then**
  print warning: mass deviation $\varepsilon(t)$ detected (typically indicates wrapping/binning errors)
end if
return $\rho(t)$ and metadata $\{\sigma, n_x, n_y, L_x, L_y, \texttt{bc}\}$

F.3 Latent LSTM ROM

Algorithm 5 Latent LSTM ROM (`LatentLSTMROM`): training and closed-loop rollout

Require: Latent trajectories $\{y_k^{(c)}\}$, lag p (`lag`), horizon H
Ensure: Closed-loop forecast $\{\widehat{\mathbf{z}}_{k_0+n}\}_{n=1}^{H}$
Windowing: construct pairs $(\mathbf{x}_k, \mathbf{z}_k)$ for $k = p{+}1, \ldots, K$, with $\mathbf{x}_k = (\mathbf{z}_{k-p}, \ldots, \mathbf{z}_{k-1})$.
Split into train/validation; initialise LSTM weights.
Training (Adam, lr $= 10^{-3}$)**:**
while epoch ≤ 300 and no early stop (patience $= 30$) **do**
  Sample mini-batch of size $B = 256$.
  Forward pass through (C.9.1); apply LayerNorm to $\mathbf{h}_p$.
  Compute $\widehat{\mathbf{z}}_k = \mathbf{z}_{k-1} + \Delta_k$ (residual mode, (6.2.2)).
  Backpropagate MSE loss; clip gradients at 1.0.
  Step `ReduceLROnPlateau` scheduler on validation loss.
end while
Closed-loop rollout:
Set window $\mathbf{x} \leftarrow \mathbf{x}_{k_0}$.
for $n = 1, \ldots, H$ **do**
  Predict $\widehat{\mathbf{z}}_{k_0+n}$ from $\mathbf{x}$ (residual mode).
  Shift window; append $\widehat{\mathbf{z}}_{k_0+n}$.
end for
Lift each $\widehat{\mathbf{z}}_{k_0+n}$ to physical space via the POD basis (Chapter 5).

F.4 Global POD restriction–lifting pipeline

Algorithm 6 Global POD restriction–lifting pipeline.

Require: Training density movies $\{\rho^{(i)}(t_k)\}_{i=1}^{M}$, grid (n_x, n_y), latent dim m, subsample stride s
Ensure: POD basis U_m, mean $\boldsymbol{\mu}$, shift data, latent trajectories $\{y^{(i)}(t_k)\}$

— **Training (restriction)** —
for each run $i = 1, \ldots, M$ **do**
  Subsample: keep every s-th frame
  Flatten: $x_k^{(i)} = \mathrm{vec}(\rho^{(i)}(t_k)) \in \mathbb{R}^d$
end for
Stack all runs: $\mathbf{X}_{\mathrm{raw}} \in \mathbb{R}^{T_{\mathrm{tot}} \times d}$ ▷ `np.vstack`
Compute reference ρ_{ref} (temporal mean of all frames) ▷ `shift_align`
Estimate per-frame shifts $(\Delta y_k, \Delta x_k)$ via FFT cross-correlation (Eq. 5.2.4)
Apply shifts: $\rho_k \leftarrow \mathrm{roll}(\rho_k, \Delta y_k, \Delta x_k)$
Transform: $\tilde{\rho}_k \leftarrow \sqrt{\rho_k + \varepsilon}$ ▷ `density_transform: sqrt`
Compute mean: $\boldsymbol{\mu} = \frac{1}{T_{\mathrm{tot}}} \sum_k \tilde{\rho}_k$
Centre: $\bar{X} = \mathbf{X}^\top - \boldsymbol{\mu}\mathbf{1}^\top$
SVD: $\bar{X} = U\Sigma V^\top$; keep U_m (m columns) ▷ `np.linalg.svd`
Restrict: $Y = U_m^\top(\mathbf{X}^\top - \boldsymbol{\mu}\mathbf{1}^\top)$
Reshape Y into per-run latent trajectories $\{y^{(i)}(t_k)\}$

— **Evaluation (lifting)** —
Forecast latent trajectory $\hat{y}(t_k)$ (MVAR or LSTM; Chapter 6)
Lift: $\hat{\tilde{\rho}}_k = U_m\, \hat{y}(t_k) + \boldsymbol{\mu}$
Inverse transform: $\hat{\rho}_k = [\max(0, \hat{\tilde{\rho}}_k)]^2 - \varepsilon$
Optional mass postprocess: apply (5.5.3) or (5.5.2) ▷ `mass_postprocess: scale | simplex`
Undo shift: $\hat{\rho}_k \leftarrow \mathrm{roll}(\hat{\rho}_k, -\Delta y_k, -\Delta x_k)$

G Additional Robustness Studies

This appendix collects two robustness studies assembled from Oscar cluster results. The noise sweep (Section G.2) tests identification and forecast stability across varying noise amplitudes, while the particle-count convergence study (Section G.3) verifies the mean-field limit.

G.1 Physical inductive biases

Section 8.7 describes six physical inductive biases in full; a compact summary is reproduced here for self-contained reference.

1. **Centred flux estimation.** The KDE flux estimator subtracts the domain-mean flux from each snapshot before convolution, preventing a spurious DC offset from aliasing into the linear drag coefficient.
2. **Biharmonic operator removal.** The biharmonic (∇^4) library term was removed: it is not part of the expected mean-field structure and in trial runs absorbed energy from the viscous term, degrading interpretability.
3. **Temporal test-function width floor.** The temporal support is floored at $\ell_t \geq 7$ grid points to prevent the width optimiser from collapsing the weak-form integral to a pointwise residual.
4. **Sign constraints on post-fit coefficients.** After each MSTLS solve, coefficients whose physical sign is known *a priori* (e.g. viscous terms must be non-positive) are checked; sign-violating survivors are zeroed and the fit re-solved.
5. **Coefficient magnitude cap.** Retained coefficients exceeding $|c| > 5.0$ are removed, guarding against inflation from near-singular library matrices.
6. **Regime-aware library construction.** The `forces.enabled` flag controls whether Morse potential terms enter the candidate library; for pure-Vicsek regimes the Morse columns are automatically excluded.

G.2 Noise robustness

The primary question is whether the *same mechanisms remain identifiable* under varying noise, rather than whether weak-form R^2 degrades. We run the full pipeline at five noise levels for each of four regimes: gas ($\eta \in \{0.15, 0.50, 1.00, 1.50, 2.00\}$, native η=1.5), blackhole ($\eta \in \{0.05, 0.15, 0.50, 1.00, 1.50\}$, native η=0.15), supernova ($\eta \in \{0.05, 0.15, 0.50, 1.00, 1.50\}$, native η=0.15), and crystal ($\eta \in \{0.05, 0.10, 0.50, 1.00, 1.50\}$, native η=0.10). We focus the full pipeline (ROM + WSINDy) on the gas and blackhole regimes, where the particle-level physics permits meaningful ROM comparison. For the supernova and crystal regimes, only WSINDy identification is reported (Table G.2). Table G.1 collects the gas and blackhole results.

Table G.1: Noise-sweep forecast R^2 for gas and blackhole regimes (best of four IC families). WSINDy $R^2_{\mathrm{wf},\rho}$ reports the weak-form fit on the density field. Native noise levels are marked with $\star$.

Regime	$\boldsymbol{\eta}$	**MVAR** R^2	**LSTM** R^2	**WSINDy** R^2_{wf}
Gas	0.15	0.995	0.972	1.000
Gas	0.50	0.993	0.972	1.000
Gas	1.00	0.995	0.983	1.000
Gas	1.50$\star$	0.994	0.977	1.000
Gas	2.00	0.994	0.806	1.000
Blackhole	0.05	0.994	0.988	—
Blackhole	0.15$\star$	0.986	0.966	1.000
Blackhole	0.50	0.993	0.251	1.000
Blackhole	1.00	0.983	0.125	1.000
Blackhole	1.50	0.992	0.857	1.000

For the supernova and crystal regimes, where the interplay of long-range forces and alignment makes ROM forecasting less effective (see Section 7.3), we report WSINDy identification only.

Across the gas sweep, MVAR remains above $R^2 \geq 0.993$ at all five noise levels; LSTM degrades gracefully from 0.983 at η=1.00 to 0.806 at η=2.00, consistent with the distribution-shift sensitivity discussed in Section 7.3. For blackhole, MVAR stays above 0.983 at all five noise levels (including 0.992 at the heaviest noise η=1.50), while LSTM drops sharply to 0.125 at η=1.00 and 0.251 at η=0.50, confirming that the black-hole's radial attractor is harder for feed-forward LSTM rollouts.

WSINDy identification is robust in every completed noise run: the continuity backbone $\nabla \cdot \mathbf{p} \approx -1$ is retained

Table G.2: Noise-sweep WSINDy-only $R^2_{\mathrm{wf},\rho}$ for supernova and crystal regimes. Native noise levels are marked with $\star$.

Regime	$\boldsymbol{\eta}$	**WSINDy** $R^2_{\mathrm{wf},\rho}$
Supernova	0.05	1.000
Supernova	0.15$\star$	1.000
Supernova	0.50	1.000
Supernova	1.00	0.999
Supernova	1.50	0.998
Crystal	0.05	1.000
Crystal	0.10$\star$	1.000
Crystal	0.50	1.000
Crystal	1.00	1.000
Crystal	1.50	1.000

wherever the Oscar jobs produced WSINDy summary files, and the density weak-form fit remains above 0.998 in all completed rows. The blank WSINDy entries correspond to Oscar jobs that produced only metadata directories and no recovered model files (η=0.05 and 0.15 for blackhole, and η=0.50 for supernova). The sole visible degradation among the completed WSINDy runs appears in the supernova regime, where $R^2_{\mathrm{wf},\rho}$ drops from 0.9998 at the native level (η=0.15) to 0.9982 at η=1.50 (Table G.2), consistent with the supernova's higher microscopic complexity (Figure G.1).

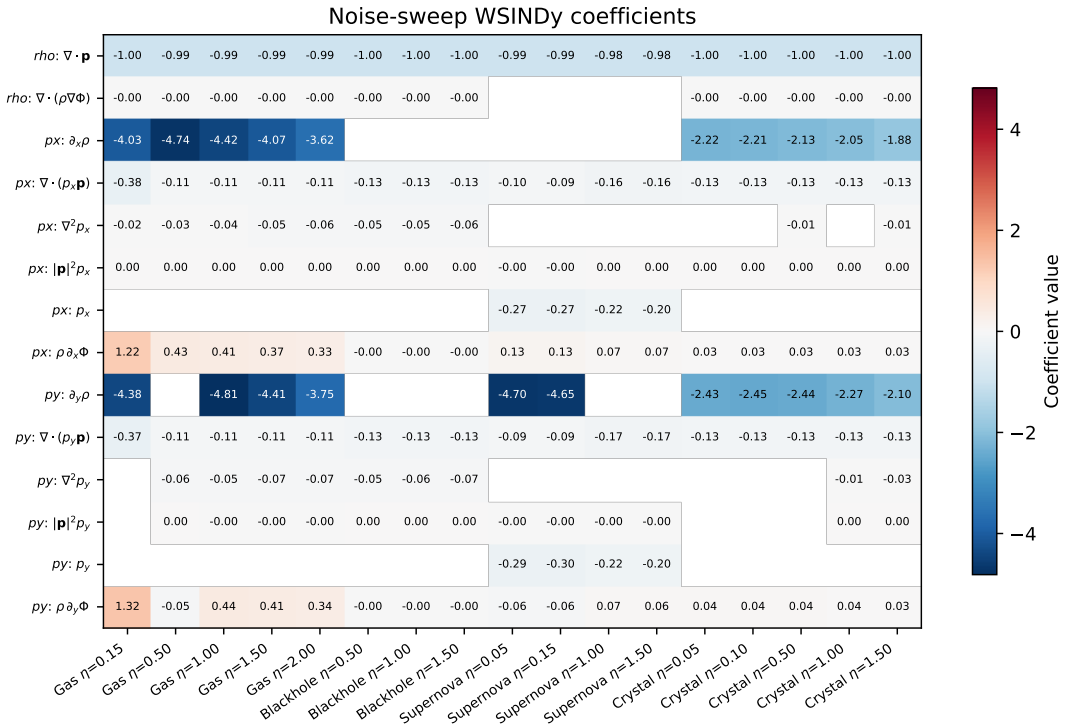


Figure G.1: Noise-sweep WSINDy coefficient stability. Rows correspond to candidate library terms; columns to the completed regime–noise configurations recovered from Oscar. White cells indicate terms absent from the active set. The continuity backbone $\nabla \cdot \mathbf{p} \approx -1$ is stable wherever the WSINDy noise runs finished; the unfinished blackhole η=0.05/0.15 and supernova η=0.50 jobs are not shown because they produced no coefficient files.

G.3 Convergence in the number of particles

To verify that the WSINDy-recovered PDEs represent a genuine mean-field limit, we repeat the pure Vicsek identification at $N \in \{50, 100, 200, 300, 500, 1000\}$. Table G.3 reports the companion MVAR and LSTM forecast quality alongside the WSINDy weak-form R^2 where available. The newly synced ROM runs extend the table to N=500 and N=1000: at N=500, MVAR drops to mean R^2=0.189 while retaining R^2=0.992 on the Gaussian-IC family, and LSTM diverges (marked †); at N=1000, MVAR rebounds to R^2=0.713, the Gaussian-IC score remains 0.992, and LSTM stays below zero. WSINDy results are available only for N=50–300; at all four completed particle counts, WSINDy recovers $R^2_{\mathrm{wf},\rho} \geq 0.9999$ with a consistent active set of $|\mathcal{A}|$=5 terms (one for ρ, two each for p_x and p_y); the reported $R^2_{\mathrm{wf},p}$ values use the p_x equation, with p_y showing comparable fits in all completed runs. Momentum fit ranges from $R^2_{\mathrm{wf},p}$=0.80 to 0.85, suggesting the mean-field limit is well-approximated even at

moderate particle counts.

Table G.3: N-convergence forecast R^2 for pure Vicsek (best of four IC families). MVAR stays above 0.98 on Gaussian ICs across all six particle counts; LSTM becomes unstable once $N \geq 200$. WSINDy weak-form $R^2_{\mathrm{wf},\rho} \geq 0.9999$ at all completed particle counts; momentum fit $R^2_{\mathrm{wf},p}$ is in the 0.80–0.85 range. Negative LSTM R^2 values indicate divergence; † marks the catastrophic N=500 failure.

N	**MVAR** R^2	**LSTM** R^2	**MVAR** R^2 (Gaussian IC)	**WSINDy** $R^2_{\mathrm{wf},\rho}$	**WSINDy** $R^2_{\mathrm{wf},p}$
50	0.635	0.229	0.982	1.000	0.801
100	0.542	0.085	0.992	1.000	0.827
200	0.647	−0.794	0.995	1.000	0.849
300	0.653	−0.914	0.992	1.000	0.834
500	0.189	—†	0.992	—	—
1000	0.713	−0.343	0.992	—	—

MVAR R^2 is stable at 0.54–0.65 for N=50–300, drops to 0.189 at N=500, and recovers to 0.713 at N=1000; however, the Gaussian-IC performance remains $R^2 > 0.99$ across all six tested counts, indicating that the forecast ceiling is limited by IC diversity rather than particle number. LSTM diverges ($R^2 < 0$) from N=200 onward, reflecting sensitivity to the growing latent state dimension. The N=500 run is the strongest failure ($R^2 = -8.12$), while N=1000 remains below zero ($R^2 = -0.34$); this trend is consistent with the shift-predictability bottleneck discussed in the main text.

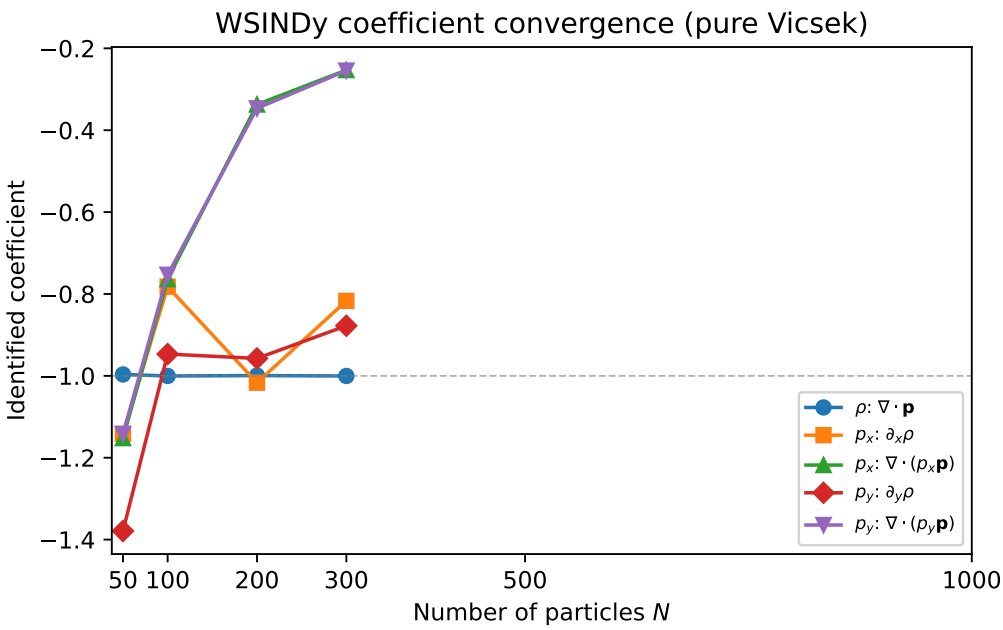


Figure G.2: N-convergence of WSINDy-identified coefficients for the pure Vicsek regime. Each curve tracks a single coefficient across the completed WSINDy runs $N \in \{50, 100, 200, 300\}$; the axis extends to N=500 and 1000 to indicate the currently missing WSINDy outputs at those counts. The continuity coefficient $\nabla \cdot \mathbf{p}$ (blue) stays near -1 on the completed runs, while the momentum pressure and advection terms show larger variation.

Summary of findings. Across both robustness studies, WSINDy identification proves remarkably stable: the continuity backbone $\nabla \cdot \mathbf{p} \approx -1$ is recovered in every completed WSINDy noise and particle-count run, with density $R^2_{\mathrm{wf},\rho} \geq 0.9982$ in all completed runs. The ROM forecasting picture is more nuanced. MVAR maintains $R^2 > 0.98$ on Gaussian ICs throughout, but its mean-R^2 degrades on non-Gaussian families—particularly for the blackhole attractor at high noise and for pure Vicsek at N=500. LSTM is the most fragile component: it degrades gracefully in the gas regime but diverges for the blackhole at $\eta \geq 0.50$ and for pure Vicsek at $N \geq 200$. These results confirm the main-text conclusion (Section 7.3) that MVAR is the more reliable ROM, while LSTM's expressivity becomes a liability when the latent dynamics shift away from the training distribution.

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